

A Message-Passing and Load-Sensitive Contention-Based MAC Protocol for Ad-Hoc Wireless Networks

by

Ariel Smoliar

This thesis is submitted in partial satisfaction of the requirements for the degree of
Masters of Science

in

Electro-Optical Engineering

in the

Faculty of Engineering

of

Ben-Gurion University of the Negev

Supervised by:

Dr. Yehuda Ben-Shimol

December 2011

Abstract

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This thesis presents a novel MAC protocol based on message passing for wireless ad hoc networks that is adaptive to the load of the stations within a cluster. The novelty of the protocol lies in its ability to resolve collisions between stations transmitting at the same time, no matter if it is a master or a slave station within the cluster and in its compatibility to new algorithms developed at the network level to form a backbone-based dominating set. The cluster head (master station) in the MAC layer is part of the network's backbone at the network layer.

In the protocol the channel access time is calculated as a function of the station load that depends on local parameters of the station, delay time of the messages and the capacity of the queue. The master station initiates a session by sending an HLIR message to all its 1-hop neighbors (slave stations). All the stations receiving the HLIR message can complete on the channel access (including master station with internal mechanism) and send back a request message which allows the master station to estimate the maximal load in the cluster and maintain the global parameters (maximal queue size and delay of data messages) within the cluster. New stations joining the cluster can calculate their load using the global parameters sent by the master station. The development of the protocol was based on a previous work which presented, for the first time, the mechanism for the estimation of a station's load. More advanced mechanisms and aspects of cross layer design between the MAC and the network layer were added to the current protocol and implemented to enable the processing of the protocol in changing topologies, joining and leaving stations to and from the cluster as a result of mobility and wake-up processes. A network simulator was used to evaluate the protocol utilization, which was found to be higher than 90% in an intra-cluster scenario.

Acknowledgements

I would like to thank my supervisor Dr. Yehuda Ben-Shimol, for opening the world of communication protocols to me and for his willingness to constantly guide through the process of the work.

I thank my parents for supporting me during different periods in my life while working on my thesis and for their endless belief in my way.

Ziva, my love, words could not describe how much I appreciate your encouragement and complete faith in my decisions during this period. I owe you the finalization of the work and everything that I am.

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List of Abbreviations and Acronyms

ACK - Acknowledge

ARQ - Automatic Repeat Request

AT - Access Time

BW - Bandwidth

CDMA - Code Division Multiple Access

CDS - Connected Dominating Set

CDSC - Connected Dominating Set Based Clustering

COL - Collision

CR - Contention Round

CSMA - Carrier Sense Multiple Access

CSMA/CA - Carrier Sense Multiple Access with Collision Avoidance

CTS - Clear To Send

DS - Dominating Set

EMR - Emergency

ESG - End Slave Grant

estLoad - Estimated Load

FDMA - Frequency Division Multiple Access

FIFO - First In, First Out

FPRP - Five Phase Reservation Protocol

HLIR - High Load Indication Request

MAC - Media Access Control

MACA - Multiple Access with Collision Avoidance

MANET - Mobile Ad-Hoc NET-work

RTS - Request To Send

WCDS - Weakly Connected Dominating Set

Keywords

Cluster, Cross-Layer, MAC, WCDS, Load-Balancing, State-Machine

Chapter 1. Introduction

An ad-hoc network is a collection of wireless mobile nodes (stations or packet radios) forming a network without the aid of any centralized administration or standard support services. The nodes form a fully distributed network and can communicate with each other either directly or, due to the limited propagation range of each mobile node's wireless transmissions, through relay nodes without relying on any pre-existing network infrastructure. Some of the main characteristics of ad-hoc networks are: peer-to peer communication, multihop propagation, dynamic connectivity, low-power consumption, and the ability to operate "anytime, anywhere".

Applications for ad-hoc networks emerge in many aspects of our life starting from conferencing, home networking, emergency services (applications in healthcare IT), embedded computer and envision applications like range extension for cellular telephones and applications in digital battlefield communication.

New Medium Access Control (MAC), unique topology (backbone) construction and its maintenance are needed for ad-hoc network to support efficient communication. Topology maintenance, mobility support and routing algorithms in ad-hoc networks are of the major and most challenging problems due to the dynamic topology of the net. One major drawback of most of the known routing solutions (network layer) is that their underlying algorithm ignores the properties of the MAC layer protocols and its abilities.

The goal of this work is to develop a unique MAC protocol with cross-layer features for wireless ad-hoc networks that will be based on load-balancing mechanism for inter-cluster scheduling and load-based contention resolution (LOCO) MAC that were developed earlier at BGU [1].

In this work we focus on the MAC layer for ad-hoc networks, where stations compete to gain access to the shared medium (a single channel) and transmit their data to other stations. We developed a protocol that is relatively robust against some of the most disturbing problems of MAC protocols for ad-hoc networks, namely, active hidden terminals, while providing fair and adaptive resource sharing among network stations. We also designed the state machines of the protocol on message passing mechanism and the protocol is designed to be compatible with new backbone and routing algorithms being developed in the field. To the best of our knowledge, the combination of MAC mechanisms we developed for achieving this were not utilized so far by any other known protocols.

The remainder of this work is organized as follows: In Chapter 2, we give a general background and literature survey. Review of MAC protocols, algorithms for forming backbones and clusters in ad-hoc networks (network approach) is presented. We also bring the motivation for development of the protocol and its strengths in intra-cluster and inter-cluster scheduling and found the base for future research based on the cross-layers features of our MAC protocol. In Chapters 3-5 we provide a detailed description of the protocol from its basis and the unique combination mechanisms which make it suitable for networks with changing topologies. We analyze different scenarios within a cluster with master and slave stations in the presence of hidden and exposed terminals, followed by detailed timing diagram examples. Chapter 6 presents the finite-state machines of the protocol for the master and the slave stations. In Chapter 7 we describe the modules in the protocol and explain how they were implemented for the station in the simulation. The interaction between the layers is explained by describing the messages and the notifications sent between them. In Chapter 8 we show results of extensive simulations of a star topology (cluster) conducted using the Omnet++ simulator and the analysis of the performance of the protocol. In Chapter 9 we summarize the work and give recommendations for future work.

Chapter 2. Background

This chapter is devoted to present the background for the research and design challenges we faced during the work on the protocol and to establish the base for future research. A wide literature review is presented under two main sections, MAC layer and network layer, in order to cover all the relevant disciplines for the design of our protocol: MAC protocols, clustering and backbone forming in wireless networks, cross-layer approach and routing. In each section, we also review briefly some of our approaches towards solutions for the design challenges -- approaches that are innovative in their way of implementation and multi-disciplinarily.

There are numerous MAC protocols that address both hidden and exposed terminals as well as protocols that attempt to achieve fairness and/or QoS. In [2] the mechanisms used by MAC protocols that may be suitable for ad-hoc networks were categorized into three categories: (a) Handshake-based schemes, (b) Signaling schemes, and (c) Time slot reservation schemes. The developed MAC in [2] integrated many of the underlying ideas proposed in these protocols and presented the idea of load balancing as a scheduling mechanism within a cluster which will be the base for our protocol. This MAC is a slot-based protocol and is not suitable for the varying nature of mobile wireless networks that we aim to be compatible with.

Moreover, a virtually structured backbone of the wireless network based on connected dominating set gains popularity in the recent years as a solution for reduction the communication overhead, for example keeping the routing tables updated in a changing topology environment, and increasing the network maintenance efficiency. Many algorithms deal with finding the smallest set and other computational considerations. In order to integrate between a network layer based virtual backbone and our MAC protocol, we had to develop new mechanisms and adapt a cross-layer designing approach.

2.1 MAC Layer

The MAC protocol is responsible for coordinating the access of active nodes in the network and on how reliably and efficiently data can be transmitted between two nodes along the path in the network. In [1] and [2], a detailed and extensive MAC protocols classification is given. We distinguish between MAC protocols at the highest level of classification by contention-free schemes (TDMA, FDMA, CDMA, etc.) where certain assignments are used to avoid contentions, more applicable to static networks and/or networks with centralized control, and random access contention based protocols (MACA, MACAW, CSMA/CA, IEEE 802.11, etc) that are aware to the risk of collisions of transmitted data.

The wireless communication channel is prone to errors and unique problems such as the hidden-terminal problem, the exposed-terminal problem, and signal fading effects. Our MAC protocol which is contention based attempts to address especially the effects of fairness in the presence of hidden and exposed nodes, which is a major problem in ad-hoc wireless networks. A hidden terminal is defined as a node that is out of the range of the transmitter, but within the range of the receiver. An exposed terminal is defined as a terminal which is out of the range of the receiver, but within the range of the transmitter. As illustrated in Figure 1, node B is within the range of nodes A and C, but nodes A and C are not in each other's range. If node A is transmitting to B, node C, being out of A's range, cannot detect carrier and may therefore send data to B, thus causing a collision at B. This is referred to as the hidden-terminal problem, as nodes A and C are hidden from each other. In another scenario, node B is transmitting to A. Since C is within B's range, it senses carrier and decides to defer its own transmission. However, this is unnecessary because there is no way C's transmission can cause any collision at receiver A. This is referred to as the exposed-terminal problem, since B being exposed to C caused the latter to needlessly defer its transmission.

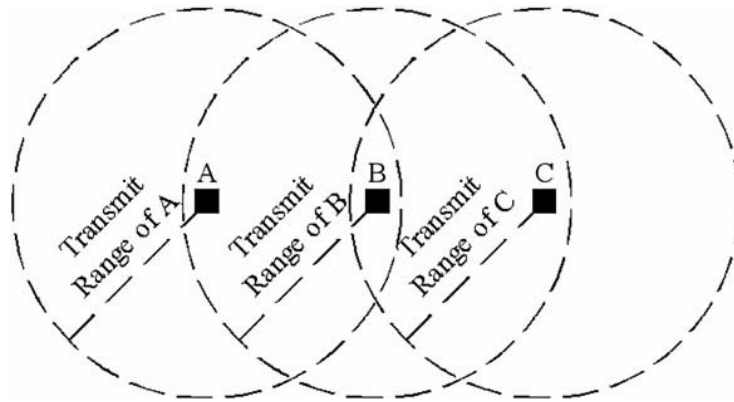


Figure 1 - Illustration of the hidden and exposed terminals problems [2].

The effect of hidden nodes on common MAC protocol mechanism such as RTS-CTS messaging (also implemented in our protocol and will be further explained), is demonstrated in the following scenario (see Figure 2). Nodes A-B, B-C, C-D are 1-hop neighbors (meaning that their transmission ranges cover each other). Node A sends an RTS packet to B, and B sends a CTS packet back to A. At C, however, this CTS packet collides with an RTS packet sent by D. Therefore C has no knowledge of the subsequent data transmission from A to B. While the data packet is being transmitted, D sends out another RTS because it did not receive a CTS packet in its first attempt. This time, C replies to D with a CTS packet that collides with the data packet at B.

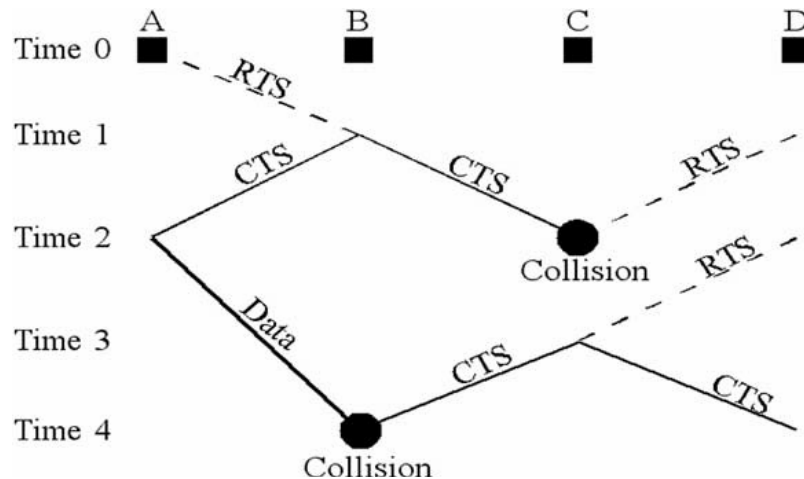


Figure 2 - Illustration of failure of RTS-CTS mechanism in solving Hidden and Exposed terminal problems [2].

Time slotted protocols are usually based on time slots that are divided into multiple frames and several inter-frame spacing slots with increasing order of lengths. The Five Phase Reservation Protocol [8] is a distributed TDMA slot assignment algorithm which was designed for dynamic slot assignments, in which the real time is divided into a series of pairs of reservation and data transmission phases. During the reservation phase, the protocol assigns the slots of the next data transmission phase to the nodes that have data to send. Thus, the actual data transmission occurs using the slots assigned in the previous reservation phase. For each time slot of the data transmission phase, FPRP runs a five-phase reservation process for a number of times (cycles) to pick a winner of each slot. The algorithm improves its assignment as it runs for more cycles. In a distributed setting, nodes do not know how many cycles are required to complete the assignment. An algorithm for slot assignment should be applied during the setup phase of the network and a maintenance mechanism for joining and leaving nodes should be operated.

In Slot-based protocols (such as DRAND [3], Z-MAC [4]) and duty-cycle MAC protocols (such as RMAC [5], T-MAC [6], and S-MAC [7]) control frames are sent along the data forwarding path to let the nodes to learn about their state (awake/sleep) and synchronize the internal clocks. On the contrary, our MAC protocol is completely based on message passing and does not require any slot or clock synchronization (will be further demonstrated in the chapter discussing protocol's state machines). The message passing concept gives the network layer more flexibility in adding new nodes and treating variable link characteristics, compatibility with backbone forming algorithms, and faster adjustment to changes in topology due to the dynamics of the mobile wireless network. From the perspective of fairness, all the mentioned protocols ensure that multiple transmissions will take place simultaneously, however they do not ensure that the most loaded ones would be the ones that reserve the slot with proportion to their load, or that a fair bandwidth share will be granted to each transmitting node.

In order to achieve efficient routing, better topology maintenance and topology which is suitable for our MAC protocol, we are basing our network design on virtual network infrastructure, which organizes nodes into hierarchy (instead of flat network). The

construction of this infrastructure is the primary application of Connected Dominating Set (CDS) in wireless networks [16] (will be explained in the network model section). Broadcasting based on different constellations of dominating sets (DSs) is a promising approach, where only nodes in the DS need to relay the broadcast packet (and border nodes in cluster based algorithms).

Our network is based on a hierarchical model with three kinds of node functionalities: master node, border (in network layer context) and slave node formed in a unique backbone that fits perfectly our MAC restrictions called Weakly Connected dominating Set [17,19]. Within the cluster a master node can also function as a relay node, however between clusters only border nodes function as relay nodes. A hierarchical concept in Connected Dominating Set Based Clustering (CDSC) can be found in [20], where every node belongs to a single cluster (with a cluster head).

We are also exploiting the advantages of clusters formed by WCDS for scheduling and maintenance. Our cross-layer protocol capabilities enable us to handle intra-cluster scheduling using our MAC advantages and handle inter-cluster scheduling using the network layer without need of clock synchronization.

The heart of intra-cluster scheduling is based on the load function in each node and its implementation in the MAC protocol. We measure the load of the nodes using a load function serving our protocol purposes well, which are to balance the loads among the nodes in the cluster (deciding on scheduling of transmission based on load value) and remain stable under partial connectivity or high traffic conditions. Each packet in the node's queue contains information on how much accumulated time this packet spent in the network (simulator time) and what is the packet length (measured in bytes). The accumulated time spent in the network can be larger than the time spent in the node's queue, since it accumulated all the relaying times as well. All the nodes in the network use the same load calculation function and the node with the highest load will be granted access to the channel by the master. To the best of our knowledge, this is the first work on scheduling based load balancing for load calculation in ad-hoc networks. While continuing to analyze the different stages of development of the protocol, we can see that having a unique scheduling algorithm inside the cluster, and probably with good

utilization performance, is not enough. Also, the inter-cluster scheduling should not be ignored.

[17] Suggests to grow several trees of nodes in parallel, but in our case we are facing real world problems, such as collisions, link failures, and joining of new nodes; thus we have to develop an algorithm keeping the utilization of the protocol as high as possible. The concept for handling multi-token in WCDS backbone was suggested to receive the highest throughput possible while taking the advantages of and coping with the restrictions forced by the MAC protocol.

We had to face many challenges during our work to keep functionality of the state machines under the varying states of ad-hoc networks. There are two major issues involved in electing masters and border nodes dynamically: one is to define what is the criteria for dynamically electing the master (even assuming that all the nodes have all the information they need and can unanimously reach a common decision), and the second is to define the algorithm for actually electing, maintaining and updating the dynamically selected masters and border nodes.

There are numerous articles in the literature that are based on a combination of different criteria for dominator (master node) selection rules [9], [16], [30]:

- Highest/smallest ID between nodes' closed neighbor set
- Node degree (1-hop, 2-hop neighbors) based rules
- Energy level based rules

However none of these criteria is implemented in a realistic MAC protocol.

In our model, we are given the network as a graph and a subset that is the weakly-connected dominating set. Each node in the network has a unique MAC address and ID. We are not dealing with network forming/initialization and the functionality of the node is derived from cluster/backbone maintenance restrictions. It must be noted that our cross-layer protocol design objective is not reducing the size of the backbone and keeping the communication cost as low as possible during topology formation/initialization, as in [14] for example.

All the nodes in the network use the same load calculation function (will be further explained) and the node with the highest load will be granted access to the channel by the master node within the cluster.

Implementing the state-machines of the slave and the master nodes was not a trivial task and complicated design and code writing were required. Due to the high complexity of the model, the complete state-machines were implemented in a network simulator and were not described merely as executed at each node. To evaluate the performances of our protocol in various scenarios, we implemented the network scenarios in OMNeT++ 3.3 with INET framework [18].

Intra-Cluster Scheduling

The load function is the core of the intra-cluster scheduling mechanism in our work. The function takes into consideration both the nodes' (slave/master) queue length and the accumulated time a data packet spent in the network. Each node in the network has two queues, one for emergency messages and the second for data and control messages. We assume that each data message in the queue contains information on how much accumulated time it spent in the network and what is the message length (measured in bytes). Each time a new message is received in the node, it is forwarded to the higher MAC and pushed to the buffer. If the new message has the highest accumulated waiting time, it is automatically placed at the top of the queue and sent immediately upon receiving a transmission grant. By knowing the two main parameters, the queue length and the longest delay, each node will be able to calculate its load and to get a grant for accessing the channel by the master. Control messages are sent in another mechanism without getting a grant, and they don't have any effect on the node's load. In order to calculate the load function, each master needs to know and distribute the maximum values of the parameters (maximum queue length and higher message accumulated time between its closed neighborhood nodes) to all its 1-hop connected neighbors. The most loaded node wins the grant for transmission in the current session of the protocol.

2.2 Network Model & WCDS Backbone

In a clustering scheme the mobile nodes in a MANET are attributed to virtual groups (clusters) according to certain rules for nodes which are included in the cluster. As previously mentioned, under a cluster structure, mobile nodes may be assigned a different status or function. Once the nodes are partitioned into clusters, there are numerous techniques to maintain the cluster organization, especially in different mobility scenarios. The main advantages of forming clusters are minimization of routing overhead and maintenance in varying topology environment.

The most common clustering schemes are:

- **DS-based clustering** - finding the dominating set in order to reduce the number of nodes participating in routing and maintenance.
- **Low-maintenance clustering** - providing a cluster infrastructure for upper layer applications with minimized maintenance cost.
- **Mobility-aware clustering** - based on nodes' mobility behavior; for example, nodes with low speed are assigned to the same cluster.
- **Energy-efficient clustering** - balancing energy consumption, very common in sensor networks.
- **Load-balancing clustering** - limiting the number of nodes in each cluster.
- **Combined-metrics-based clustering** - can include node degree, mobility, energy and size considerations.

Our ad-hoc network is based on a dominating set concept in which the intra-cluster nodes are processing the MAC protocol when the cluster head has the token to transmit. The routing information and the data messages relay on the cluster heads. The network is represented by a graph where the vertices represent the individual nodes and there is an edge between two nodes if they are within transmission range of each other.

2.2.1 Graph Domination

The method for forming clusters in our work is based on the dominating set. A dominating set of a graph is a vertex subset, so that every vertex is either in or adjacent to a vertex of . A vertex of is said to dominate itself and all adjacent vertices. The idea behind backbones based DS is to seek the dominating set and then grow it into a connected dominating set and to obtain as smaller a Connected Dominating Set (CDS) as possible, in Figure 3 we can see a network with backbone based CDS. When table-driven routing is applied, only nodes in the CDS are required to construct and maintain the routing tables. In on-demand routing, the route search space is limited to the CDS.

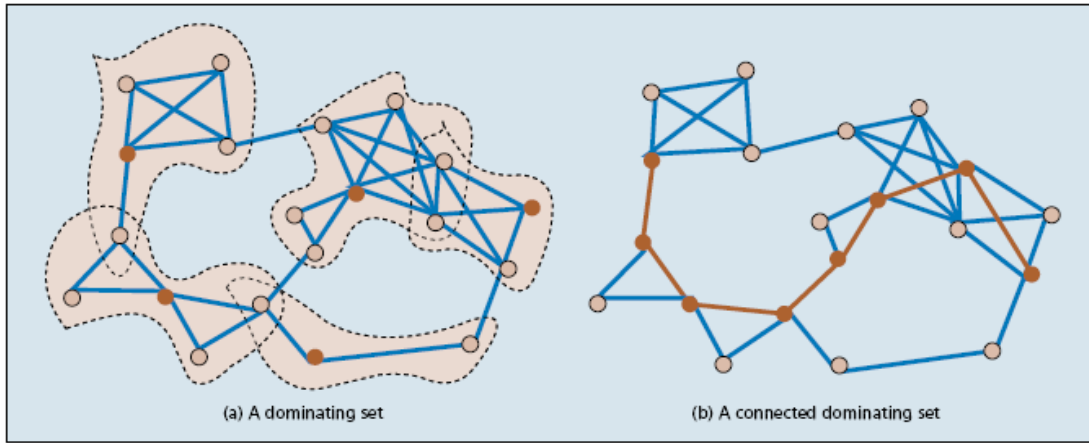


Figure 3 - Dominating set illustration [19].

The subgraph weakly induced by is the graph, is the closed neighborhood of the set. includes the vertices in and all of their neighbors as the vertex set. The edges of are all edges of which have at least one end point in. A vertex subset is weakly-connected dominating set (WCDS), is dominating and is connected.

We can see a network with backbone based WCDS with 2-hop distance between the nodes.

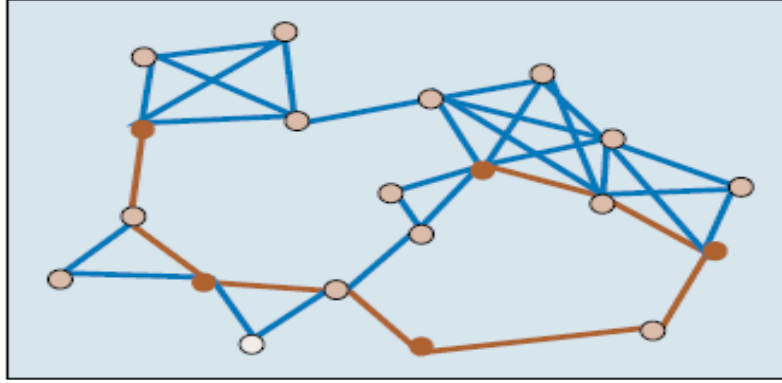


Figure 4 - Network structure form a WCDS [19].

Guha and Khuller [38] proposed two centralized greedy algorithms for finding suboptimal connected dominating sets. Das and Bharghavan [39] [40] provided the distributed implementations of [38]. Wu and Li [9] improved the distributed algorithm presented in Das and Bharghavan's work to a localized distributed algorithm. Wu and Li's algorithm works in two phases, in the first level a node marks itself as a cluster head if any two of its neighbors are not connected to each other directly. In the second level, a marked vertex v changes its mark to ordinary node if one of the pruning rules is met. In [41] two more pruning rules were added to Wu and Li's algorithm [9] and provided more reliable results. In [42] an algorithm to construct CDS with bounded diameters was proposed, the algorithm first finds a maximal independent set and then selects some other nodes as cluster heads in order to build a CDS at the end. Gao et al. [43] also proposed a CDS algorithm which uses the maximal independent set as the basis of construction of the CDS. There are various algorithms existing for clustering in MANETs using dominating sets [21]-[32].

In our work, we apply the following rules for selecting the members of the network's backbone:

Master node eligibility rule - A slave node should become a master node if it discovers, **using only information gathered from local broadcast messages**, that two of its neighbors cannot reach each other directly (1-hop); there has to be a 2-hop distance between the masters.

Border node eligibility rule - A slave node should become a border if it discovers, **using only information gathered from local broadcast messages**, that at least two of its neighbors are masters of two or more different clusters. If it is required as part of maintenance process, the border node will be defined by a message sent from one of the masters.

We are using the members of a weakly-connected dominating set (as cluster heads (masters) and the closed neighborhood of each cluster head as a cluster. Cluster heads are placed in a 2-hop distance and are joined by a border node with relay communications between them and are responsible for scheduling their transmissions by token passing. WCDS as backbone for the network in our protocol development was smoothly adapted to overcome our MAC restrictions.

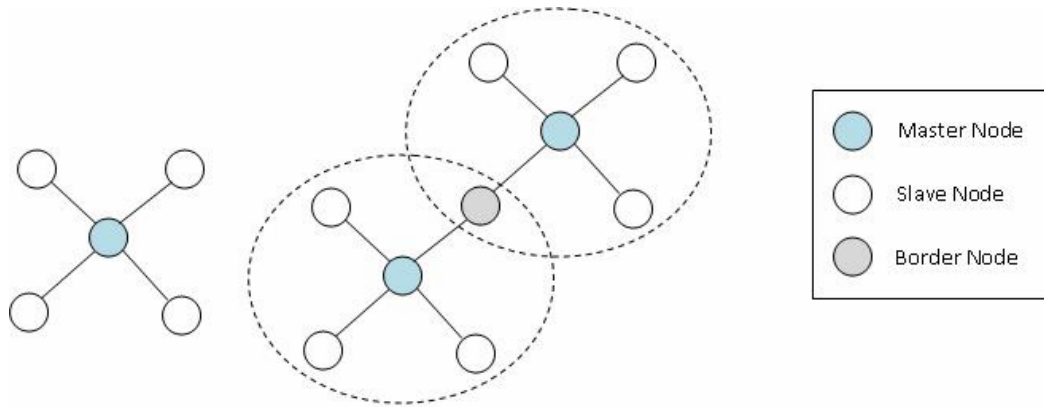


Figure 5 - Network Model - star topology with 4 hidden nodes and two formed clusters.

As an introduction to the detailed explanation on the MAC protocol in the next chapters and its cross-layer capabilities, we present a scenario that exemplifies the required interactions between the network and the MAC layers:

Renormalization of Load Parameters

Due to topology changes and continuous message passing in the network, queues are getting empty or full and the maximal parameters of the cluster, which are kept in every master node, are becoming irrelevant with time. We had to define a unique renormalization process for adjusting the parameters and getting reasonable load indication (implemented only in the MAC layer). There are three scenarios that can occur when a master node receives its token:

- A new master node receives the token and has to start estimation of the maximum load parameters by sending two HLIR messages and the third HLIR message will be sent with the estimated parameters for contention resolution process (the mechanism will be explained in Chapter 3).
- The master node receives the token, starting and completing the session (all messages fall on the access time scale between two defined threshold values).
- The master node receives the token, sends an HLIR message and the contention or distribution of messages is below the lower threshold value or above the higher threshold value. The parameters will be adjusted and an HLIR message with updated values will be sent.

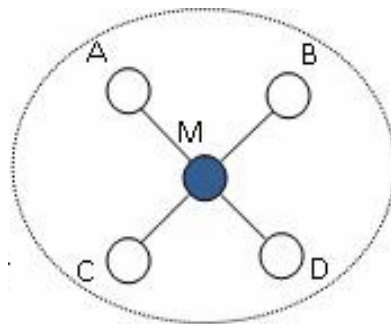


Figure 6 - Cluster with hidden nodes.

A new slave node joining an existing cluster (see Figure 6) will force the node to calculate its load with the estimated parameters from HLIR message. If its internal parameters are significantly higher, a starvation of other nodes in the cluster may occur and a renormalization will have to be performed.

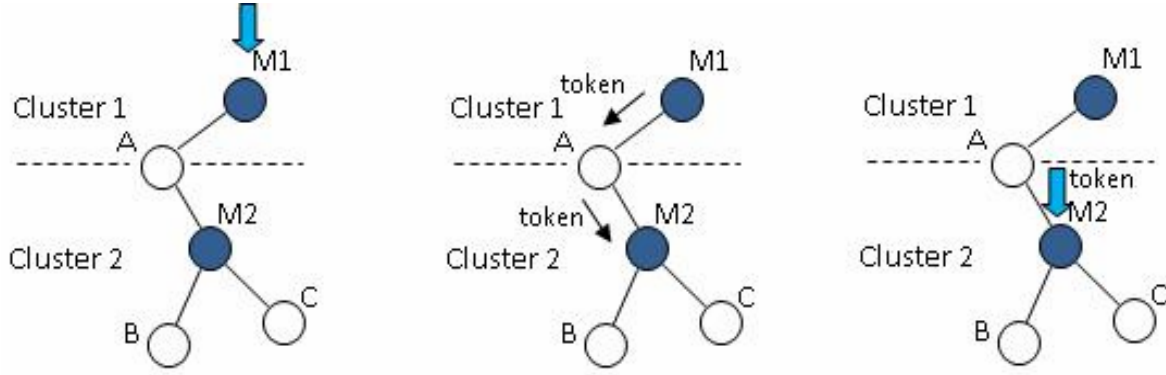


Figure 7 - Token passing between master nodes.

In token passing between master nodes (nodes M1,M2) as shown in Figure 7, a renormalization will take place at each master node receiving the token and getting enable approval from the border node if new nodes were joined and border node A's parameters were changed. The normalization criteria in the MAC layer will be explained in Chapter 4.

2.2.2 Inter-Cluster Scheduling

From [11] using the members of a weakly-connected dominating set as cluster heads and their 1-hop neighbors as a cluster, fits perfectly to our MAC topology restrictions and we can regard the network from a higher level as a graph of clusters based on our MAC protocol. Cluster heads (master nodes) have to share a neighbor that can relay communications between them and schedule their transmission periods. The cluster was divided into two layers - intra-cluster (within clusters) and inter-cluster (along cluster borders). In Figure 8 the formation of WCDS based backbone is demonstrated. The

backbone is formed, master and border nodes are selected according to the eligibility rules, and clusters are formed.

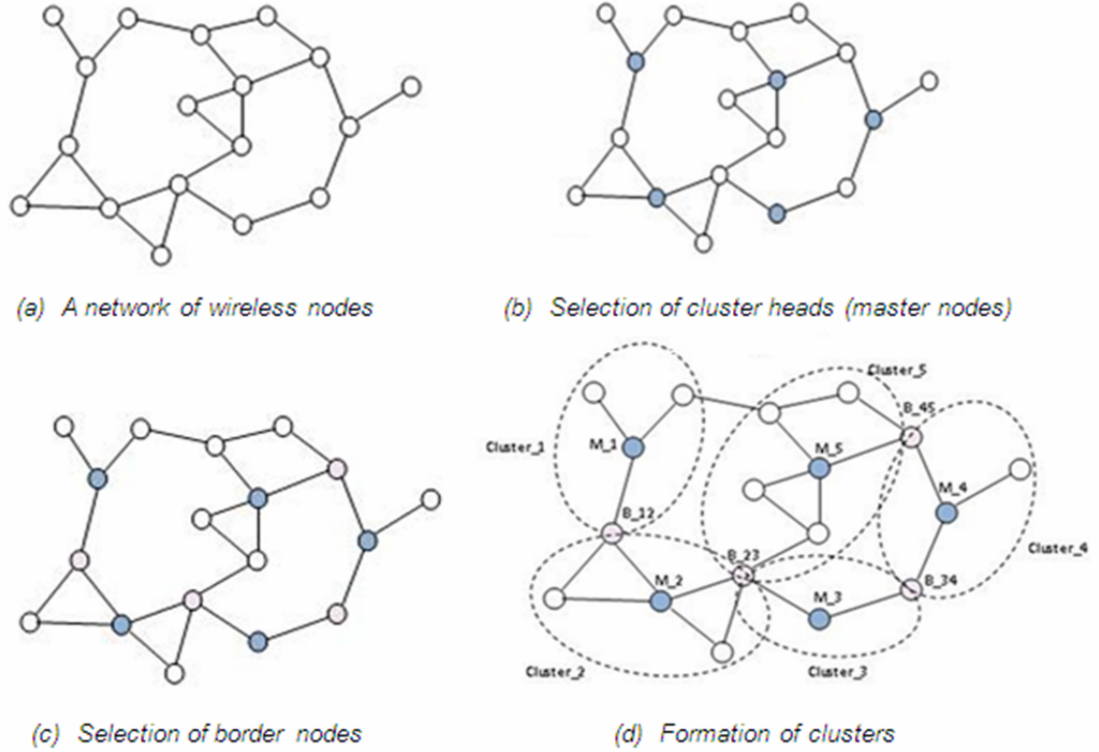


Figure 8 - The process of WCDS Backbone construction and division into clusters.

As illustrated in the WCDS backbone construction process, the network is divided into 5 clusters (noted as Cluster_1-5), in which each cluster has a master node (noted as M_1-5). When the master node receives a token, it competes with its 1-hop neighbors (slave nodes) on the intra-cluster transmission grant, and the protocol is executed during a session as will be further explained in the following chapters (master node can have several sessions). The master nodes of the clusters are located 2-hops from each other and connected via border nodes (B_12 indicates border node between Cluster_1 and Cluster_2). In the intra-cluster layer, the border nodes that are 1-hop neighbors of the master node compete on the transmission grant if they have information to send, and in the inter-cluster layer the border nodes relay communications between clusters. It can be

seen that if two adjacent cluster have a token, collision in the channel may occur. If we take as an example the scenario that master nodes M_2 and M_3 transmit simultaneously a data message, a collision will occur in node B_23 (hidden terminals problem). In order to reduce the number of collisions between clusters and increase the utilization of the network, the protocol that will be implemented in the network layer based on our MAC protocol, should contain algorithms for scheduling the transmission between clusters. A token passing mechanism in the network layer will synchronize between transmissions of the clusters formed in the network. In Table 1 all the combinations of which clusters allow internal transmission simultaneously without collisions can be seen.

Table 1 - Masters/clusters scheduling table

Enable	M_1	M_3	M_4	M_5
M_1		+	+	
M_2				+
M_3	+			
M_4		+		
M_5				+

Cluster M_1, for example, can have the token simultaneously with clusters M_3 or M_4 in order to prevent collisions in the border nodes. Cluster M_1 and M_5 are not connected via border nodes, however they have two slave nodes that are connected. If both nodes will receive a transmission grant during sessions in the clusters, a collision may occur in the nodes.

2.2.3 Maintenance

In this section, we present the strengths of our protocol and its cross-layer features in maintaining existing network based weakly-connected-dominating set backbone. We classify the possible topological changes based on [7] to edge/vertex up and edge/vertex down.

Wake Pp Process - Edge/Vertex up

Joining to a Slave Node

During a node's wake up process it performs carrier sensing. If no signal was sensed after a defined period of time, it sends an EMR message. If a slave node hears the message it treats it as unexpected data, forwards the message to its network layer and adds the node to its 1-hop neighbor list. In Figure 9, node A will remain a slave while node N will join as a master node and a notification on a new master node will have to be sent. The state of node N (slave/master) will be updated in a specific field in the EMR message. A collision (hidden node problem) may occur when M1 has a token and sends data.

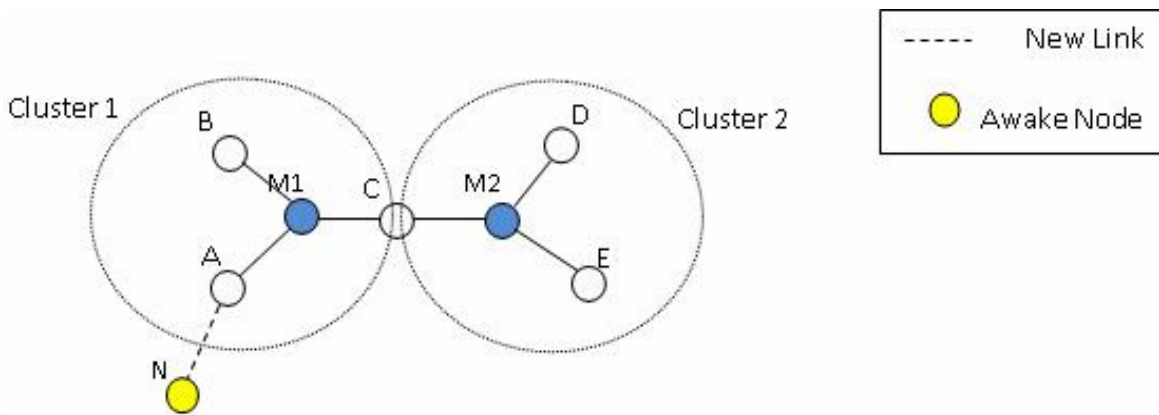


Figure 9 - Interaction between new master node and existing slave node.

Joining to a Master Node

If a master node hears the EMR, it also treats it as unexpected data, forwards the message to its network layer and adds the node to its 1-hop slave neighbor list. A node should join to a master node which is the cluster head, as a slave. Joining as a master node will lead to two cluster heads and complications with fairness of token passing between clusters. When node N hears HLIR message from M1, it adds the master node as a neighbor, changes its cluster_ID according to the master node's ID and takes part in the contention process. Figure 10 illustrates a scenario in which a node is joining to an existing cluster as a 1-hop neighbor of the cluster head.

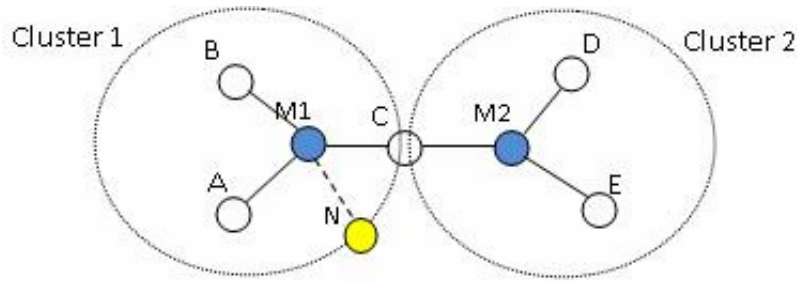


Figure 10 - Interaction between new slave node and existing master node.

Joining to a Border Node

A border node is responsible for inter-cluster scheduling by coordinating the token passing and it also functions as a relay node. When joining to a border node as seen in Figure 11, node N has to join in master mode, because it will not be able to compete for transmitting grant in 2-hop distance from a master node in our model. Node C will have to add the new node to its neighbor list and modify the token passing scheduling.

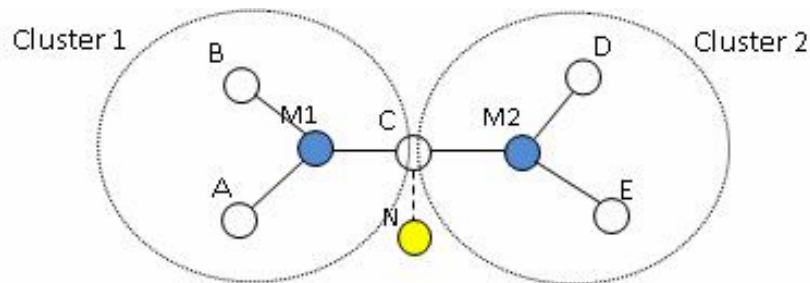


Figure 11 - Interaction between New master node and existing border node.

Combination

Joining in a more complicated scenario (as seen in Figure 12) has to take into consideration the following restrictions:

1. Maintaining WCDS topology

2. One master node per cluster
3. Defined border nodes
4. State update (master/slave/border)
5. Inter-cluster scheduling update

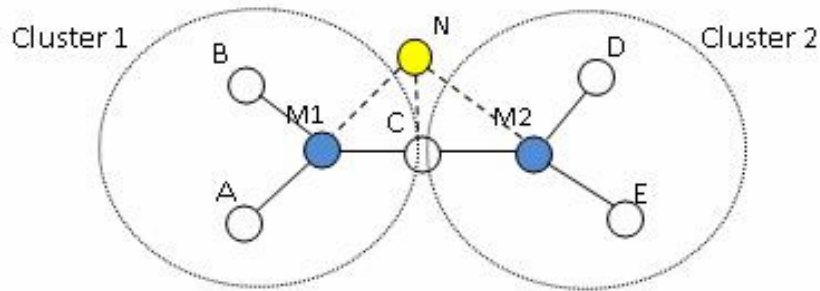


Figure 12 - Complex interaction.

Broken Link - Edge /Vertex Down

Let's look at the following star topology in Figure 13, with one master node (M) and four slave nodes (A, B, C, D), a link between two nodes indicates an active wireless channel:

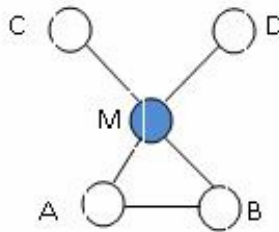


Figure 13 - Star topology with connection between two slave nodes.

List of 1-hop neighbors of the nodes in the cluster:

$$N(A) = \{B, M\}$$

$N(B) = \{A, M\}$

$N(C) = N(D) = \{M\}$

$N(M) = \{A, B, C, D\}$

When node A needs to send data to node B, it has to get a grant from the master node M and then send the data directly to node B. After receiving the data, node B returns an ACK message to node A (MAC level). The ACK message is sent from node B without getting a grant for transmission from the master node.

After sending the data, node A starts a timer for getting the ACK message. When the timer expires, it can retransmit the data to node B if it still holds the grant for transmission.

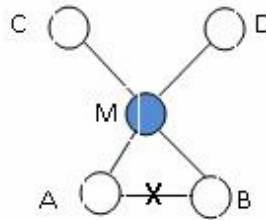


Figure 14 - Broken Link between nodes A-B.

If the link between nodes A and B is broken (as seen in Figure 14) and node A wins the grant (node B is still considered as a 1-hop neighbor of node A), after its retransmission attempts directly to node B (the number of retransmissions is limited to three), the network layer gets a failure notification, changes the destination MAC address to master node's address with destination ID and redirects the sending through the master node as a relay node.

The master node which does not have a grant (but has a cluster token) for transmitting, treats the message as unexpected data, forwards it to the network layer and sends it to node B. This mechanism guarantees a reliable message sending in the cluster, based on slave-master connectivity and cross-layer design for bypassing broken links and forwarding the data from source to destination. Nodes A and B will have to update their 1-hop neighbors list after the session.

As a summary for the review presented in the chapter, we propose a new MAC protocol that is compatible with the most recent backbone forming algorithms for MANET. The protocol attempts to address especially the effects of fairness in the presence of hidden terminals. Its intra-frame scheduling mechanism allows simple and fair joining and leaving of stations without any use of extra control messages. It should also be noted that the protocol does not require TDMA slot synchronization/assignment as most of the mentioned protocols in the review, because it is based on message passing. Robust state machines were developed taking into consideration cross-layer design advantages in order to enable the future implementation of the network layer and validation of routing algorithms (as [33]-[37]). The motivation and conceptual model for development and implementation of network layer based on the cross-layer features of our MAC protocol are presented in the chapter and can be a strong base for future research.

Chapter 3. MAC Protocol Description

3.1 Basic Definitions of the MAC Protocol

In this chapter we will summarize basic definitions of the MAC protocol such as: frame structure, fundamental parameters being used by the protocol, and messages description. We will also analyze the load function, the intra-frame scheduler, and its components.

Frame Structure

The uniqueness of our protocol is that it is based on message-passing with defined time periods, not on time slots as in [1]. Message passing enables more changeability to topology changes within the cluster. The frame is divided into Contention Rounds (CRs), each consists of the following time periods:

- Time to transmit a control message from the master to one-hop away slave (T_{MaCtl}), will be defined later as HLIR message.
- Messages propagation time (τ).

- Contention Period (CP) for station hearing the HLIR control message to compete (T_{CP}).
- LastRTS - last RTS signal that can be sent in order not to exceed the maximal contention period.

Figure 15 illustrates the structure of the frame:

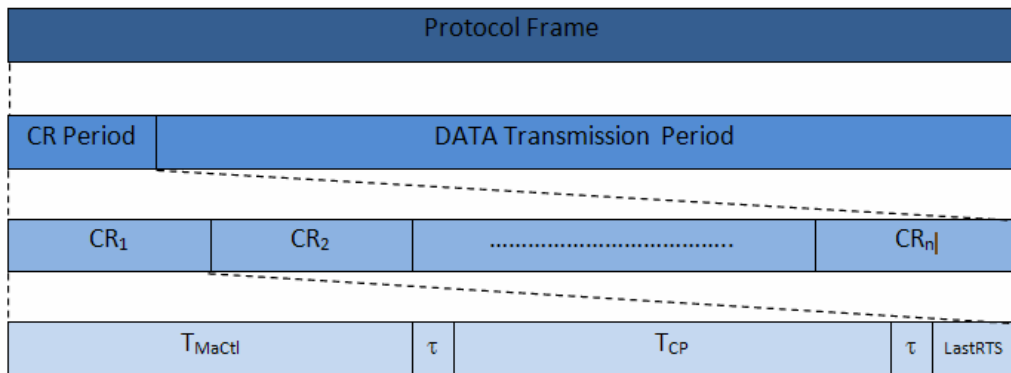


Figure 15 - Structure of a frame in the protocol.

The frame of the protocol defines a session in which the master node functions as the cluster head. During each session contention rounds take place between the nodes in the cluster. At the end of the session the most loaded stations of all the nodes or one of the most loaded (in case of collision) receives the grant for transmission. The frame of the

protocol includes the contention rounds and the transmission period. It is very important to emphasize that because the protocol is based on message-passing mechanism, the session period is not fixed and depends on collision occurrences and on the number of messages in the queues. A node changes its state to master mode from slave mode and receives the number of allowed sessions from the network layer as part of the cross-layer interactions.

The time period dedicated for contention in the protocol is called the Contention Round Period (CR) and is divided to several periods. In each period an independent contention round can occur, and it is defined as CR_i .

During CR_i all the stations in the cluster (slaves and master) that have any information to transmit compete for channel access. The transmission grant is given to the winning station based on the load criteria that will be described in more detail. If a collision occurred, the most loaded stations would compete for transmission grant in the next contention rounds. $CR_2, \dots, CR_N$ will be indicated as CR_{others} .

The following contention rounds, following the collision occurrence in the first round, are based on random access. It was shown in [1] that 5 is the upper bound for needed contention rounds in a session, until a station receives a grant for transmission from the master.

When a station wins the grant, a time period is allocated for its data transmission, defined as Data Transmission Period. The station transmits its data using a Stop & Wait policy. Depending on the state (master or slave) of the winning station the transmission period includes the control messages for transmission approval (CTS message from master node) and control message from the slave which indicates on finalizing the transmission (ESG message from a slave node). Table 2 summarizes the central parameters of the protocol that are used in CR_i to calculate the load of nodes in the cluster.

Table 2 - Central parameters of the protocol

Parameter	Description	Comments
Q_i	Queue length (bytes) of each station	Local parameters of a mobile station in the protocol, taken from its queue of messages
$D_{\max_i}$	Maximal delay (sec) of a message in the queue of a station	
$\max\{Q_i\}$	Maximal queue length (bytes) of a station in a specific cluster	Global parameters of a cluster. Estimated by the cluster head (master node)
$\max\{D_{\max_i}\}$	Maximal delay (sec) of a message in the queue of a station among nodes in a specific cluster	
	Load balancing parameter	Parameter used in the load balancing function
$load_i$	Load parameter of a station	Calculated by the load balancing function or defined as a random value after collision
T_i	Waiting time interval (Access Time) of a station before sending RTS signal	Inverse relation to the load value of the station
TS1	Estimation of the shortest waiting time (Estimated Shortest Access Time)	Defined by the master node. Indicates the beginning of receiving the first RTS signal
stLoad	Estimation of the most loaded node among the stations in a specific cluster	Defined by the master node. Relative to the TS1 parameter

3.2 Control Messages

As was mentioned in the previous sections, the protocol is based on message passing mechanism in which unique state machines handle the sending and receiving of control messages.

Definition of Control Messages

- *Highest Load Indication Request (HLIR)*

A broadcast message that is sent from a master node to its 1-hop slave nodes neighbors. The message indicates the initialization of a new session in the protocol. The message contains valid information for the contention mechanism: α (load parameter) and the maximal global parameters (maximal queue length and delay of a data message) within the cluster.

Message length: 20 Bytes

- *Request To Send (RTS)*

The slave node transmits a short RTS pulse to the master. The timing for transmitting RTS messages depends on the load function in CR_{1st} , and is randomly selected in CR_{other} .

Message length: 20 bits

- *Collision (COL)*

If multiple RTS messages from more than one slave collide in the channel, the master broadcasts a COL message. This message initiates CRs from the second round (CR_{other}). The message contains a field with the estimated load.

Message length: 20 Bytes

- *Clear To Send (CTS)*

A broadcast message that is sent from the master to its 1-hop slave nodes neighbors. Upon receiving this message, the station to which the master granted permission to transmit begins transmission of its useful data. The message contains a field with the estimated load.

Message length: 20 Bytes

- *Acknowledgment (ACK)*

ACK is a unicast message. It indicates that the data sent to the station was received correctly. The ACK message is sent by the destination station to the source station in the MAC level. The message is send without a grant for transmission.

Message length: 14 Bytes

- *End Slave Grant (ESG)*

ESG is a unicast message transmitted by the slave node to the master after the end of its transmission period or after its queues are empty. This message indicates the end of the session.

Message length: 14 Bytes

- *Emergency (EMR)*

The EMR message is sent in two cases: when a mobile station leaves its cluster and is close to a new one and for maintenance in the network layer.

(*) Mentioned in the state machine of the protocol but is not implemented in the simulator.

3.3 The Load Parameters of a Station

As was explained above, our protocol is based on the load of each station in the cluster in order to receive a transmission grant and access to the wireless channel. The loaded or one of the most loaded among the stations in the cluster will always win the transmission grant.

We had to define balanced parameters in order to prevent starvation of stations and guarantee fairness among them as much as possible. The protocol is taking into consideration local parameters (each station) and global parameters among nodes within a cluster.

In a session period, the first contention round is based on calculating the load function in each station and in case of a collision it is based random values in further rounds:

- $\text{Load}_i(\text{CR}_1) = f(Q_i, D_{\max_i})$
- $\text{Load}_i(\text{CR}_{\text{others}}) = \text{RAND}[0 \div 1]$

3.3.1 Local Parameters

Two parameters are taken into consideration when calculating the load of a station:

- Queue size
 - Calculated as the sum of all messages' length in the queue
 - Marked as Q_i
 - Measured in [bytes]
- Delay
 - Maximal system delay of a message, which is the time since the creation of the message in the simulation, in a queue of a station
 - Marked as $D_{\max_i}$
 - Measured in [sec]

For example, if messages in a queue contain the following values for length and delay: $\{[95,2],[10,1],[85,7],[90,5]\}$ the local parameters of the station will be $[280,7]$.

3.3.2 Global Parameters

The global parameters are calculated by the cluster head (master). These parameters are the maximal values among all local parameters of the stations within the cluster. The

parameters $\max\{Q_i\}$ and $\max\{D_{\max,i}\}$ respectively define the maximal queue length and maximal message delay in a specific cluster. The parameters are updated by the master and are sent in the HLIR message. Stations which receive the HLIR message from the master extract the parameters in order to use them in the load function and calculate the station's load.

3.4 The Load Function

The expression for calculating the load of the station is given in expression (1):

(1)

The load function is composed of the local parameters of the station and the global parameters in the cluster. The function represents the convex combination of the station's normalized queue size measured in bytes, and the normalized maximum delay time any packet has been waiting in the network, measured in sec. The convexity parameter α (equilibrium parameter) defines which of the two parameters is more significant. When $\alpha=1$ the queue length is the main component and when $\alpha=0$ the delay component is the main component of the function. When $\alpha=0.75$ or 0.5 this guarantees equilibrium between the two components of the function.

The range of the function is between $[0 \div 1]$, where 0 stands for the lowest weighted load and 1 for the highest weighted load the station is exposed to. To be able to normalize the queue lengths and maximal delay time of a station, each master maintains two normalizing values and that correspond to the highest values of queue length and maximal packet delay time currently known to the master (cluster head). The algorithm for updating the normalizing values will be further explained below. The normalizing values are attached to each HLIR message.

The load function was designed to balance between two demands: to give priority to the most delayed message in a cluster so it would reach its destination and to enable the station with the longest queue to release the messages from the queue.

Chapter 4. Unique Mechanisms

In the following section we will examine the 10 main mechanisms of the protocol:

- Access time calculation
- Maximum load estimation in a cluster
- Collision handling
- Estimation and updating the global parameters
- Load comparison mechanism
- Data message transmission policy
- Relay node mechanism

- ESG mechanism
- EMR mechanism
- Carrier sensing mechanism

4.1 Maximum Load Estimation in a Cluster

A station wishing to be granted access to the channel must compete for the permission with all the stations in the cluster. This is done by defining an access time period – the time a station waits before transmitting. The access time is inverse to the load of the station, meaning that a station with a higher load will have a shorter access time. At the end of each access time period a slave station (master stations, which have an inner timer, will be further explained) sends a Request-To-Send (RTS) message to the master station (unicast). We denote $TRUE_AT_i$ as the access time of a slave station i . $TRUE_AT_i$ corresponds to the time in which a slave station attempts to send a RTS message. The calculation of $TRUE_AT_i$ is:

(2)

When a collision occurs the access time is calculated differently for the following CR:

(3)

Where CR is the contention period time (fixed). Note that the access time is shorter for higher loads.

As was mentioned above, the access time is an inverse of a station's calculated load. In the following graph (Figure 16) it can be seen that a higher load leads to shorter access time. This model guarantees that the most loaded station of the competing stations in a cluster will first send the RTS signal in the contention round.

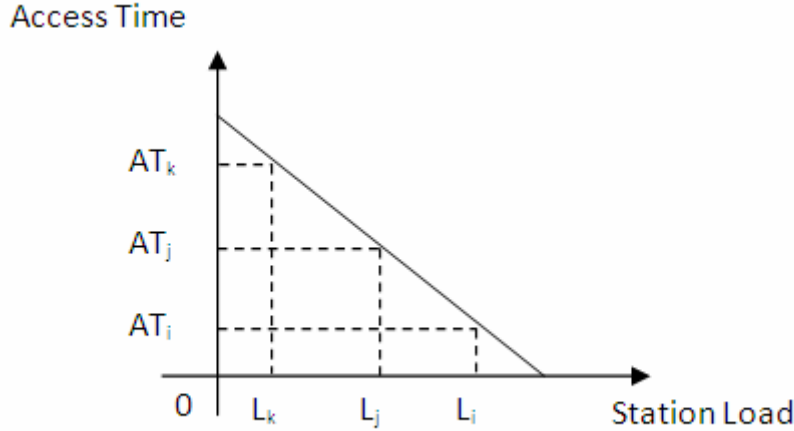


Figure 16 - Varying of access time as function of the load of the station.

Where L is the load value as function of the local parameters. The access time calculation (TRUE/RAND) depends on the contention round that took place. The RTS message contains no data. It is a short modulated energy pulse (a single OFDM symbol would suffice). The RTS pulse reaches the master station after propagation time. The following two practical assumptions are considered:

Assumption 1:

The master station is able to sense when a signal begins to be received (which is less demanding than knowing what the data is). We refer to the time since the end of the HLIR broadcast and the time of the first arriving pulse sensing as RTS_1 (will be defined below).

Assumption 2:

The master station can differentiate between a complete RTS pulse detection and two or more colliding pulses (not how many transmissions collided, but rather that a collision has occurred). The master station may receive no response from its slave station (if none of them has any data to transmit), or it may receive one or more RTS messages from its connected slave station, in which case, the master station must be able to differentiate between competing slave stations. However, since RTS messages do not contain any data, the only way a master station can differentiate between competing slave stations is according to their RTS arrival times.

4.2 Maximum Load Estimation in a Cluster

As was described, each station in the cluster that has some data to transmit (including the master with an artificial RTS signal) sends a RTS signal.

According to assumption 1, the master station is able to sense when data is received. Since $f_i(\cdot)$ is reversible, the RTS_1 can be reverse-calculated into the estimation of the load value of the slave station that sent this message.

Definition of RTS1 Parameter

For each new contention round (CR) during the session, the master station defines a parameter called RTS_1 that defines an estimation of the waiting time interval from the end of sending the HLIR message until the reception of the first bit from the most loaded station in the cluster in the first contention round (CR_1), taking in to account the propagation time, as shown in Figure 17. If a collision occurs, the RTS_1 parameter is defined from the end of sending the COL message until the reception of the first bit from one of the most loaded stations in the cluster.

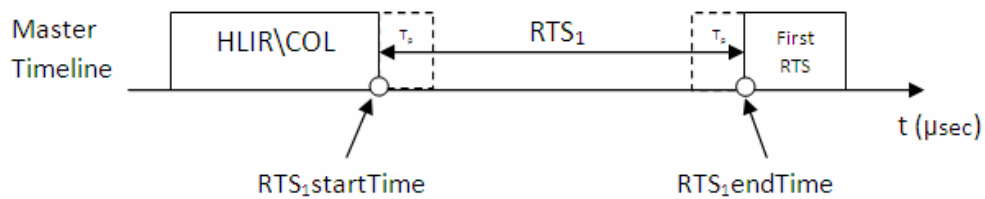


Figure 17 - Definition of the RTS1 parameter.

Therefore, based on the scheme in Figure 17, the parameter RTS_1 is defined as $RTS_1 = RTS_{1endTime} - RTS_{1startTime}$.

Definition of estLoad Parameter

The estimation of the maximal load in a cluster is performed in a master station using the expression (4):

(4)

It can be noticed that the maximal load is in direct proportion to RTS_1 and inverse to the access time as was mentioned previously.

We denote λ as the highest load among competing stations, and $\hat{\lambda}$ as the estimation made by the master station for the highest load among competing slave stations. According to assumption 2, there can be two different cases: (1) states in which the master station successfully detects the first arriving RTS pulse, and (2) states in which several RTS pulses collide with the first RTS pulse. If the master slave achieves a successful complete detection for the first RTS pulse (and its load is not the highest), it transmits, in the beginning of the next CR, a Clear-To-Send (CTS) message that contains the estimated maximal load.

The slave station whose load is higher than $\hat{\lambda}$ can successfully transmit its data. If several RTS pulses collide with the first RTS pulse, the master station transmits, in the beginning of the following CR, a collision (COL) message that also contains the value of the maximal estimated load that is related to the beginning time of the detection process. All the slave stations whose loads are higher than $\hat{\lambda}$ continue to compete in this Contention Round (CR). The following CRs (CR_{other}) are nearly the same, only the load function is replaced by a random number in the range of the contention period. In general, one may view CR_1 as a differentiation mechanism between stations whose load is the highest, and CR_{other} is designed to allow collision resolution between all the stations whose loads are similar. Note that had we randomly selected the time to transmit the RTS message in CR_1 (as we do in CR_{other}), each non-empty slave station would have had the same probability for gaining access to the channel.

Possible Problem with Load Estimation

The value of RTS_1 can sometimes be higher than the period of contention round (fixed in the protocol) due to the propagation times of messages in the following cases:

- The load in the stations is very low in CR_1
- The Lottery of the load in $RAND_AT_i$ will be 0 or close to 0 in CR_{other}

A first RTS signal transmitted at time t_i from one of the competing slave stations is detected in time $t_i + T_p$ at the master station (propagation time depends on the distance from the stations) so there is a possibility that $RTS_1 > CR$ as illustrated in Figure 18.

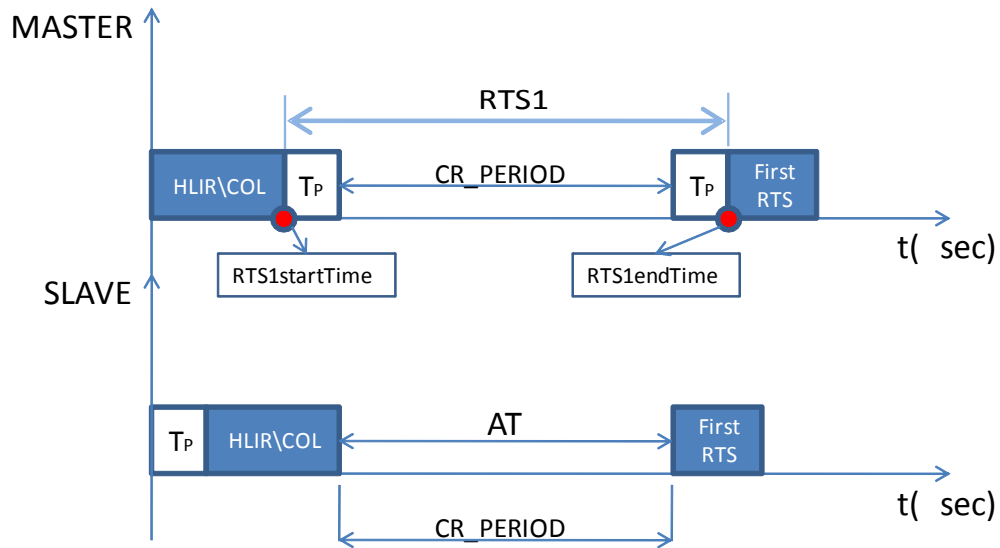


Figure 18 - Problematic scenario for calculation the RTS_1 parameter.

In a scenario when the $RTS_1 > CR$, $estLoad$ will have a negative value as demonstrated in Figure 19.

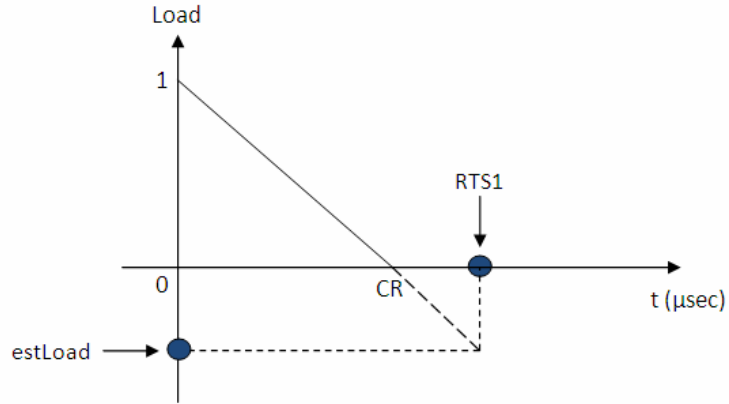


Figure 19 - Negative value in estimation of estLoad.

Therefore, the parameter estLoad that was defined in a range between 0 to will be set to 0 if it exceeds values lower than 0 during the simulation.

4.3 Collision Handling

The master station decides how to proceed when receiving the first RTS signal during a contention round in the session. The number of session per master depends on the network layer algorithm as was explained in Chapter 2.

On the one hand, first RTS signal can be received without collision detection in the channel, which indicated that there is a station in the cluster with the highest load and the station is ready for transmission. The master station sends the Clear-To-Send (CTS) message at the end of the CR. On the other hand, first RTS signal can be received with collision detection in the channel which indicated several stations with high load. If several RTS pulses collided with the first RTS pulse, the master station transmits, in the beginning of the following CR, a collision (COL) message that also contains the value of the load that is related to the beginning time of the detection process. All the slave stations whose loads are higher than the value in the load field in the COL message continue to compete in this Contention Round (CR).

For the scenario of receiving a RTS signal without collision there are two options:

- One of the slave stations in the cluster is the most loaded and it will receive that transmission grant after sending a RTS signal. The master station will reply with a CTS message with the estimated maximal load value.
- The master station (cluster head) is the most loaded station of all the nodes in the cluster and will win the grant for transmission. In this case, no control messages are sent at the end of the CR and the master station sends its information towards the destination.

There are three possible collision scenarios in relation to receiving the first RTS signal with collision detection in the channel:

- **Real collision (radio layer)** - a real collision is defined as a collision between two or more RTS signals (sent in the same time or in overlapping times) that were sent by different slave stations within a cluster. The collision is detected by the radio layer of the master station and defined in the state machine of the protocol as receiving a RTS with collision.
- **Virtual collision** - a virtual collision is defined as a collision between a RTS signal of a master station (not transmitted in the channel, self message) with a RTS signal from a slave station. The RTS signal is received at the master station without a collision in the radio layer but during RTS timer of the master. The virtual collision means that the two stations have very similar load value.
- **Real collision with virtual collision** - a real collision with a virtual collision is defined when the radio layer of the master station indicates on receiving a RTS signal with a collision at the same time of the transmission of an internal RTS signal (internal timer). This means that several slave stations and the master station have similar load values.

The real collision takes place in CR_1 and a $COL\{estLoad\}$ message is transmitted with the estimated load and a new contention begins. The following CRs (CR_{other}) are nearly the same, only the load function in the access time calculation is replaced by a random number in the range of the contention period. In general, one may view CR_1 as a differentiation mechanism between stations whose load is the highest, and CR_{other} is designed to allow collision resolution between all the stations whose loads are similar. Note that had we randomly selected the time to transmit the RTS message in CR_1 (as we do in CR_{other}), each non-empty slave station would have had the same probability for gaining access to the channel.

In case of a virtual collision, which is defined as a collision between a master station's internal RTS signal and a physical RTS signal from one of the slave stations, a toss of a coin is conducted in order to decide which station will receive the grant to transmit (saves one more contention round). If the master station wins, it transmits the data at the end of the contention round. If the slave station wins, the master station transmits a $CTS\{estLoad\}$ (with an estimation of the highest load in the cluster between the slave stations) message at the end of the contention round and the slave station is able to transmit.

Just to clarify, the internal RTS message in the master station is implemented in the communication simulator as a self-message (timer). The neighbor stations of the master are not able to sense the signal. The propagation time is taken into consideration when we analyze the timing of the messages. Each message time is expanded with twice the maximal propagation time in the system (1 μ sec for a 300 m distance between two wireless stations). A detailed analysis of the timing definition of the protocol is provided in Chapter 7.

In the following scenarios (Figures 20-25) we examine all the possibilities for receiving the first RTS signal depending on the existence of a collision and its type. The definition of RTS_1 parameters is demonstrated for each case. Internal RTS in the master is noted "RTS".

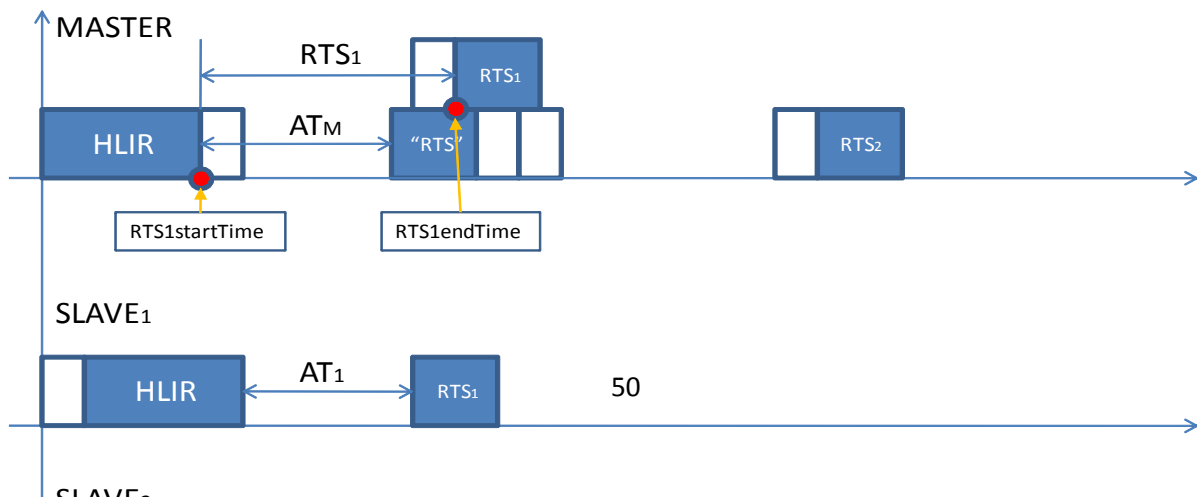
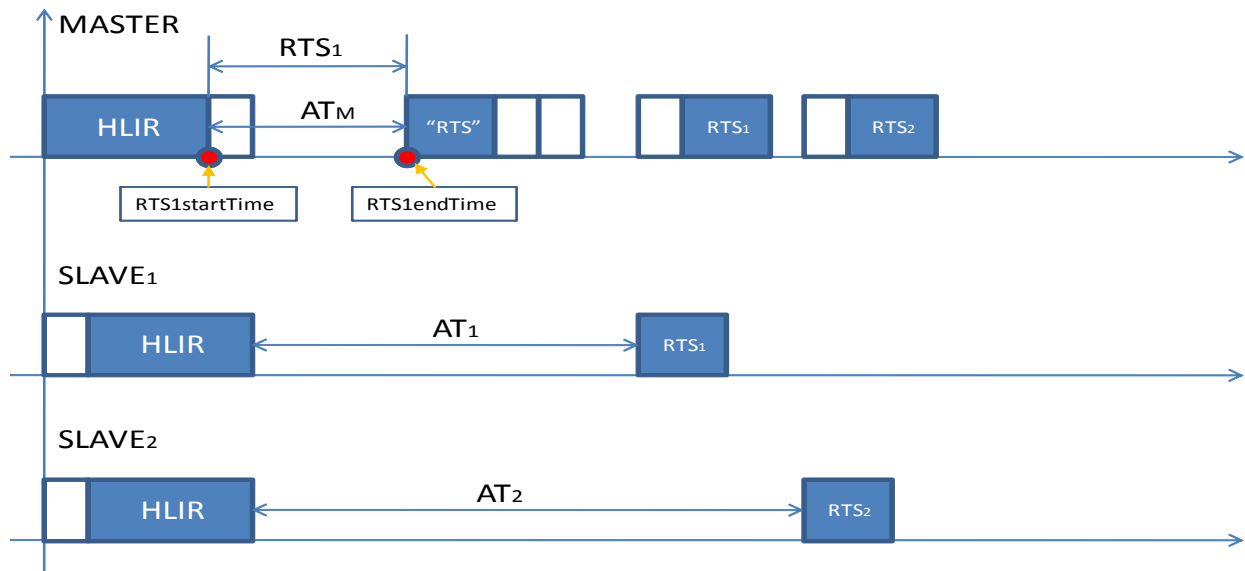
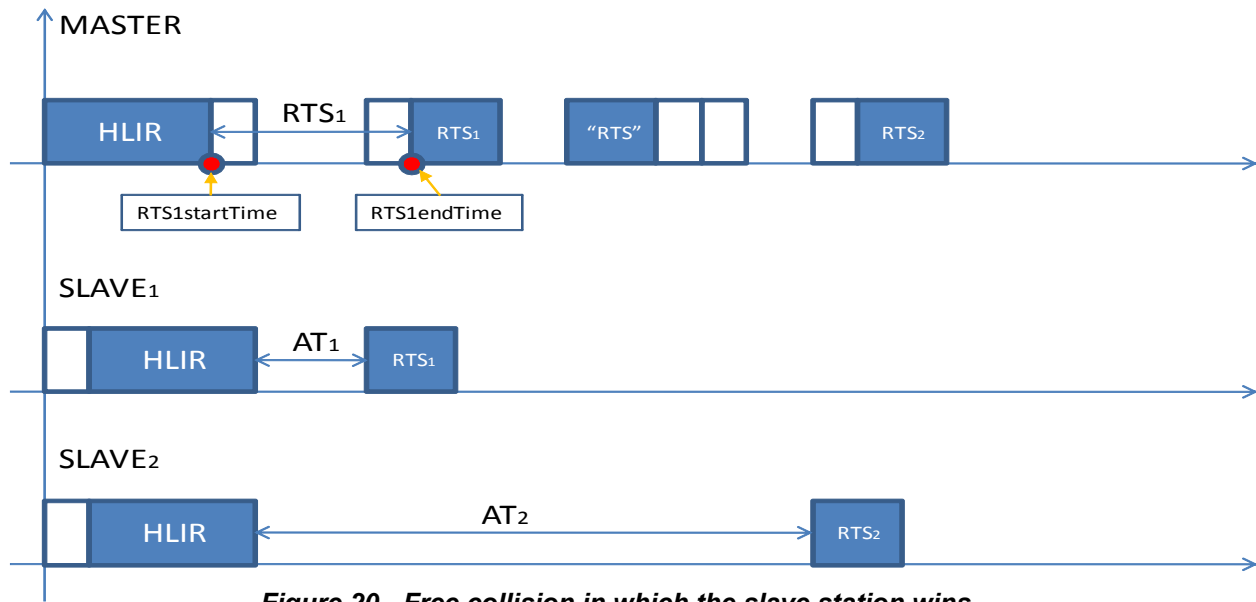


Figure 22 - Virtual collision in which the slave station wins ($p < 0.5$).

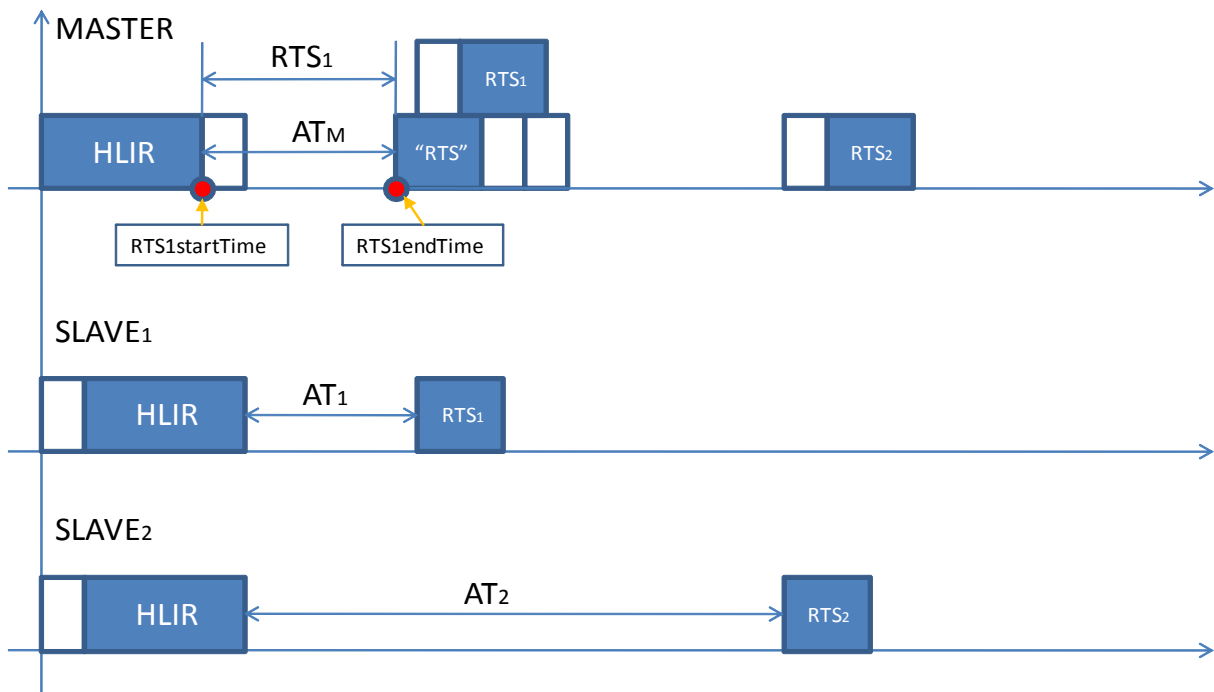


Figure 23 - Virtual collision in which the master station wins ($p > 0.5$).

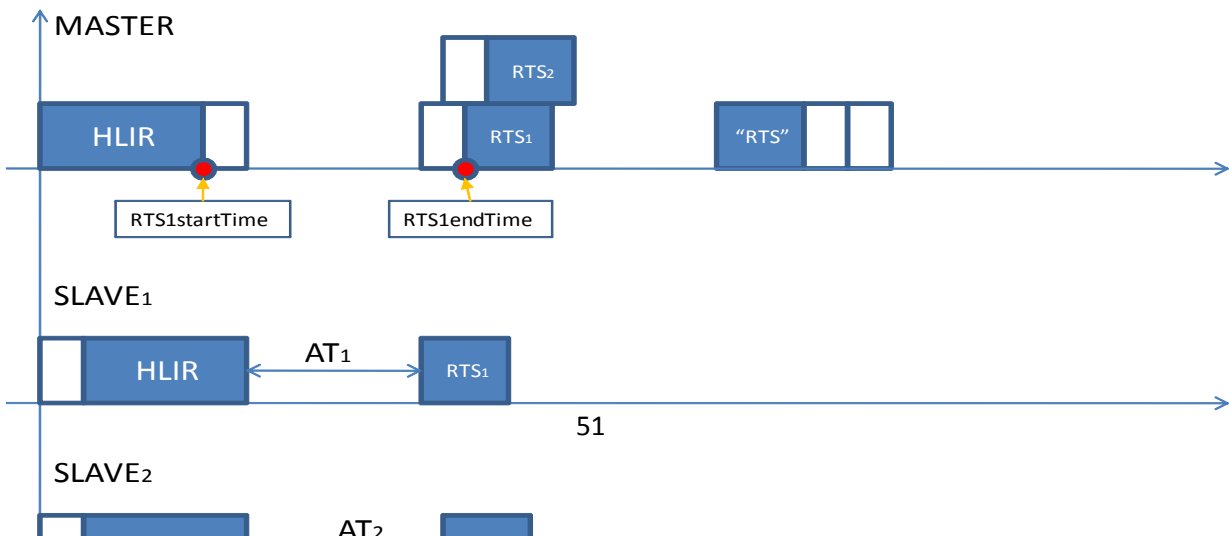
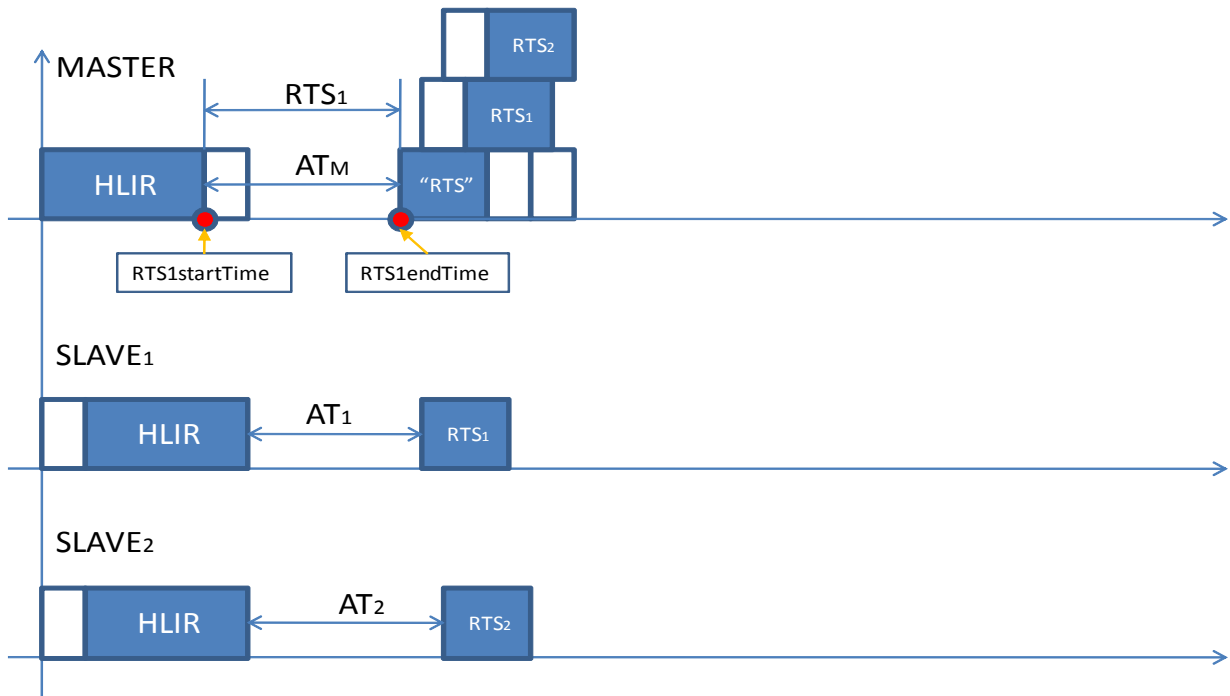


Figure 24 - Collision in the wireless channel (real) in which one of the slaves wins.



4.4 Estimating and Updating the Global Parameters

Two mechanisms were developed for estimating and updating the global parameters in the protocol (.). The first mechanism is operated in a cross-layer method, meaning that the MAC layer receives an indication from the network layer in order to execute the mechanism, and is designed for estimation of the global parameters. The second mechanism is responsible for updating and adjusting the values of the global parameters in adaptive mode during the processing of the protocol.

Mechanism for Estimating Global Parameters

An estimation of the global parameters is required in two cases: after the initialization of the network topology when nodes are noted as a master station or a slave station (not related to in the paper) and after forming a new master in an existing network (wake up process, topology changes). Usually during the execution of the protocol, a maser station sends an HLIR message with the global parameters and the equilibrium parameter (α) in the beginning of the session for each slave station to be able to calculate its internal load value. In the first session in the network or a first session for a new master, an estimating mechanism was developed (the interaction with the network layer will be explained in Chapter 7). In the first session the master station sends an HLIR message containing predefined global values with $\alpha=0$. Therefore, all the stations in the cluster will calculate their internal load based only on the delay values. The access time of each station will be calculated with the internal load value and each station that has information in its queue sends a RTS signal. The master station that receives the RTS signal can estimate the maximal load in the cluster, and through it estimates the maximal delay in the cluster. After receiving the RTS signals from the slave stations the session ends (a session is composed only of a contention round) and a new session begins in order to estimate the maximal queue length. In this session an HLIR message is sent with the updated maximal delay value, the defined maximal queue value, and $\alpha=1$. The process of estimating the internal load value in each station repeats but now the internal load is based only on the queue length. The master station that receives the RTS signal can estimate the maximal load and out of it the maximal queue length in the cluster. The session ends with one contention round without sending any data messages and in the next session (depending on whether the station still has a grant from the network layer) the complete contention based protocol will be executed. In the stage of estimating the parameters, the occurrence of collisions does not affect the process of estimation because only the RTS_i matters. As mentioned above, no data messages are sent during the two sessions of global parameters estimation. In the next session the updated global parameters are sent in the HLIR message with $\alpha=0.5$ (or $\alpha=0.75$) in order to balance the delay and the queue length.

Mechanism for Updating the Global Parameters

The decision whether to update the global parameters or not is made only at the end of the first contention round (CR_1). Based on two threshold values (upper bound and lower bound defined in the protocol) and the distribution of arrival times of RTS signals from the slave stations, the master station decides if an update is required. If most of the arrival times are spread between the two threshold values no update is required, but if they spread below the lower bound or above the upper bound an adjustment of the global values is required.

On one hand, if most of the arrival times of the RTS signals at the master station are below the lower threshold, it indicates that the stations' loads in the cluster are very high and they are competing for a very short time interval from the contention round. The following adjustment is preformed in the MAC layer:

(5)

(6)

On the other hand, if most of the arrival times of the RTS signals at the master station are above the higher threshold, it indicates that the stations' loads in the cluster are very low and they are competing for a very short interval time from the contention round. The following adjustment is preformed in the MAC layer:

(7)

(8)

By updating the values of the global parameters we expand the interval of time for competition and decrease the chance for collision in the next sessions without a change in the CR period.

In the following timing diagram of a master station (Figure 26) it can be seen when the spread of arrival times of RTS signals is a trigger for global parameters update.

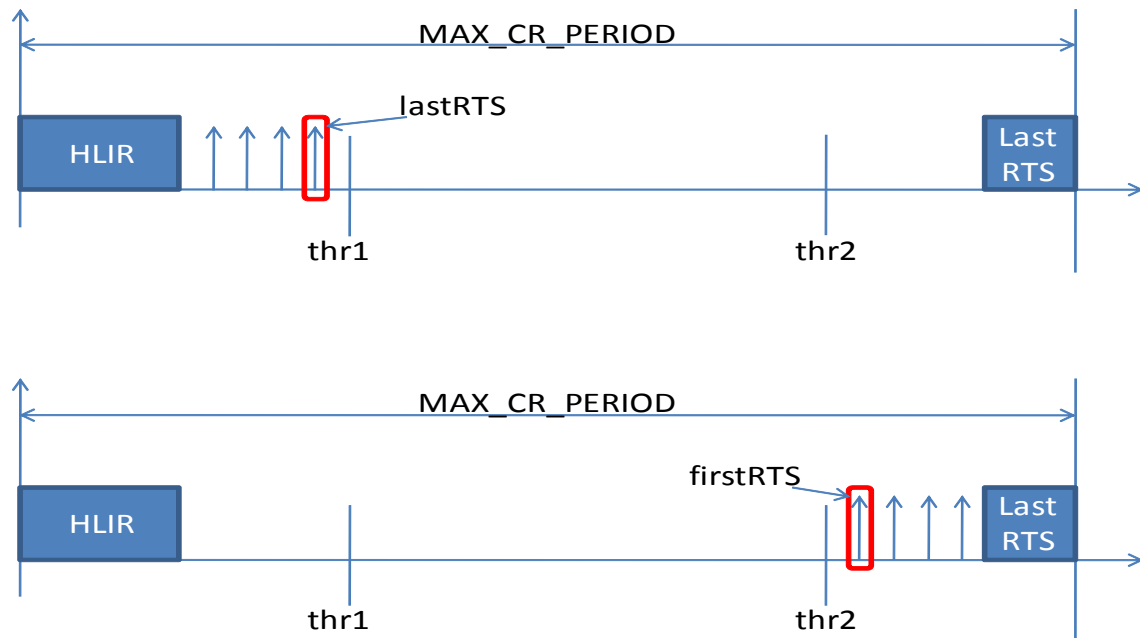


Figure 26 - Renormalization of the global parameters is required.

The adjustment of the parameters is called renormalization of global parameters - a process in which the global parameters are adjusted during the execution of the protocol without any delays on computational time.

The update of the parameters takes place at the end of CR_1 but transmitted to the slave stations only in the beginning of the next session in a HLIR message. If only one RTS signal is received at the master station, there is no need to update the parameters because the granting process is obvious. In case of detection of several RTS signals with collision (above or below the thresholds) the adjusting process takes place.

4.5 Load Comparison Mechanism

Slave stations within a cluster respond to the master station's decision based on a criterion for load comparison. Depending on the receiving of the first RTS signal at the master station, a CTS or a COL message (control messages) is sent with the estimated load value in one of the fields of the messages. If the master station is the most loaded in the cluster, it transmit the data towards the destination station after the contention period and all the slave stations change their status to IDLE before the next session. When a

slave station receives a CTS or a COL message it compares the internal load value (myLoad) with the estimated load sent by the master station (see the flow diagram in Figure 27). If a CTS message was received and the internal load value is equal to or higher than the estLoad value, the slave station receives a grant to transmit. If a COL message was received the slave stations will continue to compete for the grant in the second contention round, but based on random access. A slave station with lower internal load value changes its status to IDLE and waits for the next session.

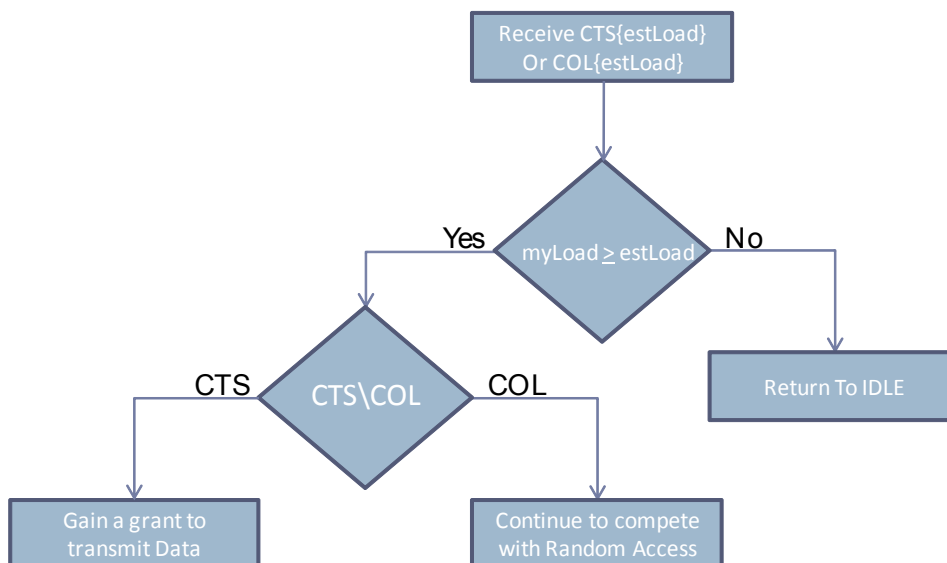


Figure 27 - Flow diagram for comparison of station's load with maximal load.

In a scenario of a collision in the cluster, it is possible that several slave stations that receive a COL message will have an internal load value higher than $estLoad$. Only these stations will compete again in the next CR of the current session. The stations generate random load value and from the random load value they generate its new access time. A new contention round starts but now the master estimates the maximal load based on the random access time of the most (or one of the most) loaded stations. It should be noted that if the granting process takes place in CR_{others} it is not guaranteed that the most loaded one will receive the grant but it can also be one of the most loaded because of the random access mechanism. In each contention round the maximal load is estimated by the master station and updated in the CTS or COL message. These mechanisms, based on CTS and COL messages, guarantee that only one station will receive the grant to transmit in each session.

4.6 Data Message Transmission Policy

A station within a cluster is granted to transmit has a fixed period of time (defined in the protocol) for the transmission of the messages in its queue. Data messages are transmitted in a Stop & Wait policy and an acknowledgement is required for each data message sent to the destination station. If the ACK message is not received after some period, a retransmission mechanism is operated and the message is sent again. If after several transmission attempts the message does not reach its destination (decreasing the counter for each attempt) the network layer is informed. The network layer can decide whether to delete the message from the queue or to change the MAC address of the message and send it through another station towards the destination (will be further explained in the network layer section). In the simulation, if a message failed to be transmitted between two slave stations in the cluster, the network layer will forward the message through the master (common neighbor of the stations).

Stop-and-Wait ARQ Schemes

The Stop & Wait policy is a basic method for an automatic retransmission mechanism also known as ARQ (Automatic Repeat-Request) Flow Control Technique. The source station sends a data message and waits for the acknowledgment message (ACK) from the destination station. The source station does not transmit any additional data messages until it receives the ACK message. A waiting period is defined in which the source station waits for the ACK, named Time Out (T.O.). This time period is defined from the end of the transmission of the data message until the beginning of receiving the ACK message (sent from the destination station) in the source station.

The main problems in this policy is that a ACK message can be lost on its way to the source station, or due to delays in the wireless link it will be received in the source station only after the Time Out period is over. These two scenarios lead to resending the data message in the source station and possible detection of the same two messages in the destination station, a situation called Duplicate Message Problem. This can be solved by adding a sequence bit (0,1) to the transmitted data message and a consecutive number in the ACK message sent from the destination station.

Three scenarios are presented in the following diagrams:

- Figure 28 - a regular operation of the mechanism is demonstrated between a source and a destination station. The first data message is sent with the sequential number 0 and the ACK message for the reception of the message is sent with a consecutive number (1) to the sender.
- Figure 29 - a data message is lost in the wireless link and therefore an ACK message is not sent from the destination to the source station. After the Time Out period the data message (kept in the queue) is sent again.
- Figure 30 - when the message is received in the destination station but the ACK is lost in the path to the source, a duplicate message problem occurs when the source station retransmit the data message. When the data message is received in the

destination station it discards the message if the same message with the same sequential number was already received.

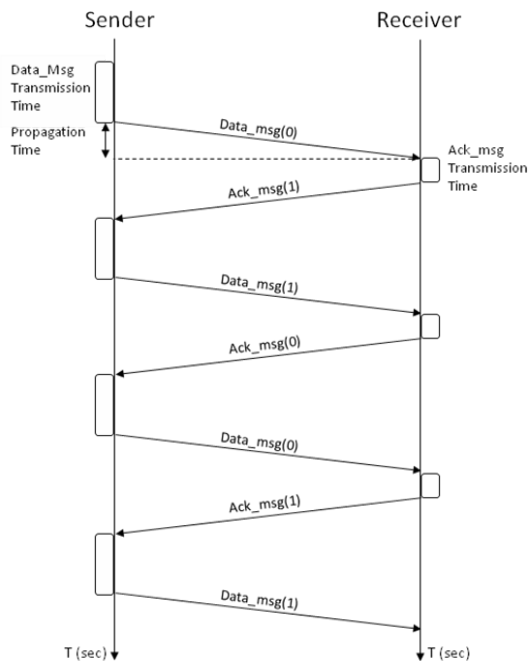


Figure 28 - Stop & Wait ARQ Flow Control Technique, step (1).

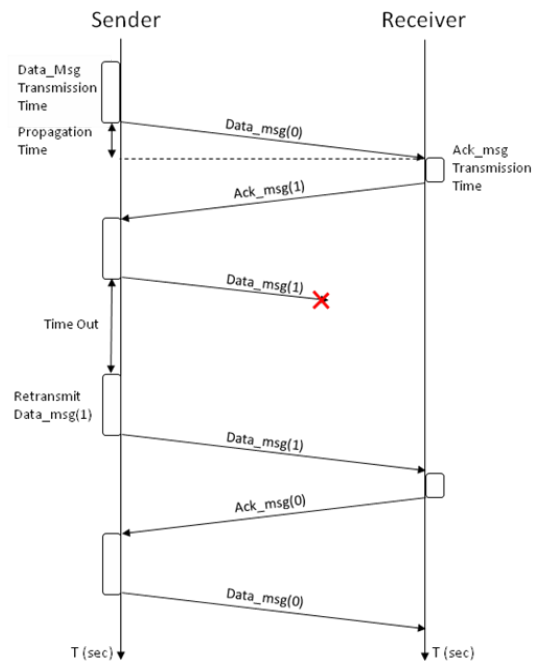


Figure 29 - Stop & Wait ARQ Flow Control Technique, step (2).

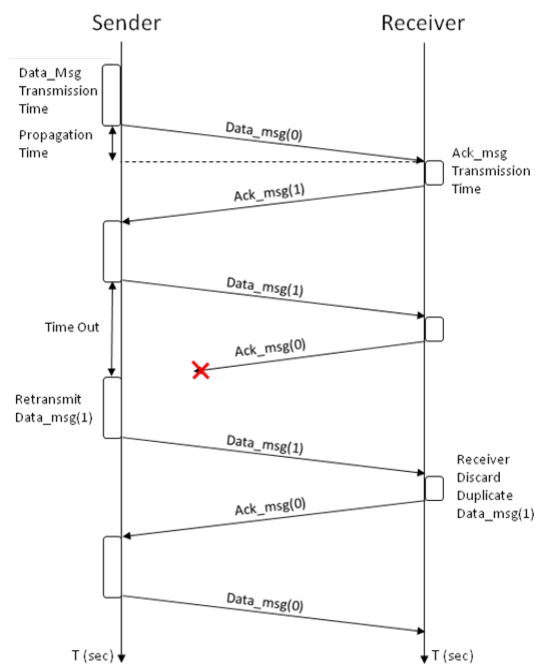


Figure 30 - Stop & Wait ARQ Flow Control Technique, step (3).

The main problem with this transmission policy is that it reduces the utilization of the wireless channel because the sender has to wait for acknowledgement before sending the next message. In our protocol, the retransmission mechanism is implemented in the MAC layer without a sequence number for the data and ACK messages. The data messages are created by a data generator.

An ACK message is sent without a transmission grant, which means that if a station received a grant to transmit the destination station within the cluster or within another cluster, it should not wait for a transmission grant to send the ACK message.

A counter with a fixed value in the protocol (which is currently 3) defines the number of retransmission attempts. As was previously explained, as part of the cross layer design of the protocol the network layer is involved in the decision of how to further process the message if all the retransmission attempts failed.

4.7 Relay Node Mechanism

In the Ad-Hoc network each station is functioning as a relay node during message transmission along a selected route. When a message is transmitted between two clusters, slave and master nodes may be part of the path of the message from its source towards the destination station. A station that receives a message checks if it was dedicated to it, based on the MAC address. If the MAC address is different from the station's MAC, the message is neglected, and otherwise it is moved to the network layer to check the IP address. If the IP of the message is the same as the station's then the message is moved for further processing in the application layer. If the IP is different the MAC address is changed to one of the station's neighbors (the IP address is not changed), all the neighbors' MAC addresses are in 1-hop levels, and transmitted to the next station until reaching the destination. The message reaching a relay node is placed into the queue of the station and is transmitted according to the mechanism of winning a grant in the

protocol. In the topology illustrated in Figure 31, node B functions as a relay node between node A (source) and node C (destination).

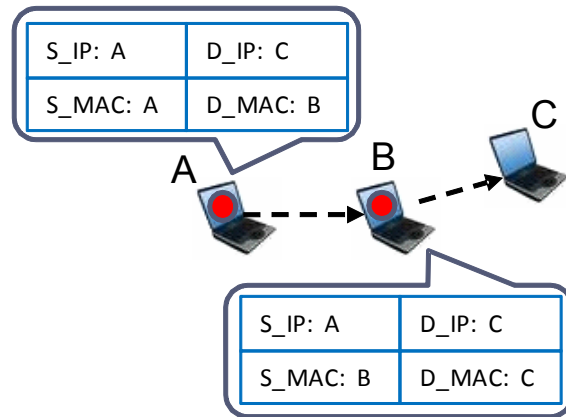


Figure 31 - Relay node mechanism.

4.8 ESG Mechanism

An End Slave Grant (ESG) mechanism is implemented to increase the utilization of the protocol. A station that wins a transmission grant had a fixed time for transmitting its data messages (the time period is defined in the protocol). During this period the station can continuously send its 100 Bytes data messages as long as there are messages in the queue. If the station completes the transmission of all its messages (empty queue) instead of waiting through the entire time period it can improve the utilization by sending an ESG message to the master station and end its timer for transmission. Another possible use for the ESG is when a station has a message to be sent in the queue but the estimated time for receiving ACK exceeds the transmission timer, so the station will end its timer for sending the ESG message to the master station. Sending the ESG message to the master station also indicates the completion of the current session. The master station has a timer for the transmission period of its transmitting slave station, and when it ends or an ESG message received, the network layer should send an indication on starting the next

session or on the end of transmission for the current cluster and passing the token to another one.

4.9 EMR Mechanism

The concept for this mechanism is described theoretically in the thesis and its state machine is explained in Chapter 6. The emergency (EMR) message is created by the network layer and sent from a special queue (exists in each station). The EMR message has a higher priority than the data message and does not require acknowledgement. The main purpose of this mechanism is for the notification of new stations joining to the existing cluster. A station that receives an EMR message will be able to update its 1-hop neighbors list and change its status (from slave to master) if it is required (depending on the cluster heads distribution in the backbone). The transmission of the EMR message is not part of the contention rounds and is based on carrier sensing. The station waits a random amount of time during which the channel is free before transmitting the message.

4.10 Carrier Sensing Mechanism

The EMR mechanism is based on carrier sensing. The main purpose of this mechanism is to decrease the probability of collision in the channel during the transmission of the EMR message. When a station has to send an EMR message (without a grant) it checks if the radio is in IDLE status and can start the transmission. In order to prevent a collision between several stations while sending EMR messages or with the processing of the protocol (contention rounds) each station conducts a lottery for a random waiting time that is defined as Random IDLE channel timer. If the channel is free during this period the station transmits its EMR message at the end of the waiting period. If there is any interruption during the waiting period such as a message being received or the channel being occupied, the station will forward the message to the upper layers and will wait for IDLE channel before initialization of the waiting timer.

Chapter 5. Timing Diagram Analysis

Message Control within a Cluster - Protocol Implementation

In this section, timing examples are given to the above description. Consider the following scenario in which a master slave is connected to three slave stations. The distance of all three slave stations to the master stations is equal, and all slave stations cannot communicate with each other directly (i.e. hidden from one another). The propagation of messages from the master station to the slave station is $\tau \mu$ seconds.

Several assumptions are taken into consideration:

- The master station knows exactly when it starts to receive a signal

- The master station senses a collision in the channel (radio layer) and distinguishes between arrival times of different RTS signals
- The master station will always receive a RTS message when the slave stations have any information to send
- The queuing system is based on a FIFO model. The existing data messages are sent from the queue, new messages that are received by the station when functioning as a relay node are inserted to the end of the queue
- The data messages are sent without any fragmentation. If there is not enough time to send the 100 Bytes data message and receive an ACK, it is not sent

The message control algorithm in the protocol defines two main cases:

- The contention takes place in the first round (CR_1) and a master station or a slave station wins the transmission grant. The most loaded station within the cluster maintains a timer for the transmission period. When it is over (with or without ESG message before) the session ends
- A collision occurs in CR_1 between several slave stations or slave stations and a master station. The contention now will take place in the second round (CR_{other} - the rounds after the first one) of the most loaded stations. The estimation of the maximal load value (estLoad) is sent in the CTS (Clear-To-Send) message and the stations compare it to their internal load value. The access time in the CR_{other} is based on a random value so it is guaranteed that one of the most loaded stations will win the grant but not necessarily the most loaded one

Three main scenarios are illustrated on a timing diagram:

- A slave station wins the grant for transmission (Figure 32)
- A master station wins the grant for transmission (Figure 33)
- Occurrence of collision in the wireless channel (Figure 34)

5.1 A Slave Station Wins the Grant for Transmission

In this scenario, station SLAVE₂ has the highest internal load; therefore its access time is the shortest of all the stations. It sends the RTS signal to the master station where the maximal load value is estimated based on RTS₁. The RTS signal is received in the master station without collision in the channel so after the end of the first contention round period the data transmission period begins. The ending of the transmission period means that the session is over; if the master station's network layer sends indication on a new session, a HLIR message will be sent in the beginning of the first contention round in the new session.

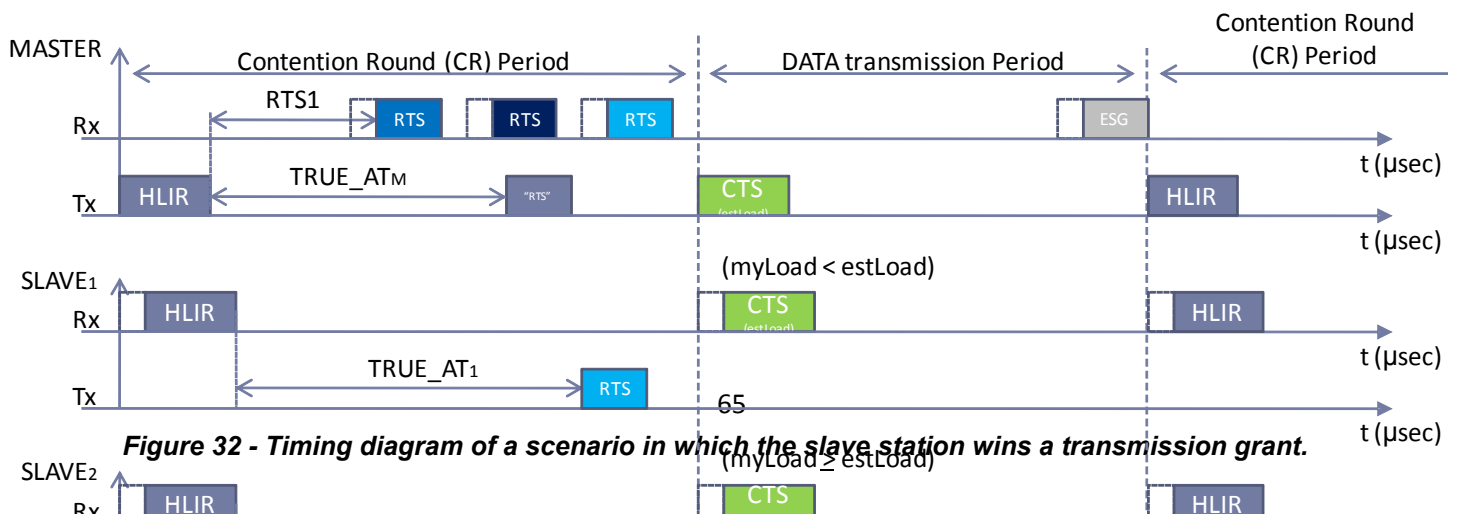


Figure 32 - Timing diagram of a scenario in which the slave station wins a transmission grant.

All the slave stations (1-hop from the master) receive the $\text{CTS}\{\text{estLoad}\}$ message and compare their internal load value to the estimated maximal load, but only SLAVE_2 has the grant to transmit in the current session. Two more mechanisms are demonstrated in the timing diagram, the virtual RTS (internal mechanism in the master station) and the sending of ESG message by station SLAVE_2 when it ends transmitting.

5.2 A Master Station Wins the Grant for Transmission

In the following scenario the master station wins the transmission grant. During its internal timer (virtual RTS) no RTS signal is received from the slave stations so after the first contention round the master station starts to transmit its data messages. The transmission period ends when there are no messages to send or when there is not enough time to complete transmission and ACK reception. A new session with HLIR message being sent from the master station will start if the master station's network layer permits continuing.

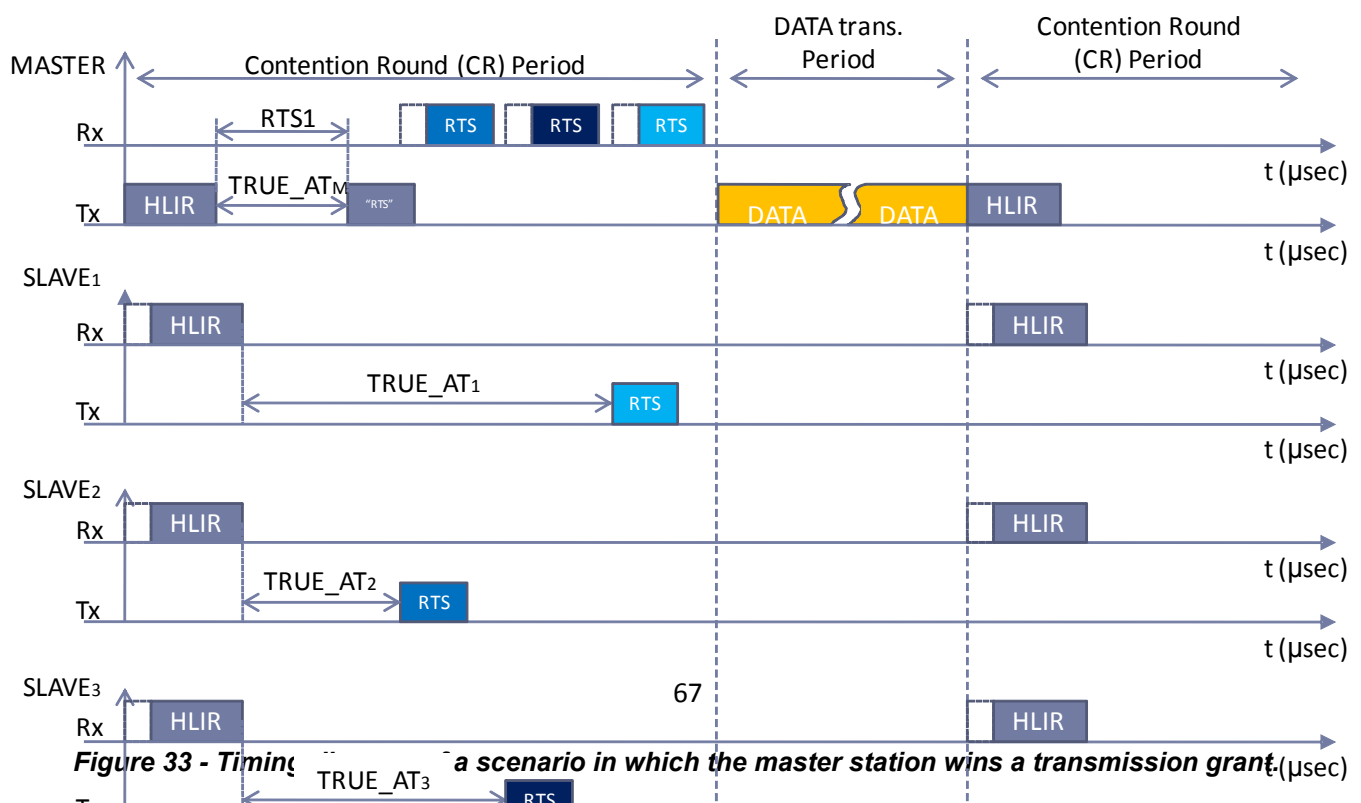


Figure 33 - Timing diagram illustrating a scenario in which the master station wins a transmission grant.

In this case, the maximal load estimation is defined by the master station's RTS signal although no CTS or COL messages are sent.

5.3 Occurrence of Collision in the Wireless Channel

If several RTS pulses collide with the first RTS pulse, the master station transmits, in the beginning of the following CR, a collision (COL) message that also contains the value $estLoad$ that is related to the detection of the first RTS signal (RTS_1). All the stations whose loads are higher than $estLoad$ continue to compete in this Contention Round (CR). The following CRs (CR_{other}) are nearly the same, only the load function is replaced by a random number in the range of the contention period. In general, one may view CR_1 as a differentiation mechanism between stations whose load is the highest, and CR_{other} is designed to allow collision resolution between all the stations whose loads are similar. Note that had we randomly selected the time to transmit the RTS message in CR_1 (as we do in CR_{other}), each non-empty station would have had the same probability for gaining access to the channel.

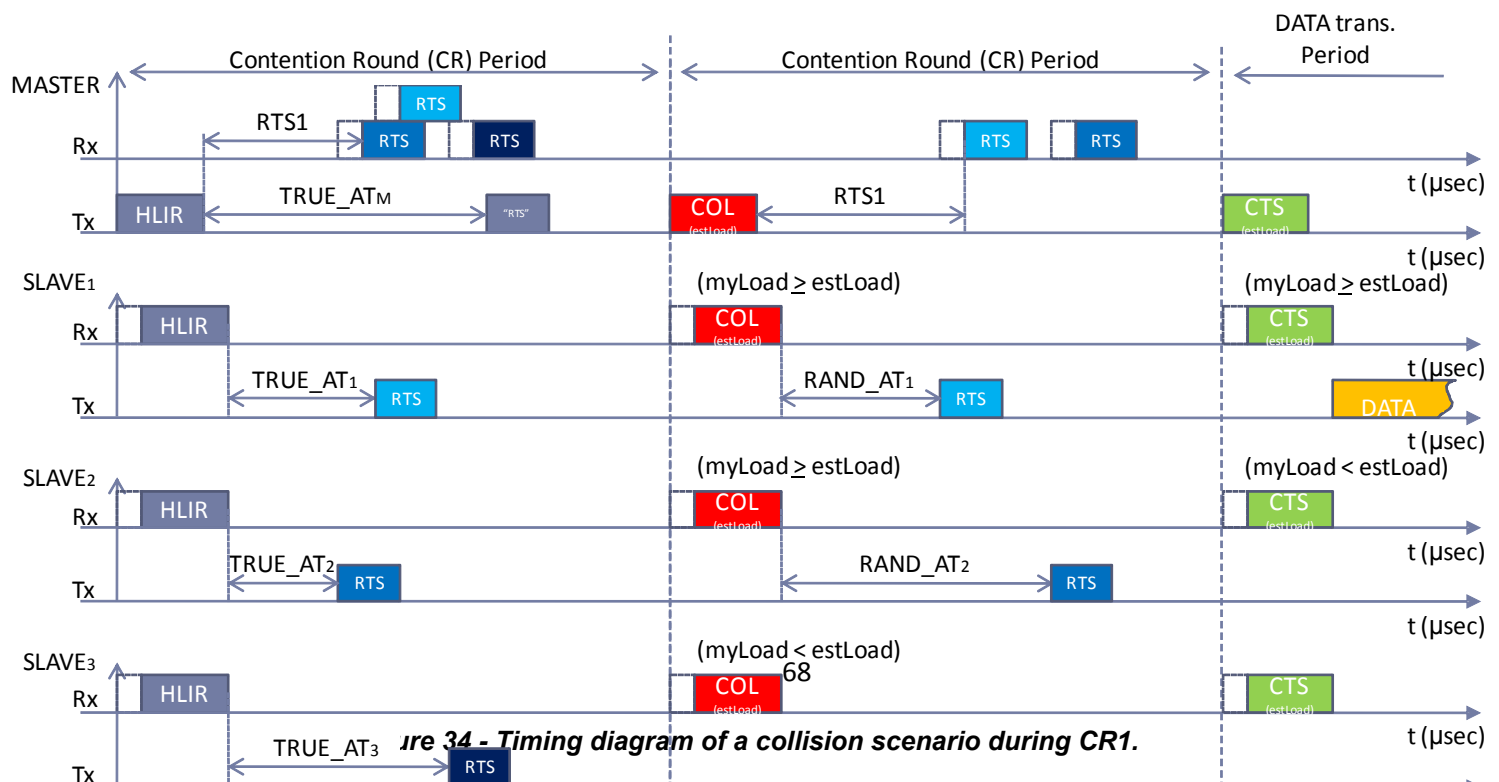


Figure 34 - Timing diagram of a collision scenario during CR1.

Only the load value of stations $SLAVE_1$ and $SLAVE_2$ are equal to or exceed the $estLoad$ value and therefore they compete again in the second contention round. The second round completes without a collision and in the CTS message sent from the master station the $estLoad$ value is defined by RTS_1 from the second round - time between the COL message and the RTS message from $SLAVE_1$. It should be noted that RTS_1 is defined according to the beginning of receiving the RTS signal in the master station. It is possible that due to distance and propagation time differences not all the stations involved in the collision will compete in CR_{other} .

Chapter 6. State Machines

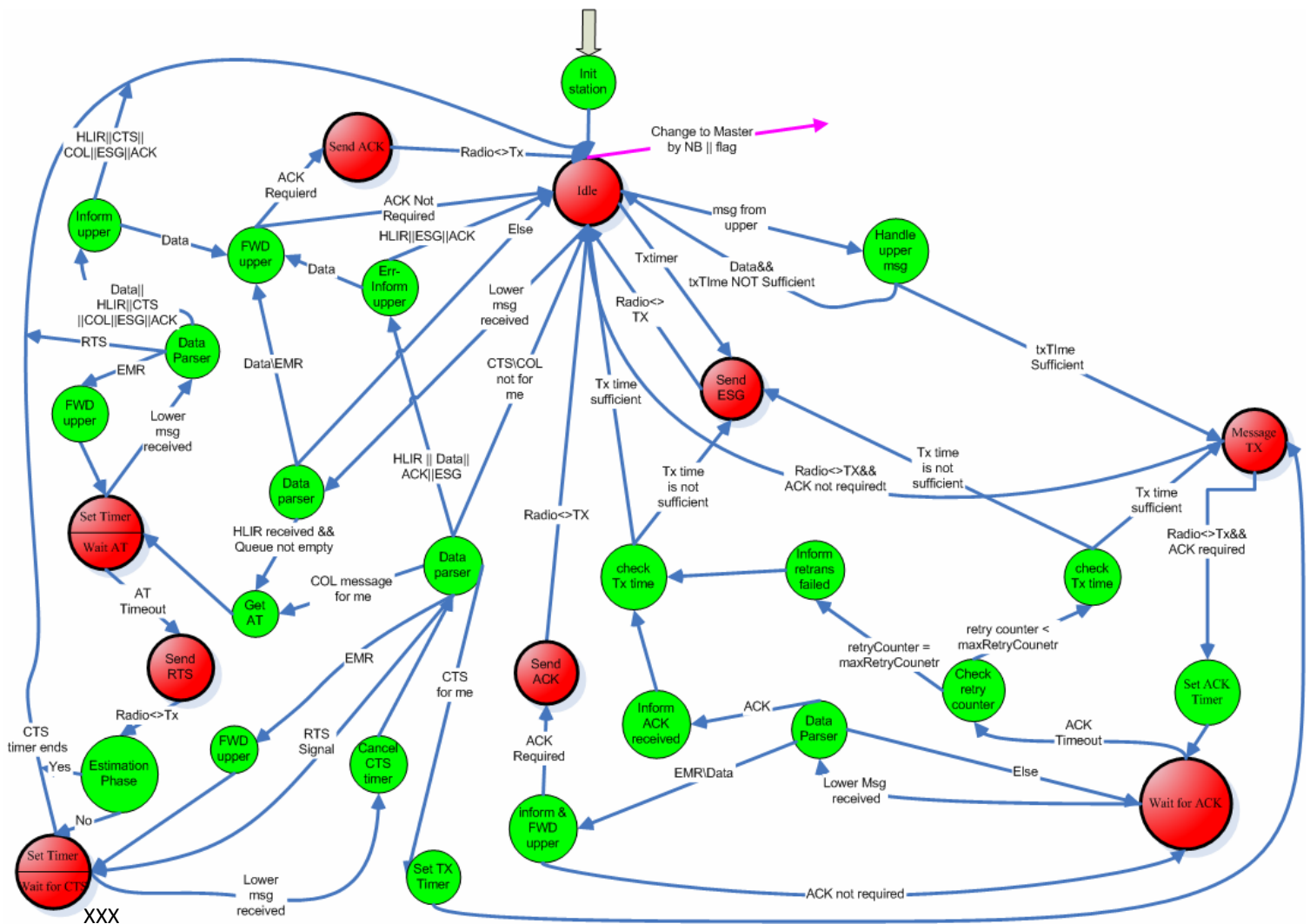
6.1 The Load Contention Resolution MAC Protocol

The Connected Dominating Set Based Clustering Algorithm (CDSC) finds a minimal connected dominating set in a MANET in a distributed manner. We developed our MAC protocol to be compatible with Weakly Connected Dominating Set (WCDS) based on Chen and Liestman [11]-[13] because it is extremely suitable for our purposes. It finds a dominating set which can be used as a backbone and uses the master stations (cluster heads) as the members of the backbone.

6.2 Finite States Diagrams of the Protocol

We assume that each cluster of nodes is composed of a master station which functions as the cluster head in the network layer backbone and a slave station connected to the master station. The maximal number of slave stations that can exist within a cluster without direct connection between them is 5 (can be easily proven by geometry). We assume that the 1-hop neighbors of the transmitting node remain constant during the period of the session. We also assume that each node has a unique MAC address (and IP in the network layer). The mode of the station (master or slave) is defined in the initialization process. We assume an already existing network of wireless nodes when the mode of the station can be changed during the simulation as a result of topology changes. There are two main finite states diagrams in the protocol, one when the mobile node is functioning as the slave station and the other when it is functioning as the master station. For each diagram the states and the events are described.

Each slave station begins in IDLE state and returns to this state at the end of the transmission time granted to the station, after sending an ESG message to the master station or after handling a message from the radio layer. A station is processing the master state machine only when it is functioning as the cluster head, but when it functions as a relay node between clusters or as a neighbor of a master station it processes the slave state machine. Figures 35-39 shows the state machines for our protocol implementation in a slave and master nodes



The definition of the protocol states for a slave station are described below:

- IDLE - initial state of the station (new in a cluster or waiting for a new session)
- Wait AT - waiting period of access time before sending the RTS message
- Send RTS - sending RTS signal
- Wait for CTS - waiting for receiving CTS message from the master station after sending the RTS message
- Msg Tx - sending a data or EMR message
- Wait for ACK - waiting to receive acknowledgement (ACK message) for the last data message sent
- Send ACK - transmitting ACK message for successful receipt of a data message
- Send ESG - transmitting an ESG message after the finalization of data transmission

Event Description

- Start Master Mode - receiving a message from the network layer indicated on changing to master mode from slave mode
- Radio=RECV - the radio is in receiving mode
- Radio=IDLE - the radio is in IDLE mode
- Radio ∇ Tx - the radio is not in a transmission mode (RECV\IDLE)
- Lower msg received - a signal or a message was received from the radio layer
- Upper msg received - a data/EMR message was received from the network layer
- Tx timer - the time dedicated for a station to transmit its messages is over (interrupt)

- ACK timeout - waiting time for an acknowledgement is over (interrupt)
- AT timeout - access time before transmission of RTS signal is over (interrupt)
- CTS timeout - the waiting time of a station to receive a CTS message from the master station is over (interrupt)
- EMR - receiving an EMR message from the network layer

6.2.2 Finite State Machine of a Master Station

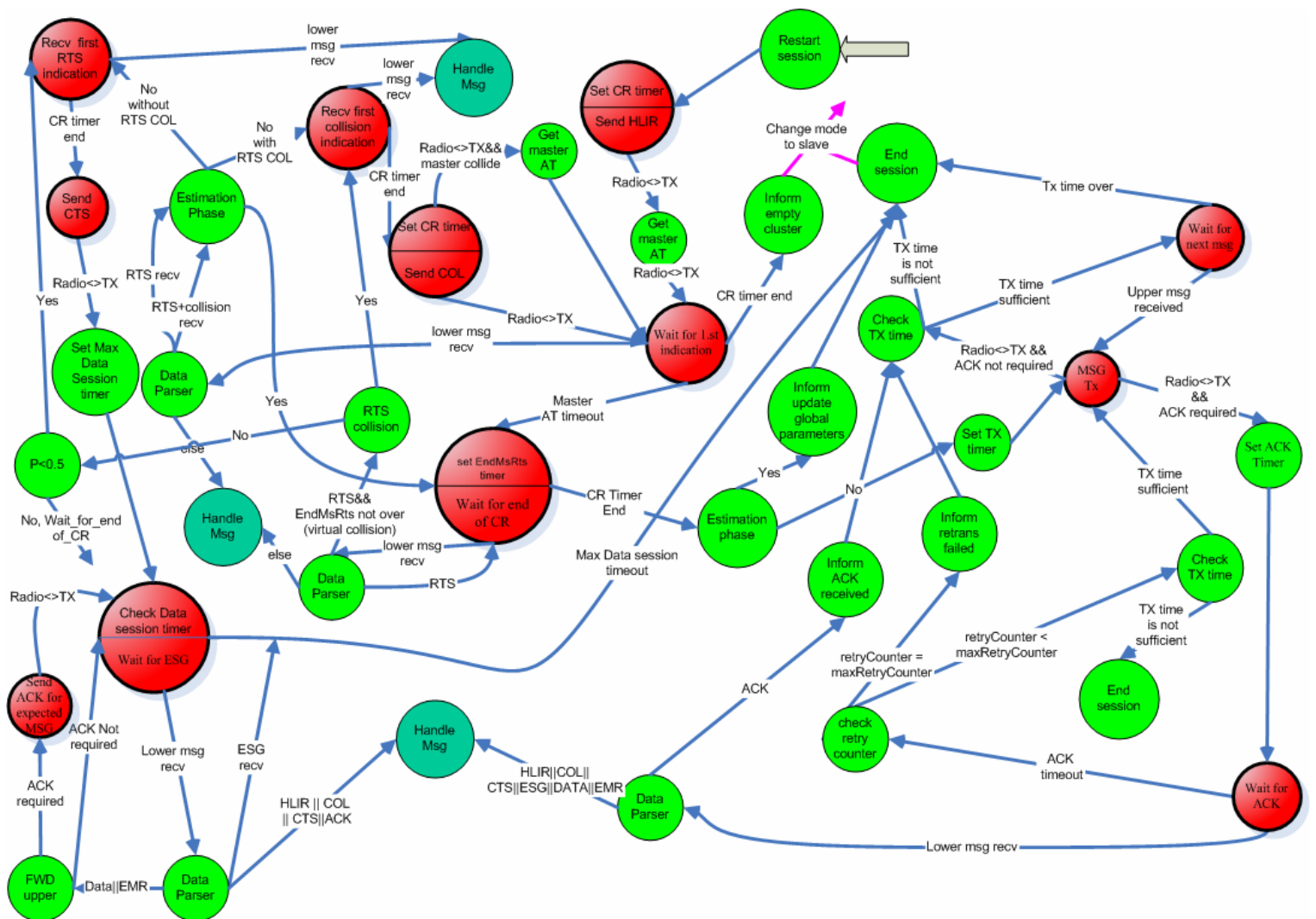


Figure 36 - Finite state machine of a master station.

The definition of the protocol states for a master station is described below:

- Send HLIR - transmitting an HLIR message in the beginning of a new session
- Wait for the first indication - waiting for the first RTS signal in a contention round
- Received first collision indication - waiting for the end of a contention round after receiving a collision indication as a result of simultaneous RTS sending
- Received first RTS indication - waiting for the end of a contention round after receiving the first RTS signal without collision from the most loaded station in the cluster
- Wait for end of CR - no RTS signal was received during the access time period of the master station, and it waits for the end of the contention round in order to transmit its data
- Send CTS - transmitting a CTS message
- Send COL - transmitting a COL message
- Wait for ACK - waiting for acknowledgement for the last data message sent
- Send ACK for expected data - sending an acknowledgement for receiving a data message on time

- Send ACK for unexpected data - sending an acknowledgement for receiving a data message when it was unexpected
- Wait for ESG - waiting for receiving an ESG message from a slave station after it completes the transmission
- Wait for next msg - waiting for a message to be transmitted (DATA\EMR) from the higherMAC
- Msg Tx - transmitting a message (DATA\EMR)

Event description

- Start Slave Mode - receiving a message from the network layer indicated on changing to slave mode from master mode
- Radio=RECV - the radio is in receiving mode
- Radio=IDLE - the radio is in IDLE mode
- Radio<>Tx - the radio is not in transmission mode (RECV\IDLE)
- Lower msg received - a signal or a message was received from the radio layer
- Upper msg received - a data\EMR message was received from the network layer
- CR timeout - the time for contention round is over (interrupt)
- ACK timeout - the waiting time for an acknowledgement is over (interrupt)
- Max Data Session Timeout - the master stations did not receive an ESG message from the slave station (interrupt)
- AT timeout - access time before transmission of "virtual RTS" signal is over (interrupt)
- Tx timer - the time dedicated for a station to transmit its messages is over (interrupt)

- End master station RTS timeout - a time period that is equal to the transmission of a RTS signal is over (interrupt)
- EMR - receiving an EMR message from the network layer

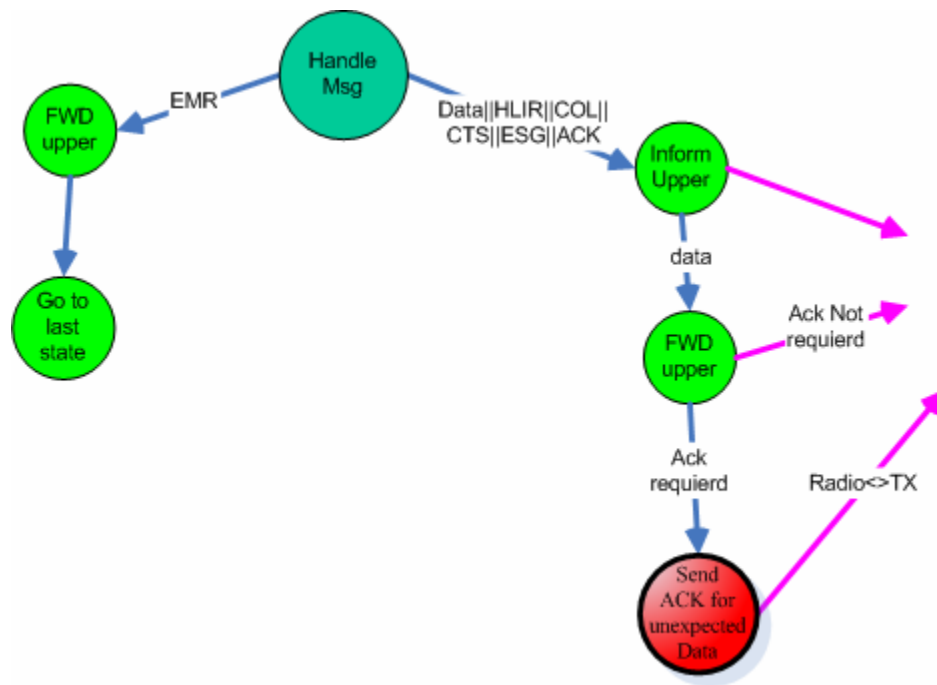


Figure 37 - Message handling in a master station.

EMR Mechanism in a Slave Station

In IDLE state the slave station checks if it has a grant to transmit. If the master station has a grant and the time for transmission is sufficient the station gives priority to the EMR message over the data message and sends it from a separate queue dedicated only for emergency messages created by the network layer. If the station does not have a grant, it uses random access to transmit the message. It senses the channel and if it is free for a random time period it can transmit the message. The EMR message will not be transmitted during the contention rounds in order not to disturb the processing of the protocol.

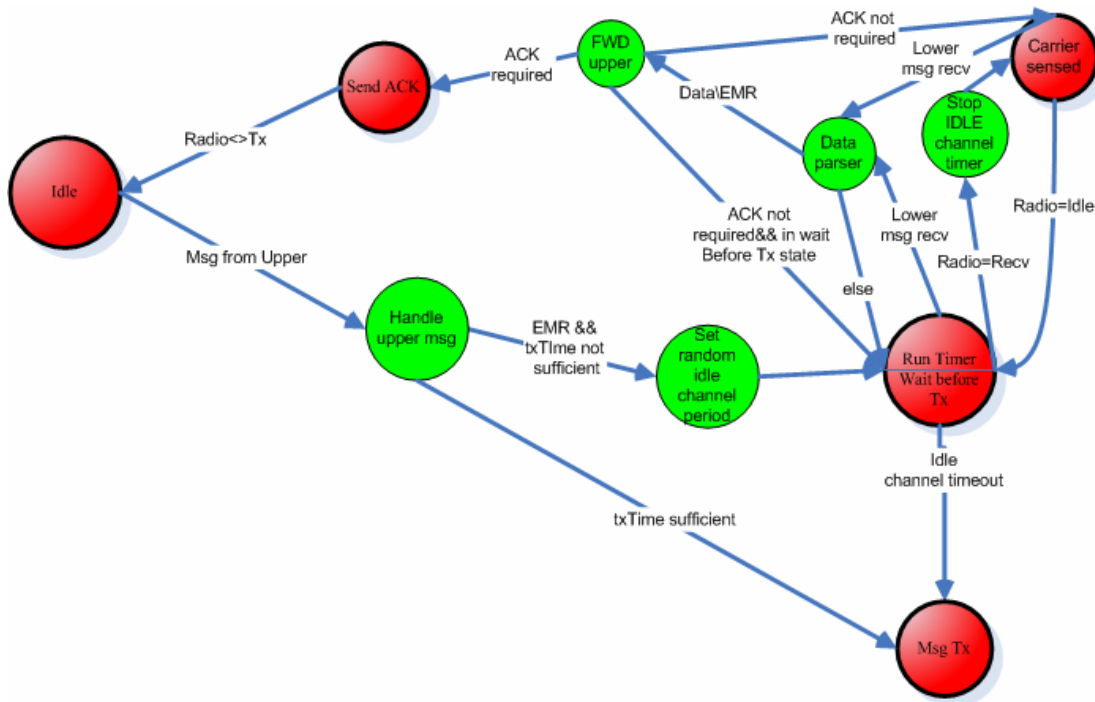


Figure 38 - Finite state machine of an EMR mechanism in a slave station.

EMR Mechanism in Master Station

In the beginning of each session the master station checks if it has EMR messages in the queue. If it does not, the session begins with transmitting a HLIR message. However, if the EMR queue is not empty the station senses the channel and will transmit the EMR message in random access if it is possible. If an EMR message is added to the queue during the processing of the contention rounds or during the data transmission periods of the master's neighbors, the master station will send the EMR message in the beginning of the next session (if it will have a token to continue) or will operate the mechanism that was explained for slave station.

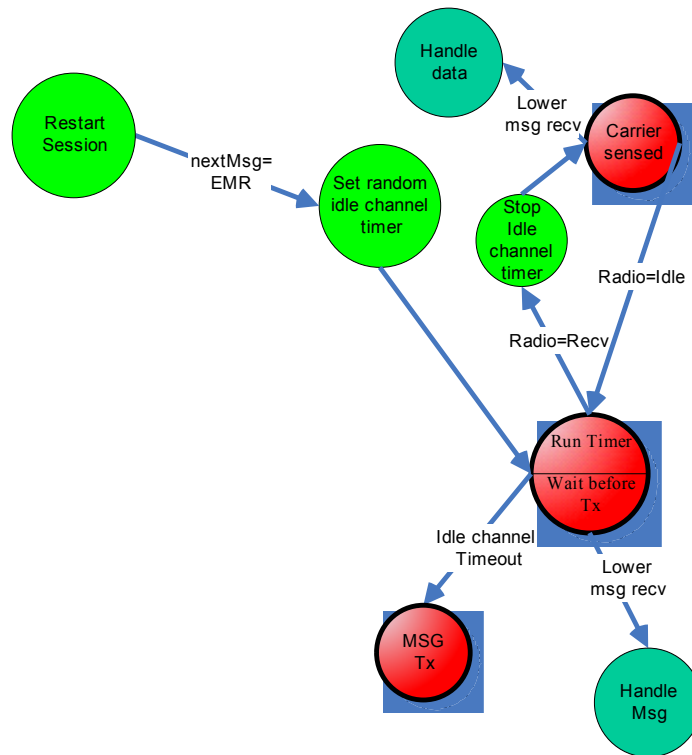


Figure 39 - Finite state machine of an EMR mechanism in a slave station.

Chapter 7. Modules of the Protocol

7.1 LowerMAC and HigherMAC Modules

The MAC layer is composed of two modules - LowerMAC and HigherMAC. Each module is separately implemented in the simulation and has its own functionalities. In Figure 40 the integration of the LowerMAC and HigherMAC in the layers architecture of each station is shown.

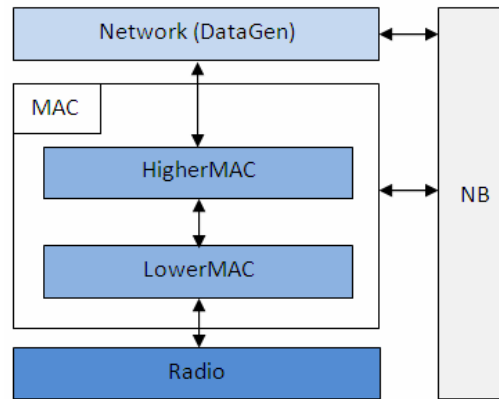


Figure 40 - Layers structure of a station.

Where a mobile station (master or slave) is implemented in simulation as described in Figure 41.

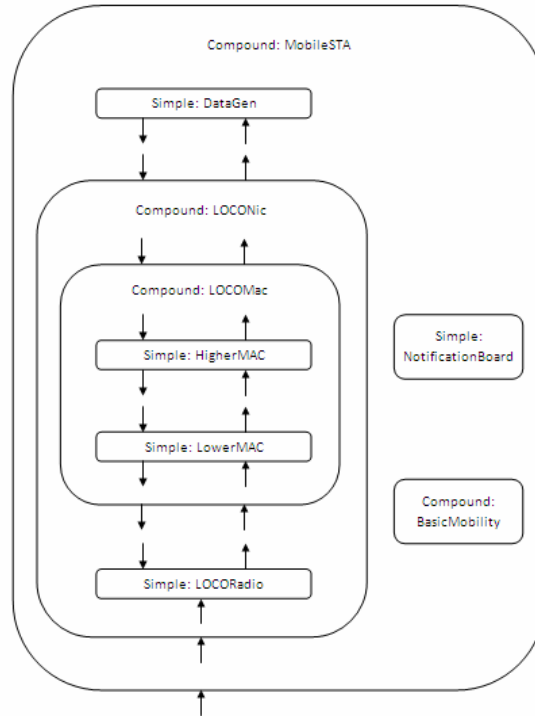


Figure 41 - Implementation of the structure of the station in the protocol.

The main functionalities of the HigherMAC module are:

- **Queue management** - data and EMR messages created in the network layer are placed in the queues and wait for their transmission. The HighMAC forwards the messages (once each time) to the LowerMAC when the station has a transmission grant or an emergency message has to be sent.
- **Calculating the load of the station** - the load function is calculated using the local parameters and queue size (Bytes) by summing the sizes of all the messages in the queue, the maximal delay of a message within the queue and the equilibrium parameter.
- **Calculating the access time** - the AT (access time) is calculated before the transmission of the RTS signal. This parameter defines the transmission time in CR_1 using the real load of the station and in CR_{other} (in case of collision in the channel) using a random access mechanism.

- **Service provider for the LowerMAC** - forwards different parameters for the HLIR/CTS/COL messages through the notification board
- **Failure in retransmission of a data message** – in case of failure in several retransmission attempts (depends on the value of the counter) the network layer should decide whether to change the destination MAC address of the message or to delete it. Both decisions will be conducted in the HigherMAC module.
- **Estimation of maximal load within a cluster** - when the station functions as a master station the HigherMAC module is responsible for the estimation of the maximal load within the cluster.
- **Global parameters estimation and update** - one more important functionality, when the station functions as a master station, these are the estimation and the update of the global parameters.

The main functionalities of the LowerMAC module are:

- Handling messages from the Radio and HigherMac modules - on the one hand, the LowerMAC module receives messages (data\EMR) from the HigherMAC module and forwards them to the radio module for transmission. On the other hand, it receives messages from the radio module and processes them as described in Figure 42 - filters unexpected messages, forwards data messages to upper layers, switches between the modes of the station (master/slave), releases the global parameters for load calculation in the HigherMAC, handles the arrival of RTS signals, and checks whether renormalization of the global parameters is required.
- **Handling notifications from the Notification Board and managing timers** - service provider for the different states in the state machines such as sending, receiving, and handling different notifications. Also provides indication on starting a timer, and at the end, on further processing.

- **Transmission time verification** - during the transmission grant, checks before the transmission of a new message whether there is enough time for transmitting the message, receiving acknowledgement, and retransmitting if necessary.
- **Updating messages fields** - updates fields of data and control messages.

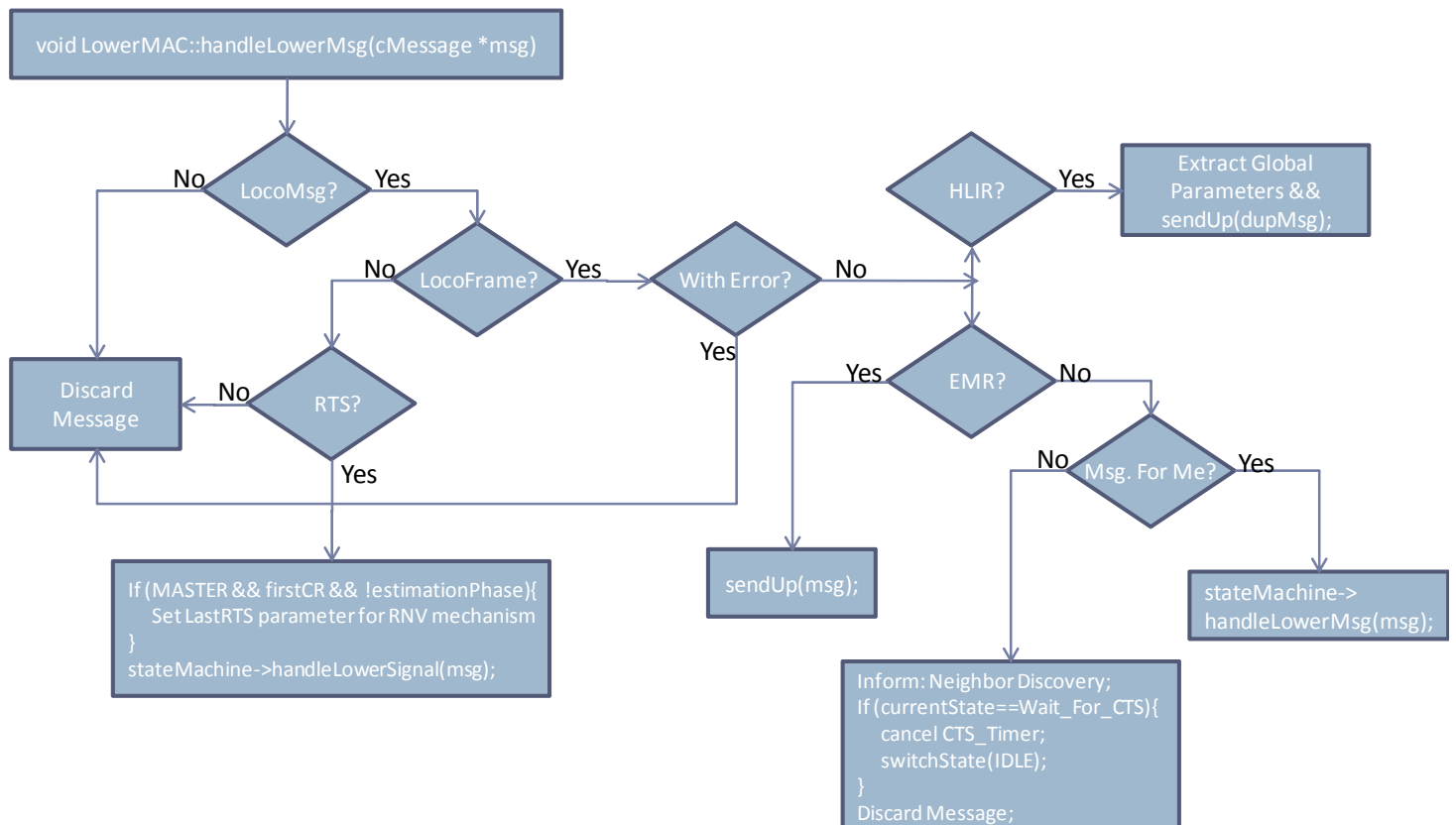


Figure 42 - Flow diagram: handling a message that was received from the radio layer.

7.2 Interaction between MAC and Network Layers

The innovation in the design of the protocol was not only the load balancing based mechanism but also the cross-layer approach that supports better implementation of backbone forming algorithms based dominating sets. Several messages are sent directly between the MAC (HigherMAC) layer and network (DataGen in the simulation) layer and also notifications which are forwarded using the NB (notification board) of the simulator between the modules. In this chapter a review of the interface between the layers and the importance of the messages and notifications in the cross-layer design will be explained. The notifications between the layers and the message types are summarized in Tables 3-4.

Messages Forwarded between the Layers

- **Master Station Parameters**

When a HLIR message sent by the master station is received by the slave station the information on the MAC address of the master station is forwarded to the network layer, it can update its 1-hop neighbor list if it is a new station, with the global and equilibrium parameters which are kept in the network layer for further processing.

- **Data Messages**

Data messages created in the network layer (DataGen) are forwarded to the MAC layer. They are inserted into the queue in the HigherMAC and are sent by the LowerMAC to the destination station. When a data message is received in the station it checks its MAC address, and if the messages are targeted at the station they are forwarded from the LowerMAC to the HigherMAC and then to the network layer. In the network layer, the IP address of the message should be checked and forwarded to the application level. In our simulation, received messages are not processed in the DataGen. When the message contains another IP address, the receiving station acts as a relay node and forwards the message to the next station in the routing path.

- **EMR Messages**

EMR messages are created in the network layer and forwarded into the emergency queue in the HigherMAC module. The processing of the EMR mechanism was described in the state machines chapter. When a station receives an EMR message, the message is forwarded to the network layer for neighbor list updating or during token passing between cluster heads and further processing that should be implemented in future work.

Notifications Forwarded between the Layers

- **End of Master Session**

This notification indicates the end of the session. The LowerMAC sends the notification to the network layer and it should decide if the station will start a new session as a master station or the token (in the network level) will be passed to another master station.

- **Retransmission Failure**

Indicates a retransmission failure after a fixed number of attempts (defined as 3 in the protocol). The notification is sent from the LowerMAC to the network layer and the network layer should decide if the message will be deleted from the queue or sent through a different path (retransmission failure decision notification).

- **Neighbor Discovery**

When a message is received in a station that is not the final destination, a notification is sent from the LowerMAC to the network layer in order to update the 1-hop neighbor list.

- **Empty Cluster**

This is an indication that the stations in the cluster do not have any information to send. The notification is sent by the LowerMAC to the network layer in the master station only. The network layer has to decide if the token should be passed to another cluster. Currently in our simulation when the cluster is empty the simulation ends.

- **Global Parameters Update/Estimation**

This notification is sent from the network layer to the HigherMAC and indicates that a global parameters estimation is required in the beginning of the simulation (or after joining a new master station) or an update of the parameters while running the simulation. The network layer also changes the value of the equilibrium parameter according the required process. Currently in our simulation only the estimation level in the beginning of the simulation is implemented.

- **Mode Change**

Indicates a mode change of the station. The network layer forwards a notification to the LowerMAC that indicates it has to change the mode (from master to slave or from slave to master) and the LowerMAC module changes the state machine to be processed.

- **Unexpected Behavior**

Indicates unexpected behavior during the processing of the protocol as receiving unexpected messages. The notification is sent by the LowerMAC to further process in a module that will be registered during the development of the state machine for the network layer.

Table 3 - Notifications between the layers of the protocol

Notification Name	Parameters	Initiator	Subscribed
-------------------	------------	-----------	------------

	Name	Value	Module	Module
NF_LOCO_AT	accessTimeType	TRUE_AT or RANDOM_AT	LowerMAC	HigherMAC
	accessTime	Waiting time	HigherMAC	LowerMAC
	staLoad	Load of the station		
NF_LOCO_CHANGE_MODE	staMode	Slave\Master	DataGen	LowerMAC
NF_LOCO_UNEXPECTED_BEHAVIOR	-		LowerMAC	-
NF_LOCO_MSG_PARAMS	msgType	HLIR\CTS\COL	LowerMAC	HigherMAC
	maxQueue	Maximal queue length (Bytes)	HigherMAC	LowerMAC
	maxDelay	Maximal delay of a message (msec)		
	alpha	Equilibrium parameter 0,0.5/0.75,1		
	estimatedLoad	Estimated load		
NF_LOCO_RTS1	RTS1	Estimated waiting time	LowerMAC	HigherMAC
NF_LOCO_ACK_RECEIVED	-		LowerMAC	HigherMAC
NF_LOCO_RETRANS_FAILED	-		LowerMAC	DataGen
NF_LOCO_RETRANS_FAILED_DECISION	retransFailedDecision	DELETE_CURRENT_MSG OR CHANGE_DEST_MAC_ADDR ESS	DataGen	HigherMAC
NF_LOCO_END_MASTER_SESSION	-		LowerMAC	DataGen
NF_LOCO_NEIGHBOR_DISCOVERY	neigDiscoveryMacAdd	MAC address	LowerMAC	DataGen
NF_LOCO_EMPTY_CLUSTER	-		LowerMAC	DataGen
NF_LOCO_EMPTY_QUEUE	-		HigherMAC	LowerMAC
NF_LOCO_ESTIMATE_GLOBAL_PARAMETERS	alpha	Equilibrium parameter	DataGen	HigherMAC
NF_LOCO_UPDATE_GLOBAL_PARAMETERS	-		LowerMAC	HigherMAC
NF_LOCO_EXTRACT_PARAMETERS	maxQueue	Maximal queue length (Bytes)	LowerMAC	HigherMAC
	maxDelay	Maximal delay of a		

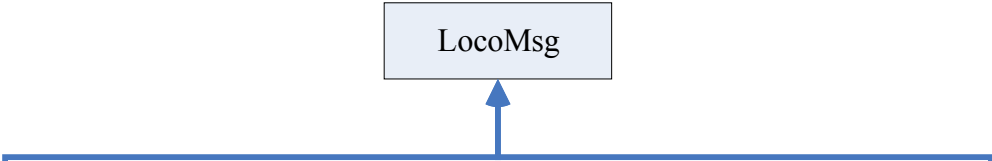
		message (msec)		
	alpha	Equilibrium parameter		
NF_LOCO_RNV_GLOBAL_PARAMETERS	handleRNV	Defines the required update	LowerMAC	HigherMAC
NF_RADIOSTATE_CHANGED	Radio state	Transmit/Recv/Idle	AbstractRadio	LowerMAC

Table 4 - Messages description in the protocol

Message	Field	Comments
---------	-------	----------

name	Name	Description	Value	
LocoMsg	type	Message type		Basic structure of a message in the protocol
RtsSignal	type	Message type	RTS_MSG	A signal and not a message. Kind field is for the radio layer
	kind	Message type	RTS_MSG	
	length	Message length (bits)	RTS_DURATION·BIT_RATE 2μsec·10Mbps = 20 bits	
LocoFrame	transmitterMacAddresses	MAC address of the source		Basic structure of a frame in the protocol
	receiverMacAddress	MAC address of the destination		
LocoAckMsg	type	Message type	ACK_MSG	Sent from the source to the destination station
	byteLength	Message length	14 (Bytes)	
LocoColMsg	type	Message type	COL_MSG	Sent by the master station, Indicates a RTS signal with collision was received. Includes the maximal load estimation
	byteLength	Message length	20 (Bytes)	
	estimatedLoad	Estimated load		
	receiverMacAddress	MAC address destination	ff:ff:ff:ff:ff:ff broadcast message	
LocoCtsMsg	type	Message type	CTS_MSG	Sent by the master station, includes the estimation of the maximal load within the cluster
	byteLength	Message length	20 (Bytes)	
	estimatedLoad	Estimated load		
	receiverMacAddress	MAC address destination	ff:ff:ff:ff:ff:ff broadcast message	
LocoDataMsg	type	Message type	DATA_MSG	The delay of a data message is calculated in the following way simTime()-creationTime The sequence number is added for the
	byteLength	Message length	100 (Bytes)	
	creationTime	Creation time of the message	Random or simTime()	
	reTxCounter	Retransmission counter	0	
	ackRequired	Indicates of ACK is	true	

		required		network layer processing
	sequenceNumber	Sequence number per message		
	transmitterIpAddress	IP address source		
	receiverIpAddress	IP address destination		
LocoHlrMsg	type	Message type	HLIR_MSG	Sent by the master station in the beginning of the session, includes the global and equilibrium parameters
	byteLength	Message length	20 (Bytes)	
	receiverMacAddress	MAC address destination	ff:ff:ff:ff:ff:ff broadcast message	
	maxQueue	Maximal queue length	$\max\{Q_i\}$	
	maxDelay	Maximal message delay	$\max\{D_{\max_i}\}$	
	alpha	Equilibrium parameter	$[0 \div 1]$	
LocoEsgMsg	type	Message type	ESG_MSG	Sent by the slave station at the end of the transmission period to the master station (cluster head)
	byteLength	Message length	14 (Bytes)	
LocoEmrMsg	type	Message type	EMR_MSG	Created by the network layer for topology maintenance and inter-cluster communication
	byteLength	Message length	20 (Bytes)	
	transmitterIpAddress	IP address source		
	receiverIpAddress	IP address destination		



In Figure 43 we can see the inheritance of the protocol's messages as implemented in the network simulator.

Figure 43 - Inheritance of message types in the protocol.

7.3 Time and Constant Parameters Definitions in the Protocol

In this section all the relevant time periods and parameters of the protocol will be defined and explained. The design considerations of several protocol parameters will be demonstrated in timing diagrams. All the parameters can be found in the file LOCOMacConsts.h.

Transmission Time of Messages in the Protocol

(9)

A summary of the sizes and transmission times of the protocol's messages is given in Table 5.

Table 5 - Sizes and transmission times of messages in the protocol

Message	Size [Bytes]	Transmission Time [μ sec]
Data_Msg	100	80
HLIR_Msg	20	16
RTS_Msg	2.5	2
CTS_Msg	20	16
COL_Msg	20	16
ACK_Msg	14	11.2
ESG_Msg	14	11.2
EMR_Msg	20	16

Timing Parameters

Now we will examine the protocol design considerations that led to the definition of the following parameters. Figures 44-48 use master-slave and source-destination diagrams to illustrate how the parameters are defined.

- MAX_PROPAGATION_TIME (MaxTp)

Defines the maximal propagation time of a message in the channel for a distance of 300 m between two stations.

- MAX_IDLE_CHANNEL_PERIOD

Part of a channel sensing mechanism for EMR message transmission.

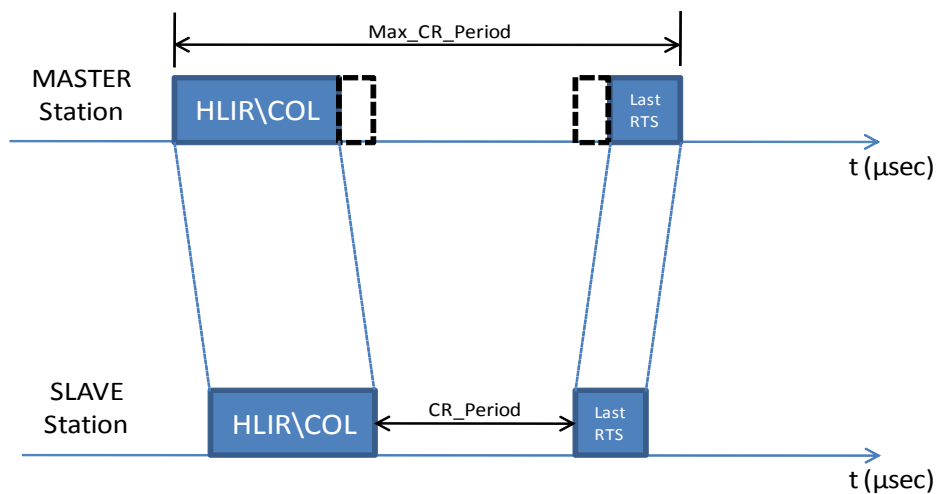
- CR_PERIOD

Defines the time period for contention between stations within a cluster - the time between receiving the HLIR/COL message and the transmission of the last RTS signal.

- MAX_CR_PERIOD

Defines the maximal time period for a contention round (CR_i) in the protocol. This parameter includes the control messages sent by the master station, HLIR or COL, the contention period as defined in the previous item, and the receiving time of the last RTS signal possible in the protocol.

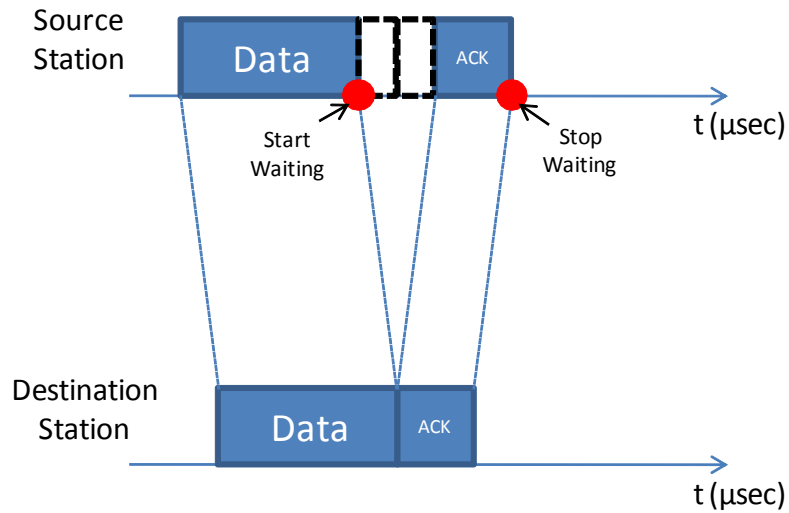
(10)



- MAX_WAIT_FOR_ACK_PERIOD

Defines the maximal waiting time (Time Out) period of a transmitting station (source) for acknowledgement (ACK message) from the destination.

(11)



- MAX_WAIT_FOR_CTS_PERIOD

Defines the maximal time period a slave station waits after transmitting its RTS signal for receiving a CTS message from the master station.

(12)

It should be noted that and are constant components whereas is a varying component.

It can also be expressed as:

(13)

It should be noted that t_{CR} and t_{CTS} are constant components whereas t_{ATi} is a varying component.

Therefore the value of this parameter depends on the access time of each station:

(14)

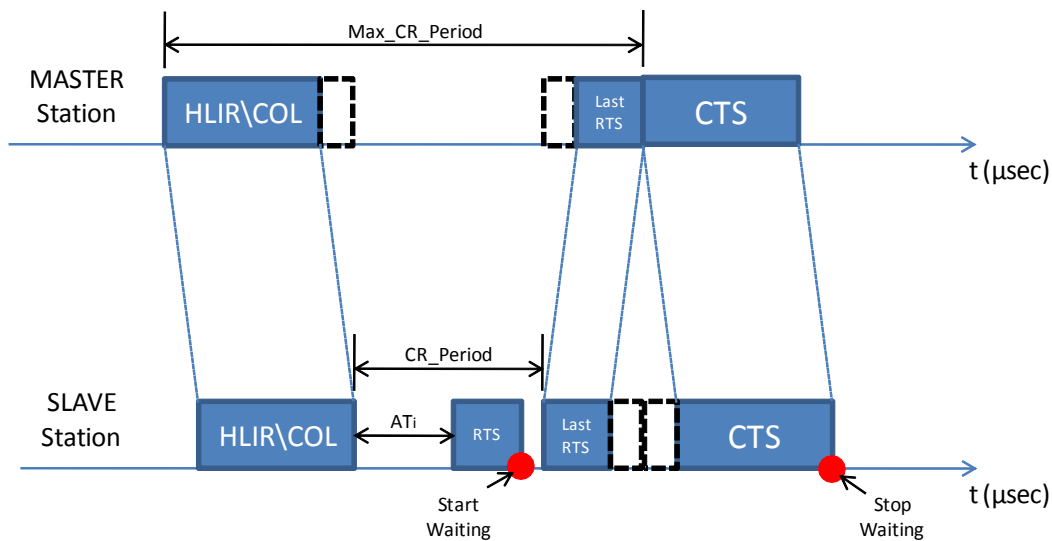


Figure 46 - Definition of MAX_WAIT_FOR_CTS_PERIOD parameter.

The red mark in the diagram indicates on the end of receiving the message. Only then the stations recognize what kind of message was received.

- **MAX_TX_PERIOD**

Defines the maximal time period a station within a cluster has for transmitting its data messages after receiving the grant. The parameter was defined after

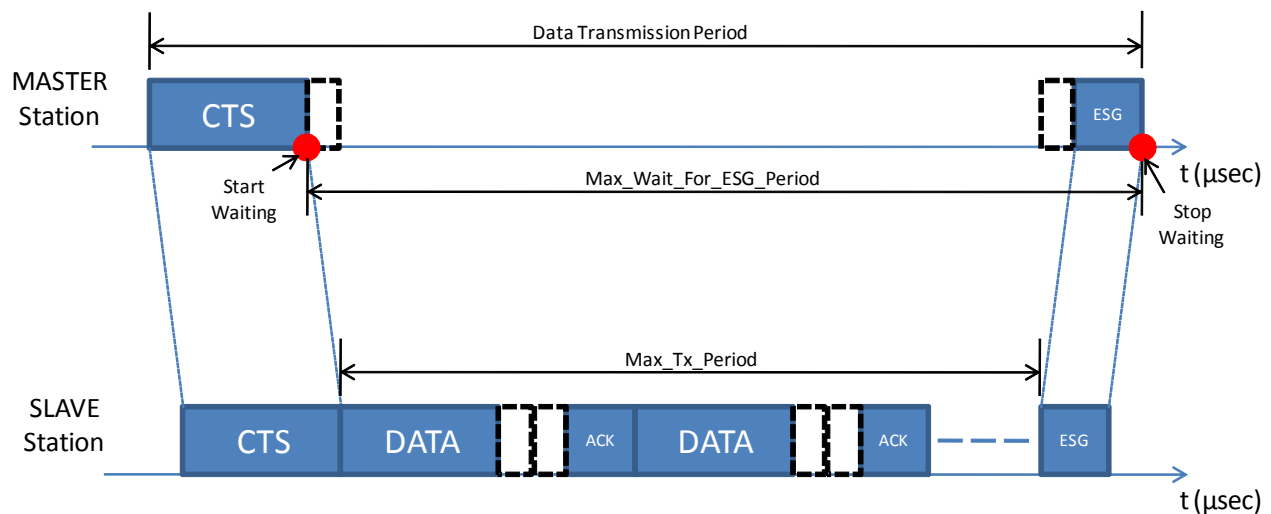
examination of the utilization of the protocol and is optimized for transmission of 10 data messages (and receiving acknowledgement) per station.

(15)

- MAX_WAIT_FOR_ESG_PERIOD

Defines the waiting time period a master station waits for an ESG message from the slave station that received the transmission grant. This period is defined from the completion of transmitting the CTS message by the master station until the receiving of the ESG message in the station.

(16)



In the simulation, timers were defined based on the parameters:

- CR_TIMER

- END_MS_RTS_TIMER
- CTS_TIMER
- ACK_TIMER
- MAX_DATA_SESSION_TIMER
- TX_TIMER

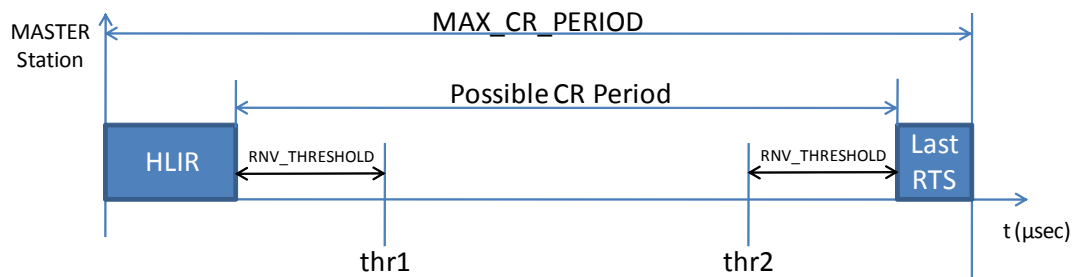
- RNV_THRESHOLD

Defines the values of the thr1 and thr2 threshold parameters that were defined as part of the renormalization mechanism for updating the global parameters. The value of the parameter is defined as a percentage (25% in our design) from the total possible contention time in the protocol, the time period when the master station can receive the RTS signals within its cluster.

(17)

The threshold parameters, thr1 and thr2, are defined in the following way:

- thr1 - relative to the time of completion of the transmission of the HLIR message in the master station and the addition of the parameter RNV_THRESHOLD.
- thr2 - relative to the beginning of receiving the last possible RTS signal in the master station.



Chapter 8. Simulation and Performance Analysis

8.1 Theoretical Background

This section includes a theoretical review for the performance analysis of the protocol in the MAC layer within a cluster with four wireless stations. The following parameters will be explained in the context of a cluster: utilization, minimal message delay, and fairness in resources allocation.

As was explained in the previous sections, a session of the protocol is based on two parts: the first one for the contention and the second one for the data transmission of the winning station. There are two unique sessions in the initialization process and in a wake up / joining process when the master station has to update the global parameters, so it runs two sessions only with the contention rounds.

Regarding the convergence of the protocol (number of contention rounds before transmission of data), [1] describes the probability of collision resolution for up to 50 stations at the beginning of the slotted version of the protocol, after 1 to 6 contention rounds. It is revealed that after 6 contention rounds the probability for contention resolution is almost 100% regardless of the number of stations within the cluster.

The contention round period is fixed in the protocol. The maximal transmission period is defined as the time for transmitting 10 data messages of 100 Bytes. This parameters define the utilization of the protocol in the following way:

(18)

As long as the contention period is shorter than the data transmission period, the utilization of the protocol will be higher.

We designed specific mechanisms in the protocol (which were explained in Chapter 4) such as global parameters update and handling virtual collision in the master station in order to reduce the time spent on contentions, connectivity maintenance and updating as much as possible.

The main consideration that was in front of our eyes while developing the protocol was to keep on a minimal delay for a data message (delay time is calculated from the time the

message was created in the source station until the message is successfully received in the destination station). However, keeping on minimal delay in one station can cause a starvation in another station with data messages in its buffer within the same cluster. Therefore, we defined a maximal number of data messages that can be sent during the transmission period in order to provide fairness between the members of the cluster. Two mechanisms, end slave grant (ESG) and empty cluster indication, were defined in order to minimize the time that the wireless channel is not used in order to improve the utilization and enable faster transmission of other stations.

We examined the utilization of the protocol in a star topology (single cluster) while relating to the contention period in the session, transmission period and the accumulated delay time for a master and slave stations. The difference between the state machines of the winning station (no need in CTS and ESG messages if it is a master station) affects the utilization of the protocol.

The transmission of 15 messages in contention rounds 1 to 4 (one session) was examined and summarized in the following figures:

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
CR1	0.592122	0.743815	0.813264	0.853089	0.878914	0.897016	0.91041	0.920721	0.928904	0.935555	0.941068	0.945713	0.949679	0.953104	0.956094
CR2	0.484407	0.652661	0.73812	0.789831	0.824487	0.849332	0.868015	0.882576	0.894243	0.903801	0.911775	0.918528	0.924321	0.929345	0.933743
CR3	0.40985	0.58141	0.675689	0.735306	0.776408	0.806461	0.829392	0.847465	0.862076	0.874132	0.88425	0.892863	0.900282	0.906741	0.912414
CR4	0.355183	0.524184	0.622995	0.687823	0.733627	0.76771	0.79406	0.815042	0.832143	0.846349	0.858339	0.868593	0.877462	0.88521	0.892037

#Data Msg's	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
CR1	0.726989	0.841915	0.888748	0.914174	0.93014	0.941097	0.949084	0.955163	0.959945	0.963806	0.966987	0.969655	0.971924	0.973877	0.975576
CR2	0.571078	0.726989	0.799771	0.841915	0.869403	0.888748	0.903101	0.914174	0.922975	0.93014	0.936085	0.941097	0.945381	0.949084	0.952316
CR3	0.470232	0.639671	0.726989	0.780243	0.816112	0.841915	0.861368	0.876558	0.888748	0.898746	0.907096	0.914174	0.920249	0.925521	0.93014
CR4	0.399657	0.571078	0.666349	0.726989	0.768977	0.799771	0.823322	0.841915	0.856968	0.869403	0.879849	0.888748	0.896419	0.903101	0.908973

Figure 50 - Calculation of utilization for a master station that received a transmission grant.

- The utilization of the protocol is higher in a master station
- As long as there are more contention rounds in the session the utilization of the protocol decreases (see Figure 51)
- As long as more data messages will be transmitted during the transmission period the utilization of the protocol will increase but it will cause starvation and delay increase in other stations.

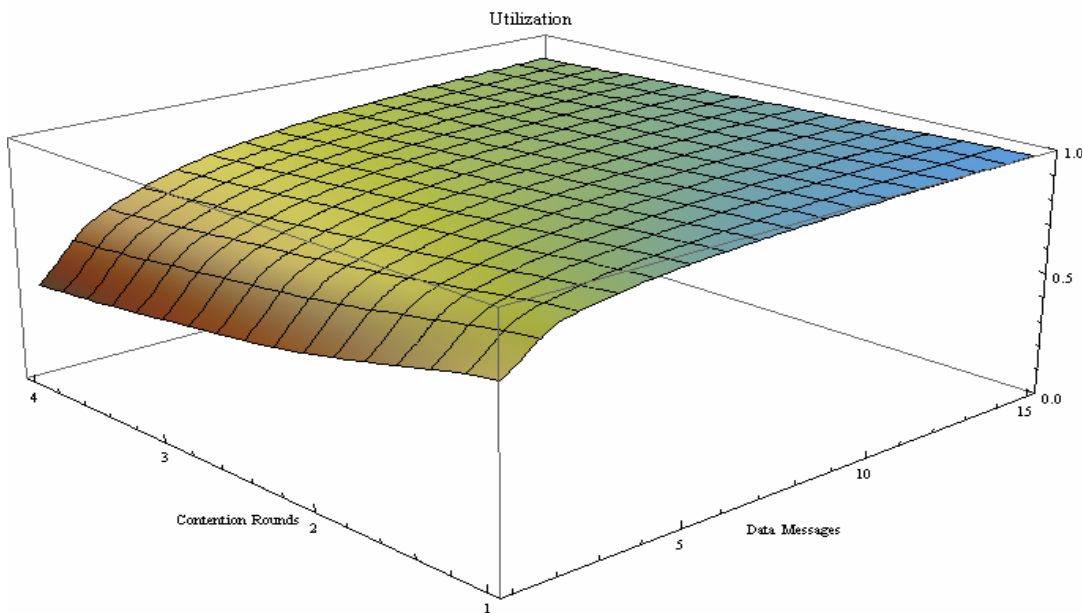


Figure 51 - Utilization graph of the protocol.

The time that is required for transmission of 10 data messages of 100 Bytes each (acknowledgment is received for each message) is seen in Figure 52.

Data message	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
TxTimer(μ Sec)	93.2	186.4	279.6	372.8	466.	559.2	652.4	745.6	838.8	932.	1025.2	1118.4	1211.6	1304.8	1398.

10 messages were defined as the maximal numbers of messages to be transmitted by the station in a session and therefore, the maximal transmission time is 932 μ sec as was defined in timing parameters (Chapter 8).

Fairness

The uniqueness of the protocol in guarantying fairness between stations within the cluster is based on the load balance function and special mechanisms. The real/virtual collision mechanism ensures that only the most loaded stations will continue to compete for the transmission grant (based on random access) and the timing definitions of the protocol keep on low accumulated message delay as much as possible.

8.2 Simulation Description

8.2.1 Measured Parameters

In order to verify the correctness of the model under multiple hop scenarios and using routing protocols, traffic generators and transport layer protocols, we implemented the protocol using OMNeT++ simulator.

For each simulation the following parameters were taken into consideration:

- Number of bytes in a queue per station and maximum in the network.
- Message's delay time measured from the packet's generation time until it is received by its final destination. This parameter is measured per station. We examine the maximal delay of a message in a station compared to the maximal delay of a message within the cluster.

- Convergence of the protocol - number of contention rounds until successful transmission.
- Utilization of the protocol - was defined as the accumulation of the transmission period of the stations in the cluster divided by the total simulation time.

8.2.2 Topology

The protocol was examined on a star topology, a cluster with a master station and four slave stations as seen in Figure 53. Since our model is supposed to cope well with topologies with hidden terminals, we decided to test the worst-case star topology scenario. Therefore, we examined a case where there are 4 slave stations and all their traffic is sent to one central master station. The master station is within communication range of all slave stations, while the slave stations cannot communicate with each other. It is somewhat difficult to construct a star topology in a wireless network environment, since the propagation model usually depends mainly on the distance between stations. This implies that in most wireless propagation models it is almost impossible to construct a star topology of more than 5 slave stations.

Each station processes the relevant state machine based on its mode. The stations are placed by defining spatial coordinates (X, Y) and the SAT parameters that defines the transmission radius. The spatial coordinates can be part of a path whenever mobility scenario is implemented. The normalization parameters (thresholds) were set to 25% of the total contention period.

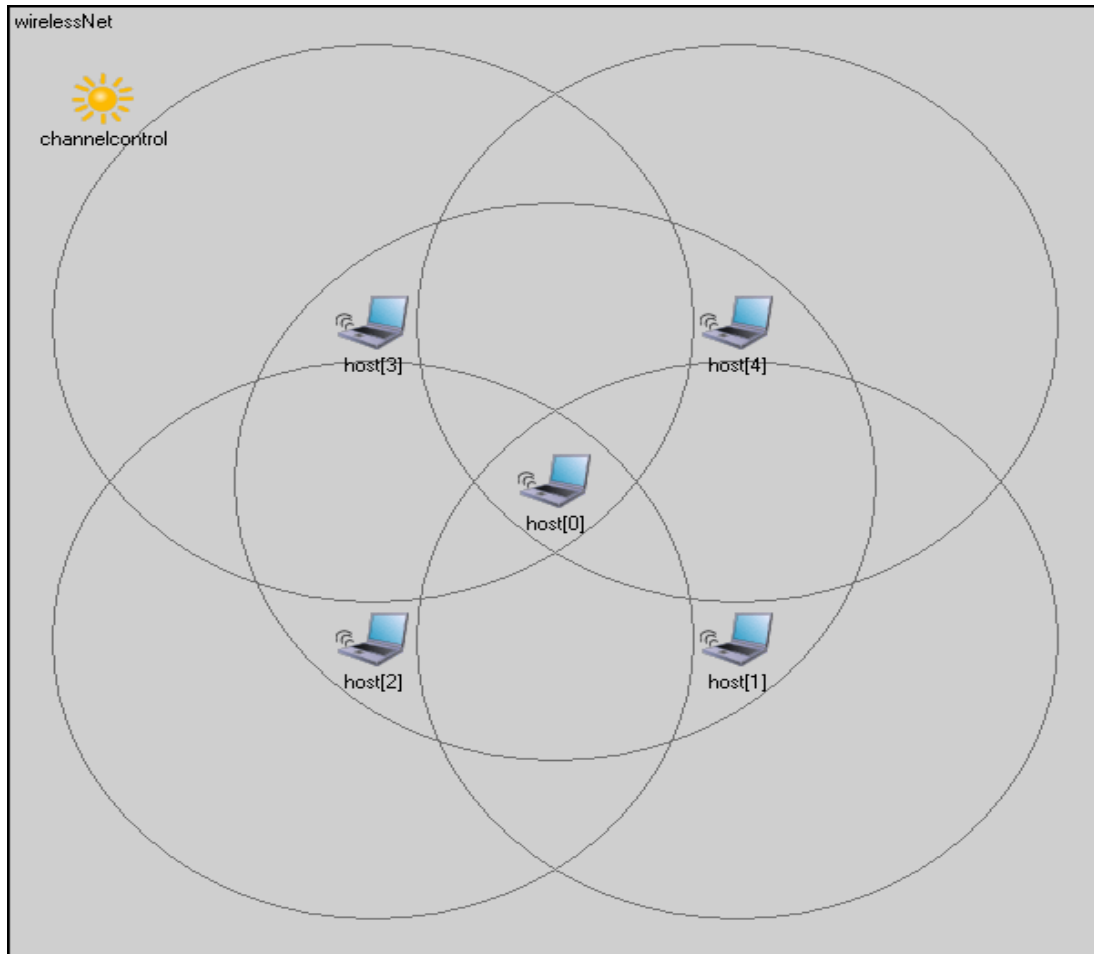


Figure 53 - Star topology of a single cluster.

8.2.3 Generation of Data Messages

In our study, a maximal data message length is 100 bits and a buffer that can contain 40 different data messages (maximum 4,000 bytes). For each station, a random number of data messages was created. Each data message generated in the network layer contains

the following fields: sequential number, identification number of the station, MAC and IP addresses of the source and the destination, and the creation time of the message, defined as creationTime. The message is sent to the HigherMAC and is placed in the queue, if the queue is not full. The delay time of a data message is an important component in the load function and is based on the value of creationTime and of the current time in the simulation, $\text{delayMsg} = \text{simTime}() - \text{creationTime}$.

8.2.4 Running the Simulation

A number of network initialization steps take place in the beginning of the simulation. The main step responsible for forming the hierarchies in the cluster comes into effect when $\text{host}[0]$ is defined as the master station of the cluster by internal notification, while the other stations remain in slave mode. At the end of the session, $\text{host}[0]$ returns to slave mode and waits for notification from the network layer whether to proceed to the next session in a master mode. In a complete simulation handling inter-cluster processing, a token will be passed to another cluster head (master station) if the current master station does not continue to the next session.

In the following steps we will summarize the processing of simulation:

- The initial values of the global parameters at the master station are:
 $\max\{Q_i\} = 4000$ [Bytes]; $\max\{D_{\max,i}\} = 0.1$ [sec]
- The first HLIR message is sent with the initial global parameters at simulation time of $5\mu\text{sec}$ by the master station in order to update the values of the parameters as mentioned in the explanation on the mechanism.
- For the longest location of a slave station from a master station (300 meters), a correction factor can be added to the estimated load.
- The estimation of the global parameters is conducted during the first two sessions of the simulation (and further, depending on the decision of the network layer). The value of α , the equilibrium parameter, depends on the processing stage in the protocol - during estimation/update of the global parameters the value of the parameter will be 0 (for $\max\{D_{\max,i}\}$ estimation) or 1 (for $\max\{Q_i\}$ estimation).

During the contention rounds the value of the parameter will be 0.5 or 0.75 as seen in the AirFrame fields sent from the master station in Figure 54.

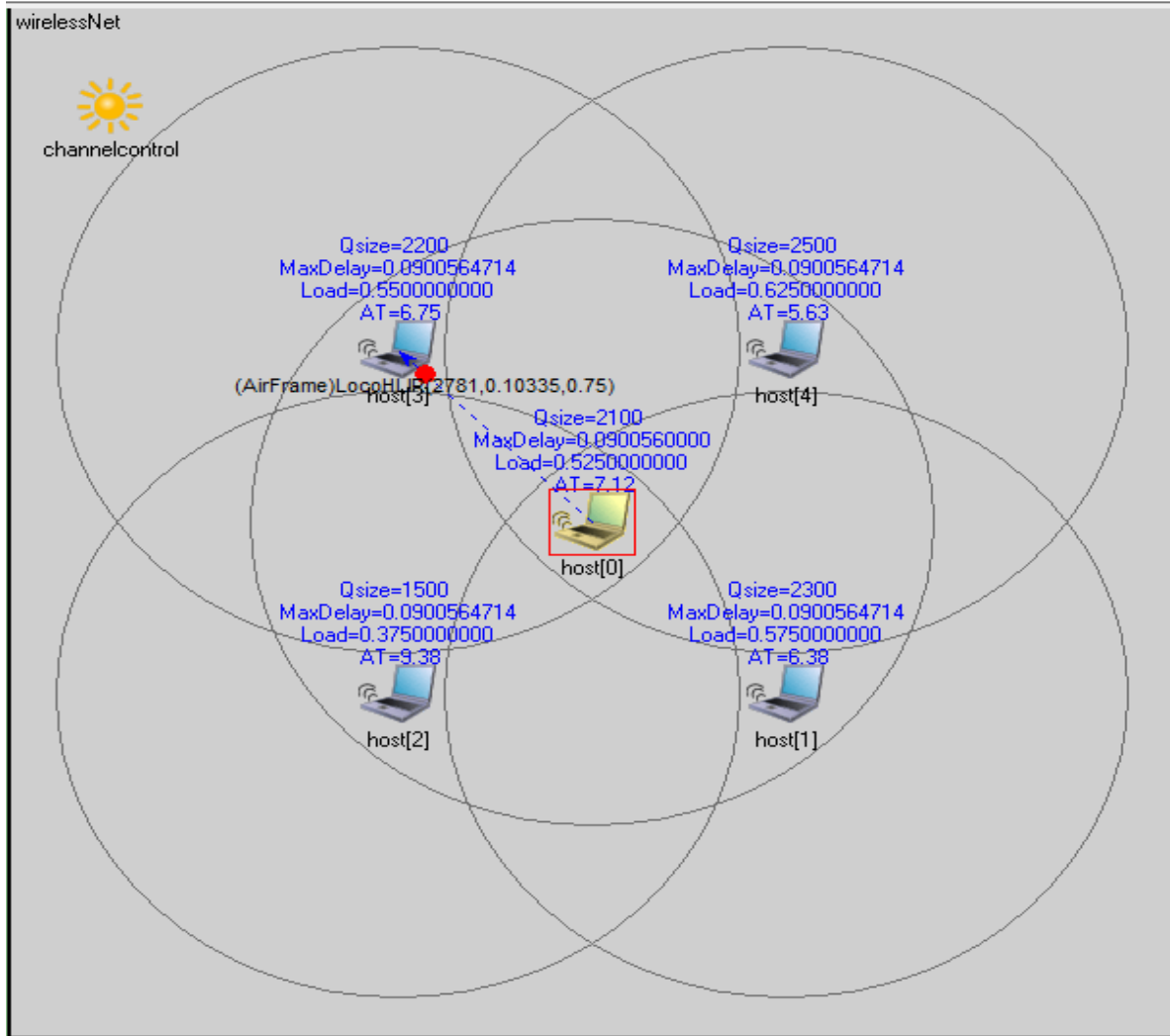


Figure 54 - Single cluster during the execution of the protocol.

- From now on, every time an HLIR message is sent, it will contain the estimated global parameters unless the renormalization mechanism is operated.
- As seen in the previous figure, each station maintains the following parameters throughout the simulation: queue size (Qsize), maximal delay of a message in the queue (MaxDelay), internal load (Load), and the access time (AT) for

transmitting the RTS signal. The parameters are being updated with each receipt of HLIR message or COL message, as well as during data transmission.

- In each session the station with the shortest AT is to be the first to transmit the RTS signal in CR_1 .
- The master station estimates the maximal load in the cluster, detects a collision (if occurred) and its type (real or virtual) and decides on the loaded station among its 1-hop neighbors.
- Depending on the distribution of RTS signals arrival times in CR_1 , the master station decides if the global parameters should be updated (renormalization mechanism). If the update took place, the new global parameters will appear only in the HLIR message sent during the next session.
- If the master station is the most loaded station in the cluster, it transmits its data without sending a CTS message.
- If a collision (virtual or real) occurs, the collided stations only continue to compete in the next contention rounds (CR_{other}) using a random access mechanism for transmitting the RTS signal until convergence to only one winning station.
- After transmission of a data message the source station waits for an ACK message to be received from its 1-hop neighbor (based on MAC address). In case the ACK message was not received, a retransmission mechanism is operated. If after certain number of retransmission attempts the message still doesn't reach its destination, the network layer decides if it will be erased from the queue or its MAC address will be changed to the MAC address of the master station, and it will function as a relay node.
- At the end of the transmission period of the slave station an ESG message is sent towards the master station. A new session starts if the master station has the token (or counter for sessions).

- Empty cluster - when all the queues of the stations in the cluster empty out, as presented in Figure 55, the queue size, message delay, and internal load are zeroed, while the value of the AT parameter becomes 15μsec and the simulation terminates.

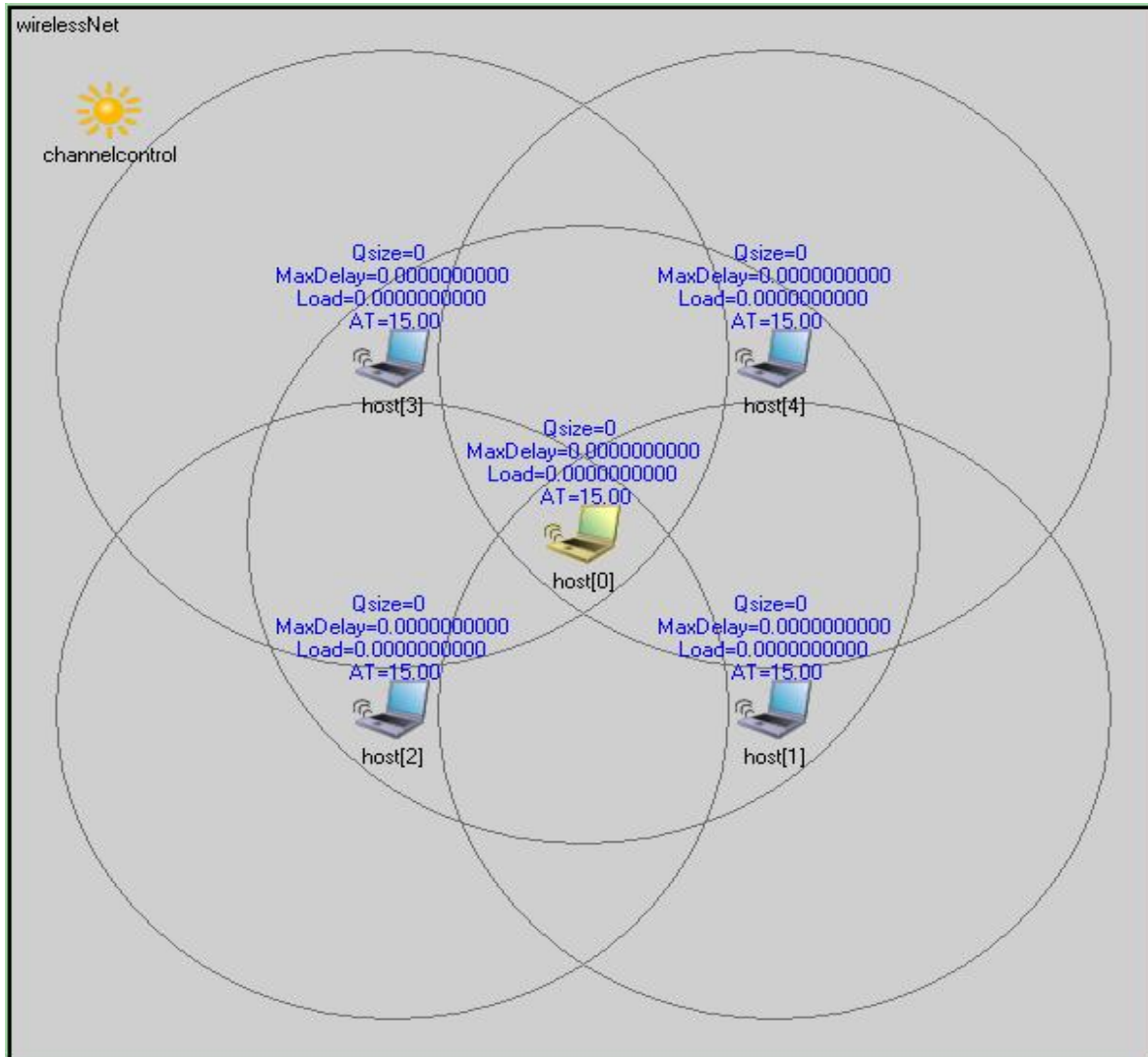


Figure 55 - Empty cluster in the simulation.

8.3 Performance Analysis

In this section we will examine 3 different scenarios in a star topology:

1. Varying loads between the stations within the cluster
2. High load for all the stations within the cluster

3. Low load for all the stations within the cluster

For each scenario the internal values after load calculation in CR_1 (of the first session) will be presented, as well as the utilization of the protocol, distribution of number of contention rounds before transmission (convergence) in sequence of sessions, fairness criteria, and an analysis of maximal delay of a message within the cluster.

Varying Loads

The stage after parameters estimation is described in the following figure. It can be seen that host[3] is the most loaded, thereby has the shortest access time, while host[1] is the less loaded station in the cluster.

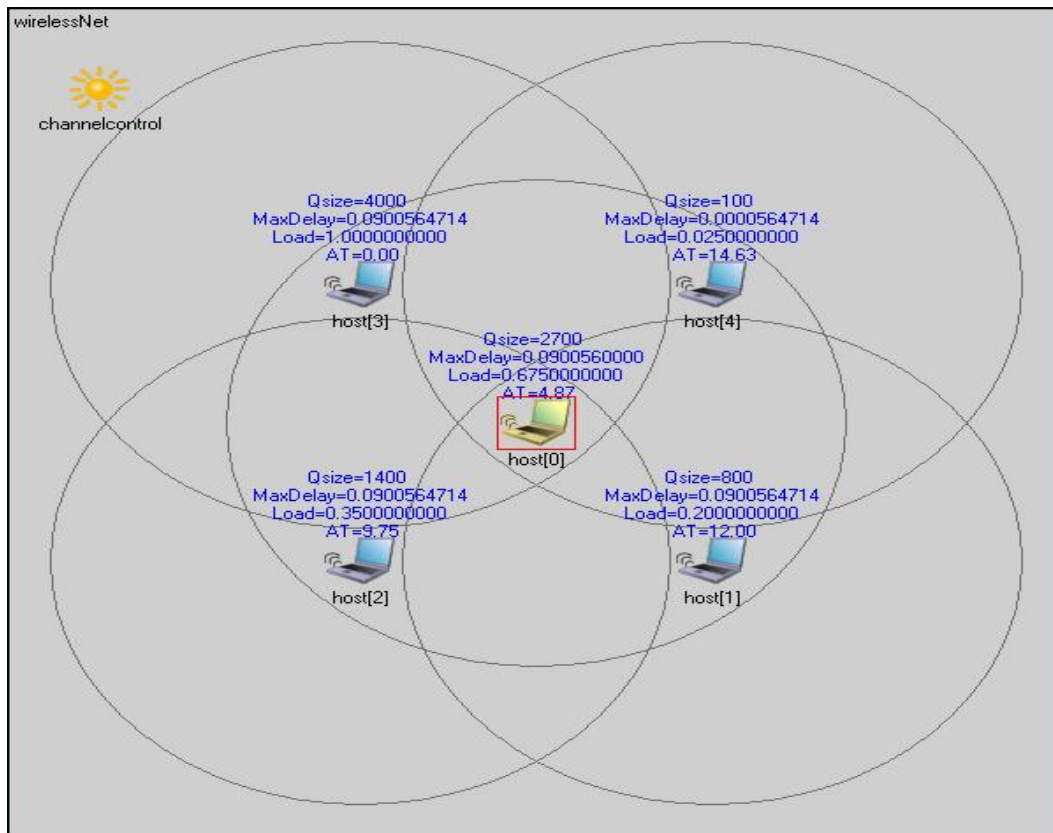


Figure 56 - Varying loads scenario.

Utilization of the Protocol

We define the utilization of the protocol as the accumulation of the time periods allocated for the stations to transmit their data divided by total time of the simulation as described in the following expression:

(19)

The transmission period of a station starts from the moment it receives the CTS message until the occurrence of one of the following three scenarios: (a) the station has nothing to send; (b) no time is left for more messages to be transmitted and receiving ACK; (c) the time dedicated for the transmission period is over.

The term *transmission period* as taken into consideration in the expression above, refers to the total transmission period of a station during a session (after receiving the grant) and not to the actual time of transmission of the messages; meaning that it includes the time necessary for the ACK mechanism, propagation times of messages, and processing of the retransmission mechanism.

By analysis of the log time from the simulation, we received a high value for the utilization of the protocol, in a single cluster star-based topology, 91.48%.

Contention Rounds in a Session

We examined the distribution of the number of contention rounds required for the convergence of the protocol along simulation of 22 sessions. It can be seen from the following graph that in 13 out of 22 sessions, only one contention round was required, meaning that no collision occurred, to allocate the transmission grant for the most loaded station. In 8 out of the 22 sessions, a collision occurred in the second round and the random access mechanism was operated.

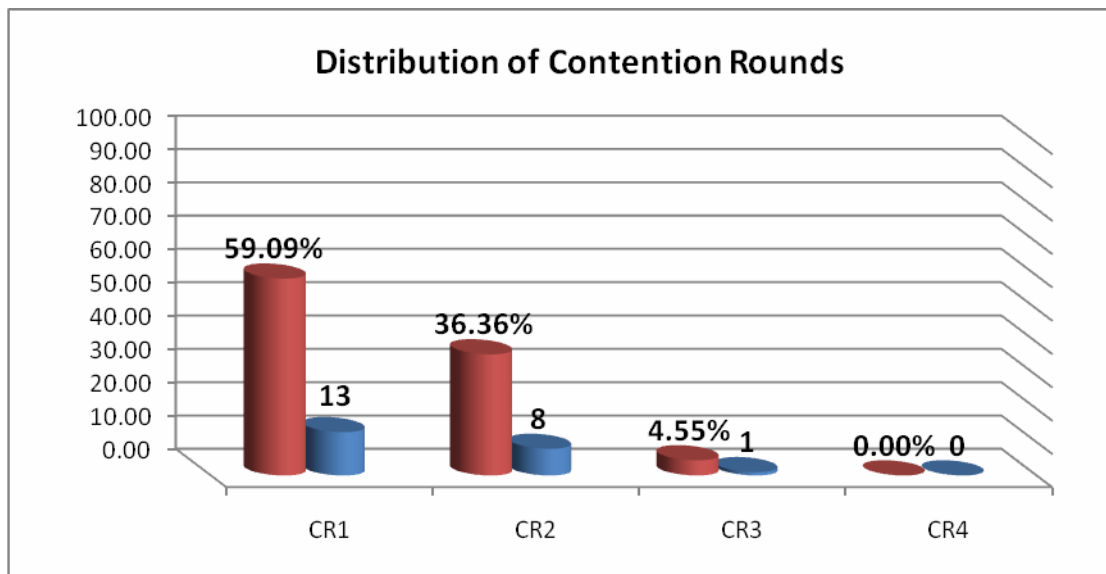


Figure 57 - Number of contention rounds per session along 22 sessions.

Maximal Delay of a Message within the Cluster

The destination station of the generated messages in the stations was set to host[2]. In the following graph we can see the longest period a data message stays in the queue of the station (from the moment of its generation) compared to the maximal delay of a message in the cluster, meaning the total time from creation until it is received in the destination station. Host[4], which is the less loaded station, has the highest message delay in the cluster due to its low priority during the contention on the transmission grant. Host[0] emptied out the first and continued to function as a relay node.

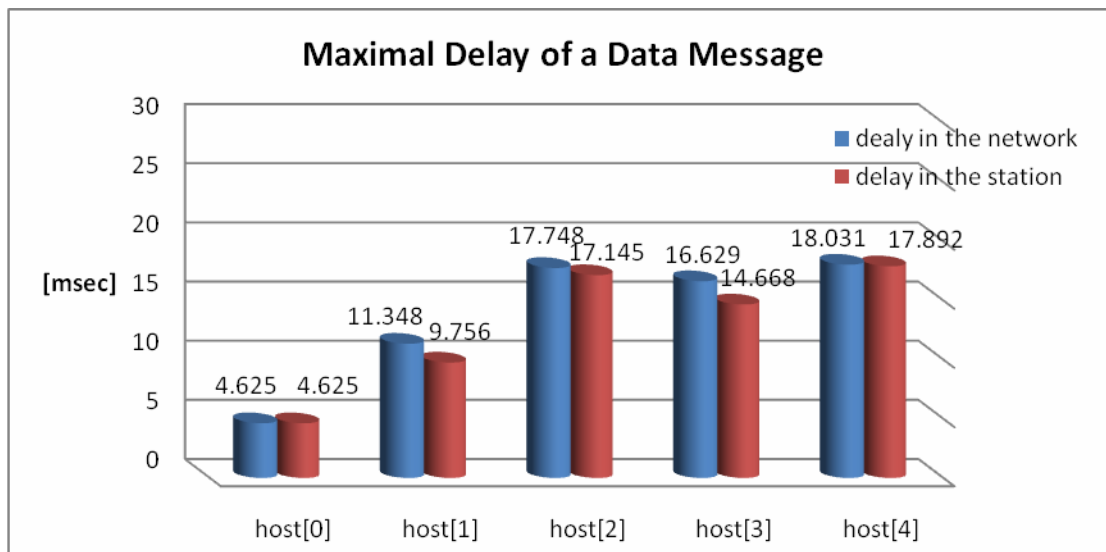


Figure 58 - Maximal delay of a data message along 22 sessions.

High Loads

In the second scenario we run a simulation with 39 sessions when each station had a high load, the length of the queues is close to the maximum per station, as presented in Figure 59.

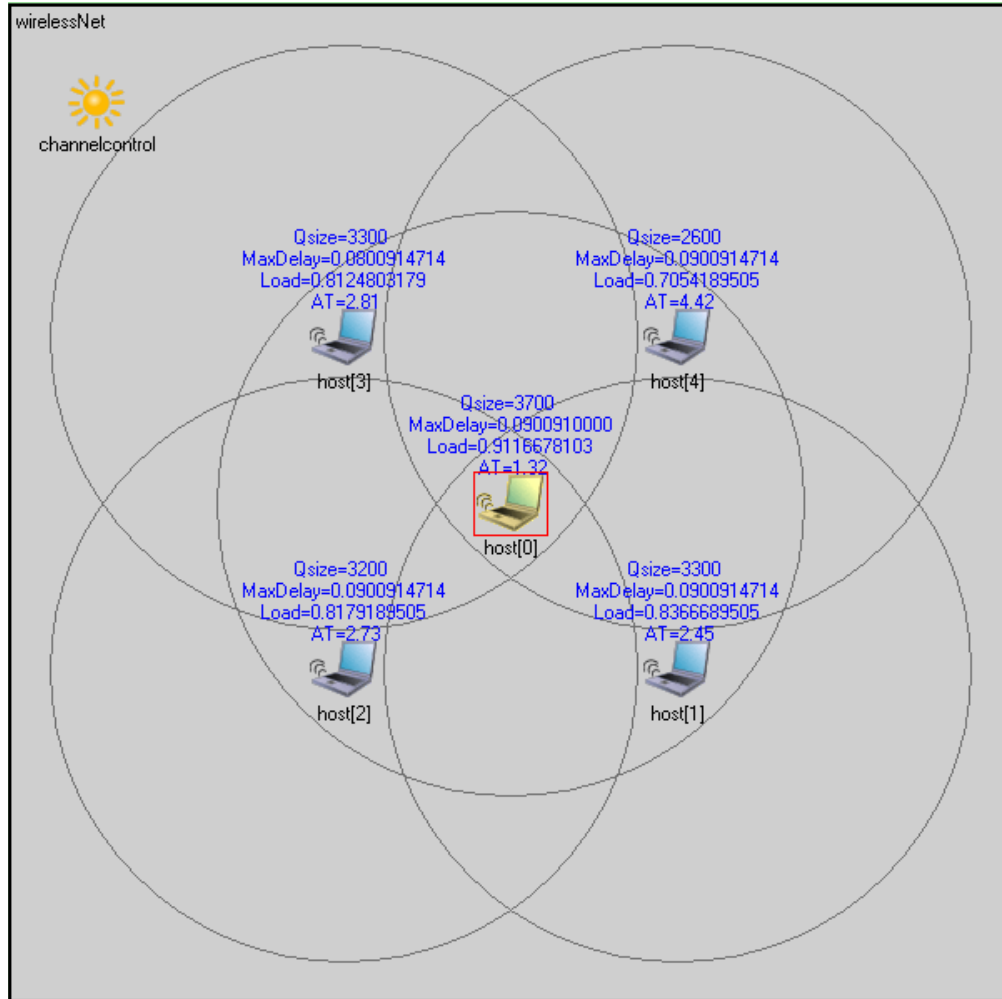


Figure 59 - High loads scenario.

Summary of the Results of the High Loads Scenario

The utilization of the protocol in this scenario is 90.91%

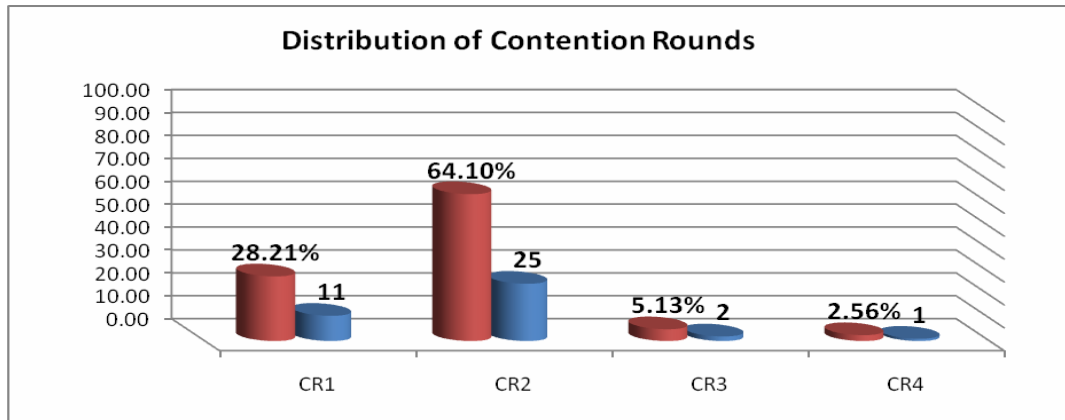


Figure 60 - Number of CRs per session along 39 sessions in high loads scenario.

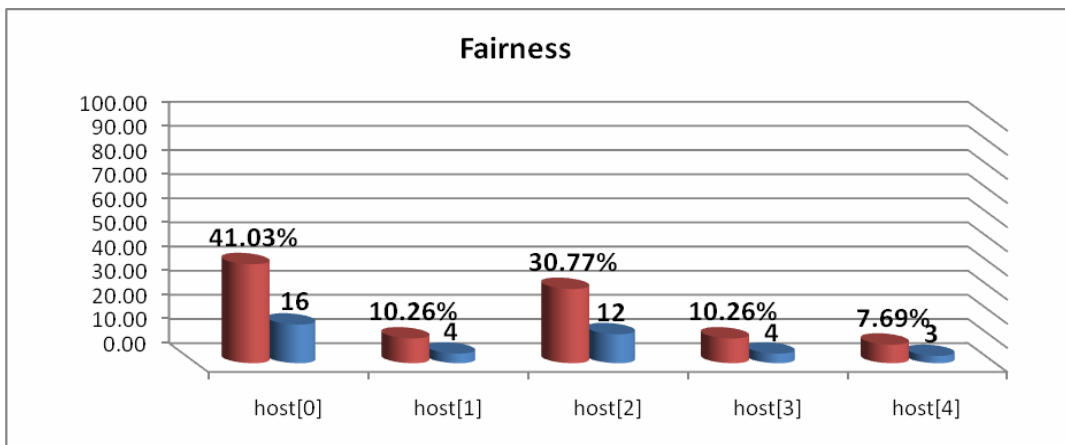


Figure 61 - Fairness criterion per each node in the cluster in high loads scenario.

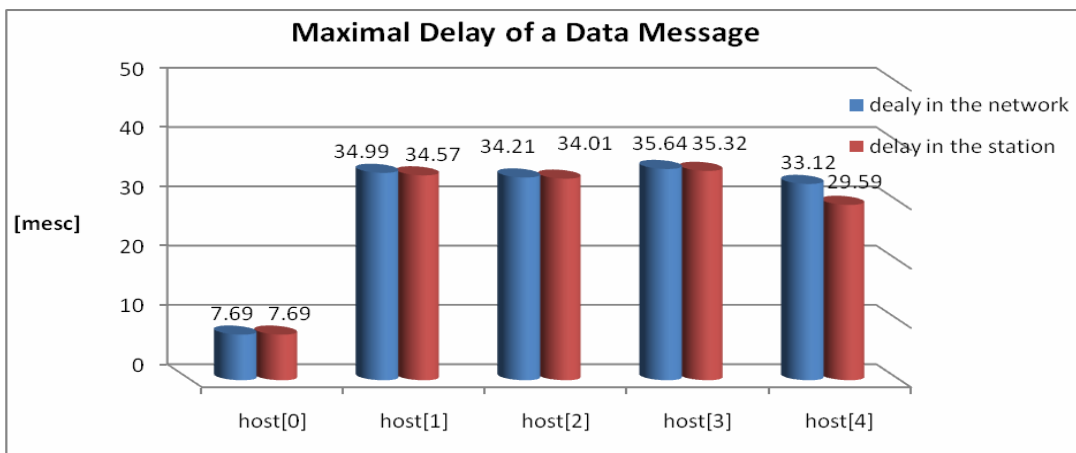


Figure 62 - Maximal delay of a data message per station/cluster in high loads scenario.

Low Loads

In the third scenario as seen in Figure 63, we ran a simulation with 11 sessions, while each station had a low load, and 5-11 messages per station were generated only.

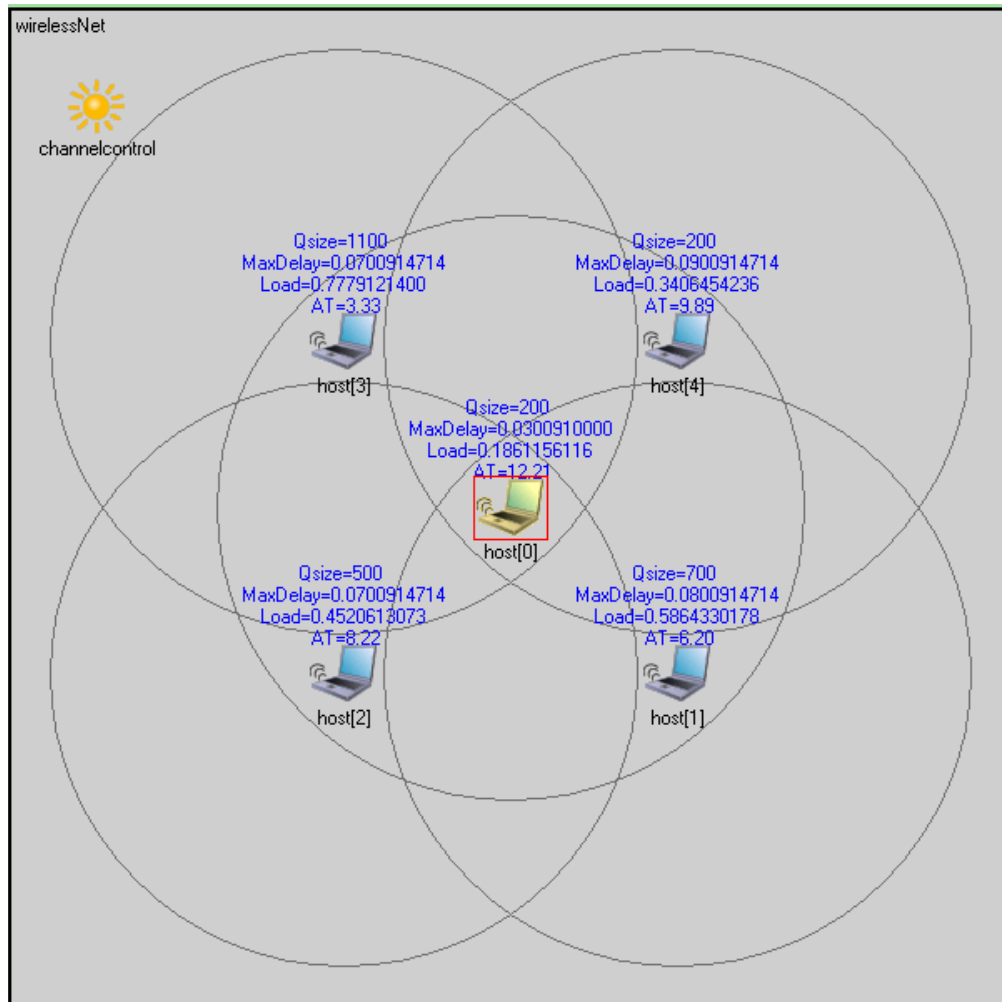


Figure 63 - Low loads scenario.

Summary of the Results of the Low Loads Scenario

The utilization of the protocol in this scenario is 87.56%.

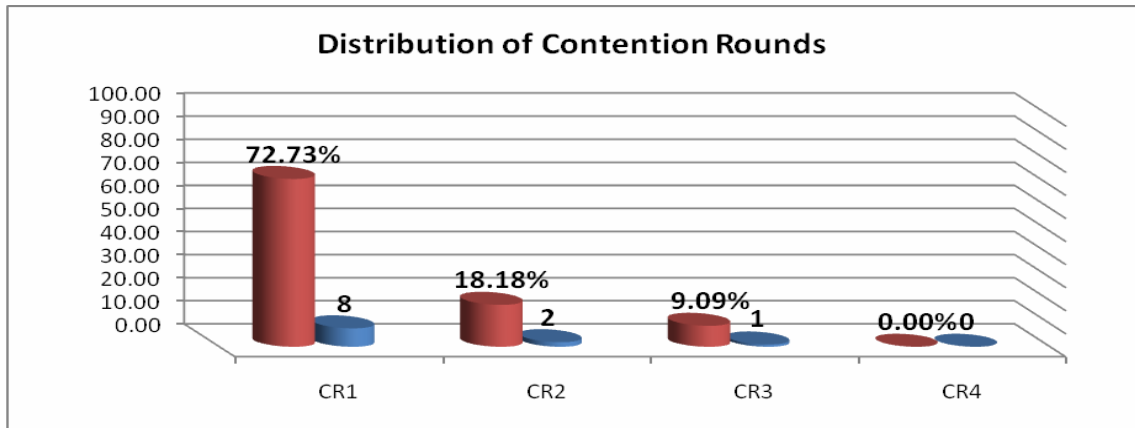


Figure 64 - Number of CRs per session along 11 sessions in low loads scenario.

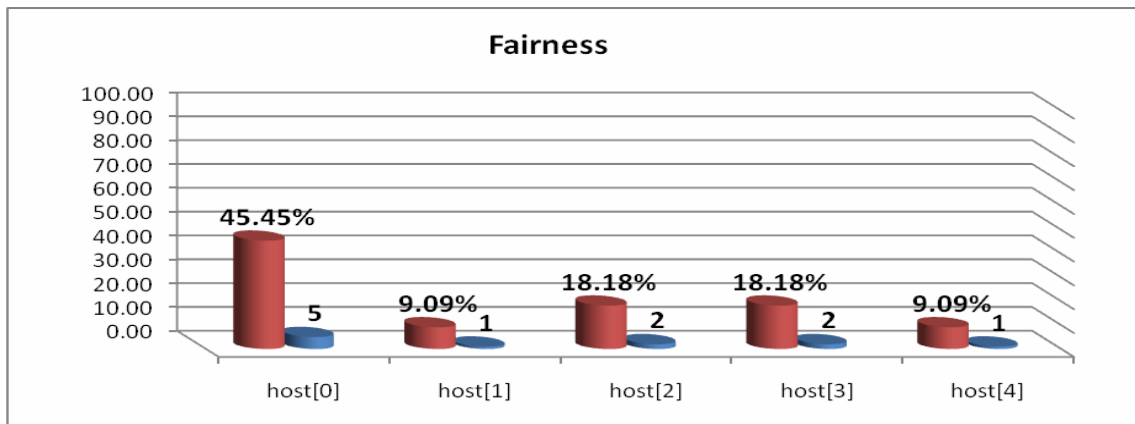


Figure 65 - Fairness criterion per each node in the cluster in low loads scenario.

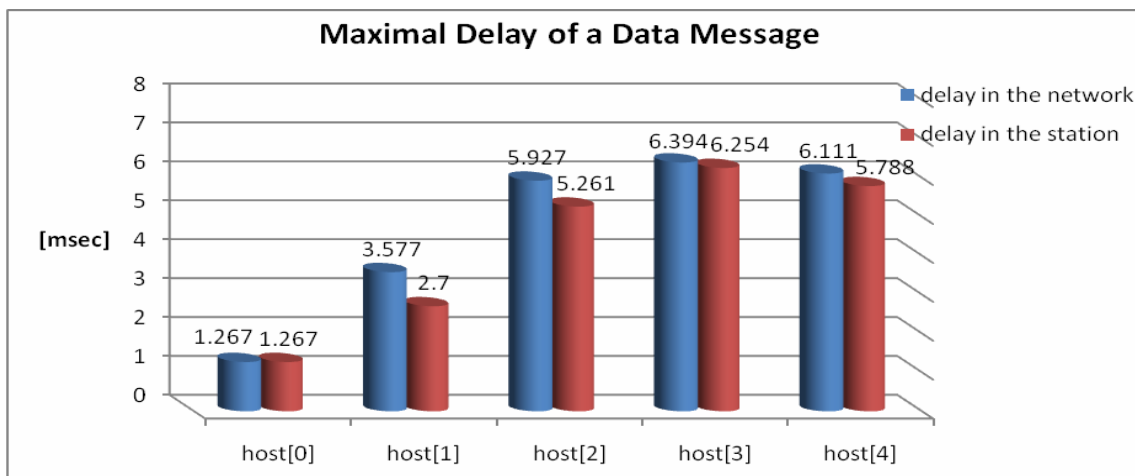


Figure 66 - Maximal delay of a data message per station/cluster in low loads scenario.

Conclusions

The utilization of the protocol being achieved in the two scenarios of high and low loads is pretty high: 91% and 87.6% respectively. In the high loads scenario, it can be seen in the distribution of contention rounds that in most of the first contention rounds in the sessions, a collision occurred and the transmission grant was given only in the next contentions, while the random access mechanism was operated. Due to the high capacity of queues in the cluster, a station that receives a transmission grant utilizes the whole period of time dedicated for transmission. In the low loads scenario, it can be seen that most of the contentions were resolved in the first contention round, which is a reasonable result for this scenario. The tradeoff between the short contention round period (35 μ sec) and the longer transmission period (932 μ sec) should be investigated more deeply for extreme scenarios in a future work. Moreover, in the low loads scenario, the cluster reached an empty cluster status very fast and an estimation mechanism should be developed in order to enable neighbor clusters to start their sessions earlier when considering inter-cluster performance analysis.

Chapter 9. Future Work

In this paper a novel fast message passing and adaptive load-based contention resolution protocol was developed, analytically analyzed and extensively simulated. The contention resolution is achieved by using time-constant, short modulated energy pulses. In our work, we were the first to develop a message passing protocol-based contention which is compatible with dominating sets (DSs) algorithms that are very popular in current ad-hoc networks research (to the best of our knowledge).

From [1] the main contributions of the slotted protocol version were that it was found stable and provided fair bandwidth share among several competing stations at high throughput regardless of the network connectivity conditions, the number of competing stations, or their loads. The MAC did not lose any of its effectiveness under hidden terminal conditions (including the worst case star topology), and consistently achieved stable and high throughput of about 80% and goodput of about 90%, even when the traffic load momentarily exceeded the maximum network capacity.

Our protocol enables high utilization within a cluster, and as long as stations have data to transmit the most loaded stations will win the grant and would be able to transmit their data messages. The master and the slave stations participate in the contention process and if the master station is the most loaded it can start transmitting without spending time on contention rounds. The master station can gain knowledge on its new 1-hop neighbors while detecting collisions in the contention process. One enhancement that must be made in order to make the protocol fit mobile ad-hoc networks is to implement the suggested EMR mechanism. New stations, whether during their mobility or after the wake up process, will be able to join clusters even if the cluster head does not have a token (network level) at that moment. A detailed state machine description is presented in Chapter 6 for master and slave stations.

One more mechanism that should be examined is the global parameters update.

However, it should be noted that the purpose of this thesis was not to devise an algorithm for network initialization, and the global updating mechanism was added to the protocol as part of the unique interaction design between the MAC and the network layers. We explore novel MAC mechanisms and examine their behavior and performance in order to be compatible to current backbone forming and routing algorithms based on dominating sets (DSs) in their different forms, especially weakly connected dominating set (WCDS). There are two major issues when investigating and developing the network layer over the proposed MAC protocol:

One is to develop a robust inter-cluster token passing mechanism (we handle only intra-cluster issues in this thesis) and the other is to define the algorithm for actually selecting, maintaining and updating the clusters, and in particular the cluster heads (master stations). In the network layer the basis for maintenance of the backbone-based WCDS and the handling scenarios of different interactions of nodes and clusters was formed. The state machines of the MAC layer were designed to support cross-layer interaction with the network layer and a state machine for the network layer can easily be developed and utilize the interfaces with the MAC.

In this sense, a lot of work can be done to extend the current scope of the protocol to allow it to better fit continuously changing connectivity or mobility conditions. It will enable performing a huge verity of simulations with different topologies in different mobility scenarios, and deeper investigation of the performances in inter-cluster environments. In conclusion, we believe that the mechanisms introduced by the protocol can also be used in any master-slave based network environment as well.

Additional smaller topics that could be further developed are to include improvements and refinements to the load function to better support criteria other than maximum queue lengths and maximum network packet delay (e.g. jitter control, BER, etc.). One might also consider developing a new routing algorithm through the advantages of the cross-layer designed protocols. Extending the protocol to support multiple channels, and adding more initialization features to the protocol can also serve as a good baseline for future work.

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תקציר

A Message-Passing and Load-Sensitive Contention-Based MAC Protocol for Ad-Hoc Wireless Networks

מאת

אריאל סמוליאר

בעבודה זו מוצג פרוטוקול MAC חדיש מבוסס על מנגנון העברת מסרים עבור רשתות Ad-Hoc, שמתאים את עצמו באופן אדפטיבי לפרמטר העומס של התחנות באשכול (Cluster). החדשנות בפיתוח הפרוטוקול היא ביכולת לפתור במהירות התנגשויות בין תחנות המשדרות בו-זמנית, בין אם זאת תחנת ה-master או תחנת ה-slave בתוך האשכול ובהתאמתו לאלגוריתמים חדשים המפותחים ברמת הרשת (Network) ליצירת backbone בין התחנות הניידות ומבוססים על Dominating Sets. זמן הגישה לערוץ בפרוטוקול הוא פונקציה של העומס הנוכחי של התחנה שמחושב כתלות בפרמטרים הלוקאליים בכל תחנה, כלומר בזמני ההשהיות ובנפח המידע הקיים בתור. תחנת ה-master מתחילה את ה-session על-ידי שליחת הודעת HLIR ובהתאם להודעות בקשה לשידור ששולחות תחנות ה-slave ניתן לשערך את עומס התחנות ששידורו את הבקשה אפילו אם התרחשה התנגשות בערוץ. הקצאת הגישה לערוץ ניתנת לתחנה בעלת העומס הרגעי הגבוה ביותר.

תחנת ה-master מחזיקה את הפרמטרים הגלובליים (מספר בתים בתור והשהייה מקסימליים בין כל התחנות) של האשכול ותחנות חדשות שמצטרפות לאשכול יכולות לחשב את העומס בהתאם לפרמטרים האלה. פיתוח הפרוטוקול התבסס על עבודה קודמת שבה הוצג לראשונה מנגנון לשיערוך עומסים כשבעבודה הנוכחית פותחו מכונות מצבים לשכבת ה-MAC תוך תכנון המשלב בין שכבות ה-MAC וה-Network (Cross Layer Design) על-מנת לאפשר לפרוטוקול לפעול בטופולוגיות משתנות, הצטרפות ועזיבה של תחנות, כתוצאה מניידות ותהליכי התעוררות. באמצעות סימולטור רשתות בוצע אימות ליכולות הפרוטוקול להגיע לניצולת גבוהה מ-90%.

בעבודה זאת נבחן את הפרוטוקול ברמת אשכול בודד (Intra-Cluster) וניפתח פתח למחקר רב ברמת רשת מרובת אשכולים (Inter-Cluster).